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the unforgetable year

CHAPTER 4

CAPITAL RATIONING

Question 1

Adetula Plc is considering selecting from the fallowing independent projects:

PROJECT	Initial Outlay	NPV
A	N140,000	N 32,000
B	N260,000	N 65,000
C	N 35,000	N 15,000
D	N340,000	N135,000
E	N110,000	N 28,000
F	N 65,000	N 21,000
G	N 95,000	N 36,000
H	N275,000	N 92,000

The company has a budget limit of N400,000. Calculate the Profitability Index of each project. Hence, advise the company on the acceptable projects assuming projects are divisible.

Question 2

The Sanusi Company is considering identical projects as follows:

	PROJECTS					
	A	B	C	D	E	F
Initial Outlay (N)	32,000	25,000	40,000	35,000	22,000	50,000
Annual Net Cashflow (N)	12,000	9,000	15,000	10,000	8,000	21,000

Each project is scheduled to have a life of five years.

The company hopes to be able to rationing N100,000 out of the N160,000 budgeted for capital expenditure. The cost of capital is 16% which combination of projects will maximize the value of the firm, assuming the projects are:

1. Divisible
2. Indivisible.

Question 3

Abalti Plc is experiencing a shortage of funds for capital investment in the current year, when only N24 million is available for investment. No shortage, however, is foreseen thereafter, the following independent projects have been presented to the board of directors of Abalti Plc.

Project	A	B	C	D	E	F
Cost (N'000)	10,000	2,800	4,400	8,400	6,800	5,400
Annual Cash Receipt from the project (N'000)	2,900	1,200	1,800	2,800	2,800	1,400
Expected life (years)	6	5	3	8	4	7

The cost of capital of Abalti Plc is 16%.

Which projects would you recommend the company to undertake under the following alternative assumptions:

(i) Projects are divisible?
(ii) Projects are indivisible?

Question 4

Gabbage Ltd., is an all equity finance company, the economic policies of the recent time do not favour the company, hence the management diversification campaign. Mr. Bisi who is the company's Investment Manager has identified the following areas of diversification:

(1) Buying shares in WIN Nig. Plc.
(2) Establishing a video club.
(3) Palm Oil Plantation.

The Investment Manager has been able to generate these investigation ideas with the associated cash flows - one major set back is that no one can be delayed for the other.

The retained earning of the company stands at N250m. Also dividend due from a subsidiary company amounting to N160m has been lodged in the company's bank account. The only funds at the company's disposal are as stated above, that is the company's retained earnings and dividend just received.

For a corporate company like Gabbage Ltd., to invest in the shares of WIN Nig. Plc a sum of N140m is required. From the past records, Gabbage Nig. Plc. will receive a constant annual dividend of N30m for the next 10 years when the shares will be sold off for 666⅔%. Whereas, to establish a video club a sum of 180m will be required immediately. The video club will generate a net annual sales of N35m for the next 10 years when the video club will be scrapped for N45m.

On the other hand, Palm Oil plantation will cost N20m more than the cost of video club by year 0; and because of the long gestation period, sales of palm products will start in year 4 as stated below:

	Sales	Cost of Sales	Admin Cost
Year 4	N110m	N20,000,000	N2,000,000
Year 5	120m	20,000,000	4,000,000
Year 6	150m	26,000,000	4,000,000
Year 7	160m	30,000,000	2,000,000
Year 8	180m	30,000,000	6,000,000
Year 9	190m	28,000,000	10,000,000

The Company's cost of capital is 12% and all projects are deemed divisible and undelayable.

Required: Based on the above information, which project should be accepted?

Question 5

Nejo Ltd. has estimated the expected investment outlay and returns for four feasible projects as follows:

	PROJECTS			
Year	A N'000	B N'000	C N'000	D N'000
1999	(12,000)	(8,000)	(3,000)	(20,000)
2000	4,000	2,000	1,000	10,000
2001	5,000	2,500	1,100	8,000
2002	6,000	3,000	1,300	7,500
2003	6,500	4,000	1,800	6,000
2004	4,000	10,000	2,00	8,500

The company cost of capital is 20%.

Required:

Which projects should be accepted if the projects are independent and divisible, and the company has N20 million to invest.

Question 6

Mention Five management adjustable causes of Capital Rationing.

Question 7

What are the limitations of Benefit Cost Ratio?

Question 8

Toyosi Gbemisola Limited is currently having a capital budget limit of N160,000. The company is also envisaging a capital budget limit of N140,000 in year one. The following independent project are now considered.

PROJECT	YEAR				
	0	1	2	3	4
A	(100)	(40)	80	60	40
B	(60)	(80)	60	62	64
C	(40)	(60)	60	20	46
D	(60)	(30)	40	42	44
E	(10)	(20)	20	10	15

All the values are in N'000 and the company cost of capital is 10%.

Required:
Formulate the linear programme that will enable the company to select an optimal projects.

Question 9

Bello Limited is considering the following independent project:

	CASH OUTLAY		CASH RECEIPT			
Project	YEAR 0	YEAR 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4
	N'000	N'000	N'000	N'000	N'000	N'000
A	2,000	4,000	–	3,000	4,060	3,600
B	10,00	–	4,00	4,600	4,800	4,400
C	8,000	–	3,600	3,500	3,000	3,400
D	3,500	2,500	–	3,200	3,400	3,000

The company is currently facing the budget constrain as follow:

(i) N12 million in year 0.
(ii) N4 million in year 1 plus any cash receipt in year one.

Required:
Assuming the company cost of capital is 12% formulated a linear programme problem that will enable the company to select an optimal combination of project.

Question 10

Bolaji Ltd. has estimated the expected investment outlay and returns for five feasible projects as follows:

	PROJECTS				
	A (N'000)	B (N'000)	C (N'000)	D (N'000)	E (N'000)
1998	(8,000)	(10,000)	(20,000)	(4,000)	(15,000)
1999	(3,000)	(4,000)	(2,000)	(1,000)	(6,000)
2000	2,000	6,000	10,000	800	7,000
2001	3,000	7,000	8,000	1,400	8,000
2002	5,000	8,000	8,600	1,600	9,000
2003	10,000	3,000	6,000	2,000	9,600

The company's cost of capital is 15% and the projects are divisible. The company has a maximum of N20 million to invest now (1998) and a further N5 million in one year's time (1999).

Required:
Formulate, but do not solve, the linear programming problem to select the optimal combination of projects.

CAPITAL RATIONING

SOLUTION 1

PROJECT	PROFITABILITY INDEX	RANK
A	$\frac{172,000}{140,000} = 1.23$	8TH
B	$\frac{325,000}{260,000} = 1.25$	6TH
C	$\frac{50,000}{35,000} = 1.43$	1ST
D	$\frac{475,000}{340,000} = 1.40$	2ND
E	$\frac{138,000}{110,000} = 1.25$	6TH
F	$\frac{86,000}{65,000} = 1.32$	5TH
G	$\frac{131,000}{95,000} = 1.38$	3RD
H	$\frac{367,000}{275,000} = 1.33$	4TH

DIVISIBLE PROJECTS

Fund available	N400,000
Less Cost of Project C	50,000
	350,00
Less Cost of Project D	340,000
	10,000

Fraction Part of Project G = $\frac{10000}{95000}$ = 0.11

PROJECT	FRACTION UNDERTAKEN	INITIAL OUTLAY N	NPV N
C	1.0	50,000	15,000
D	1.0	340,000	135,000
G	10/95	10,000	3,789
		400,000	153,789

The company should accept Project C, D and 10/95 of G with a total net Present Value of N153,789.

SOLUTION 2

PROJECT A

Year	CASHFLOW N	DCF 16%	PV N
0	(32,000)	1.0000	(32,000)
1–5	12,000	3.2743	39,2916
			7,291.6

PROJECT B

Year	CASHFLOW N	DCF 16%	PV N
0	(25,000)	1.0000	(25,000)
1–5	12,000	3.2743	29,468.7
			4,468.7

PROJECT C

Year	CASHFLOW N	DCF 16%	PV N
0	(40,000)	1.0000	(40,000)
1–5	15,000	3.2743	49,114.5
			9,114.5

PROJECT D

Year	CASHFLOW	DCF 16%	PV
0	(35,000)	1.0000	(25,000)
1–5	10,000	3.2743	32,743
			(2.257)

PROJECT E

Year	CASHFLOW N	DCF 16%	PV N
0	(22,000)	1.0000	(22,000)
1-5	8,000	3.2743	26,194.4
			4,194.4

PROJECT F

Year	CASHFLOW N	DCF 16%	PV N
0	(50,000)	1.0000	(50,000)
1-5	21,000	3.2743	68,760.3
			18,760.3

PROJECT	INITIAL OUTLAY	NET PRESENT VALUE	EXCESS OF PRESENT VALUE INDEXES	RANK
A	32,000	7291.6	0.23	3RD
B	25,000	4468.7	0.18	5TH
C	40,000	9114.5	0.23	2ND
E	22,000	4194.4	0.19	4TH
F	50,000	18760.3	0.38	1ST

(i) Assuming the Projects are Divisible

	N
Funds available	100,000
Less Cost of Project F	50,000
	50,000
Less Cost of Project C	40,000
	10,000

Fractional Part of Project A = $\frac{10,000}{32,000} = \frac{10}{32}$

Project	Fraction Undertaken	INITIAL OUTLAY N	Net Present Value N
F	1.0	50,000	18760.3
C	1.0	40,000	9114.5
A	10/32	10,000	2278.6
		100,000	30153.4

Comment

The company should accept Project F, C and 10/32 of A with a total net Present Value of N30,153.40.

(ii) Assuming the Projects are Indivisible

Project Combination	Total Initial Outlay N	Total Net Present Value N
C, F	90,000	27,874.8
A, F	82,000	26,051.9
B, C, E	87,000	17,777.6
B, E, F	97,000	27,423.4

Comment

The best combination of Projects is C and F with a combine net Present Value of N27,874.80.

Workings

(i) Excess Present Value Indexes = $\frac{\text{Net Present Value}}{\text{Initial Outlay}}$

Project A = $\frac{7291.6}{32000}$ = 0.23

Project B = $\frac{4468.7}{25000}$ = 0.18

Project C = $\frac{9114.5}{40000}$ = 0.23

Project D = $\frac{4194.4}{22000}$ = 0.19

Project E = $\frac{18760.3}{50000}$ = 0.38

SOLUTION 3

$$NPV = \sum_{t=1}^{n} \frac{CFt}{(1+k)} - IO$$

NPV_A	=	2,900,000 (3.6847)	–	10,000,000	=	N685,630
NPV_B	=	1,200,000 (3.2743)	–	2,800,000	=	N1,129;160
NPV_C	=	1,800,000 (2.2459)	–	4,400,000	=	(N357,380)
NPV_D	=	2,800,000 (4.3436)	–	8,400,000	=	N3,762,080
NPV_E	=	2,800,000 (2.7982)	–	6,800,000	=	N1,034,960
NPV_F	=	1,400,000 (4.0386)	–	5,400,000	=	N254,040

PROJECT	NPV	Initial Outlay	Excess Present Value Indexes	RANK
A	685,630	10,000,000	0.07	4TH
B	1,129,160	2,800,000	0.40	2ND
D	3,762,080	8,400,000	0.45	1ST
E	1,034,960	6,800,000	0.15	3RD
F	254,040	5,400,000	0.05	5TH

DIVISIBLE PROJECTS

Fund available		24,000,000
Less Project	D	8,400,000
		15,600,000
	B	2,800,000
		12,800,000
	E	6,800,000
		6,000,000

Fraction Part of Project A = $\frac{6,000,000}{10,000,000}$ = 0.6

PROJECT	FRACTION UNDERTAKEN	INITIAL OUTLAY	NET PRESENT VALUE
D	1.0	8,400,000	3,762,080
B	1.0	2,800,000	1,129,160
E	1.0	6,800,000	1,034,960
A	0.6	6,000,000	411,378
		24,000,000	6,337,578

Comment

The company should embark on Project D, B, E and 0.6 of A with total Net Present Value of N6,337,578.

INDIVISIBLE PROJECTS

Project Combination	Total Initial Outlay N	Total Net Present Value N
B, D, E	18,000,000	5,926,200
A, B, D	21,200,000	5,577,170
B, D, E, F	23,400,000	6,180,240

Comment

The best combination of Projects is B, D, E and F with a combine net Present Value of N6,180,240.

Workings

(i) Discounted Cashflow Factors (DCF)

Project

A = $\sum_{t=1}^{6} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-6}}{0.16}\right] = 3.6847$

B = $\sum_{t=1}^{5} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-5}}{0.16}\right] = 3.2743$

C = $\sum_{t=1}^{3} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-3}}{0.16}\right] = 2.2459$

D = $\sum_{t=1}^{8} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-8}}{0.16}\right] = 4.3436$

E = $\sum_{t=1}^{4} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-4}}{0.16}\right] = 2.7982$

F = $\sum_{t=1}^{7} \frac{1}{(1.16)^t} = \left[\frac{1-(1.16)^{-7}}{0.16}\right] = 4.0386$

(ii) Excess Present Value Indexes is defined as $\frac{\text{Net Present Value}}{\text{Initial Outlay}}$

Project A = $\frac{685,000}{10,000,000}$ = 0.07

Project B = $\frac{1,129,160}{2,800,000}$ = 0.40

Project D = $\frac{3,762,080}{8,400,000}$ = 0.45

Project E = $\frac{1,034,960}{6,800,000}$ = 0.15

Project F = $\frac{254,040}{5,400,000}$ = 0.05

SOLUTION 4

Funds Available	N
Retained Profit	250,000,000
Dividend from subsidiary	160,000,000
	410,000,000

Capital Required	N
WIN Ltd.	140,000,000
Video Club	180,000,000
Palm Oil Plantation	200,000,000
	520,000,000

From the above computation, capital required is N520,000,000 whereas only N410,000,000 is available thereby creating insufficiency of (N520,000,000 – N410,000,000) = N110,000,000, thus there is need to ration capital.

WIN NIG. PLC

Items	Year	Cashflow N	DCF 12%	PV N
Initial Capital	0	(140,000,000)	1.0000	(140,000,000)
Annual Dividend	1–10	30,000,000	5.6502	169,506,000
Disposal	10	200,000,000	0.3220	64,400,000
			NPV ⇒	93,906,000

Benefit Cost ratio = $\frac{\text{NPV}}{\text{Initial Capital}}$

= $\frac{93,906,000}{140,000,000}$

= 0.67

Working

Disposal = 30,000,000 x $\frac{666.67}{100}$

= N200,000,000

VIDEO CLUB

Items	Year	CASHFLOW N	DCF 12%	PV N
Initial Capital	0	(180,000,000)	1.0000	(180,000,000)
Net Annual Sales	1-10	35,000,000	5.6502	197,757,000
Disposal	10	45,000,000	0.3220	14,490,000
				32,247,000

Benefit Cost ratio = $\frac{\text{NPV}}{\text{Initial Capital}}$

= $\frac{32,247,000}{180,000,000}$

= 0.18

PALM-OIL PLANTATION

Computation of Net Sales

Year	SALES N	COST OF SALES N	ADMINISTRATION COST N	NET SALES N
4	110,000,000	20,000,000	2,000,000	88,000,000
5	120,000,000	20,000,000	4,000,000	96,000,000
6	150,000,000	26,000,000	4,000,000	120,000,000
7	160,000,000	30,000,000	2,000,000	128,000,000
8	180,000,000	30,000,000	6,000,000	144,000,000
9	190,000,000	28,000,000	10,000,000	152,000,000

Items	Year	CASHFLOW N	DCF 12%	PV N
Initial Capital	0	(200,000,000)	1.0000	(200,000,000)
Net Sales	4	88,000,000	0.6355	55,924,000
Net Sales	5	96,000,000	0.5674	54,470,400
Net Sales	6	120,000,000	0.5066	60,792,000
Net Sales	7	128,000,000	0.4526	57,932,800
Net Sales	8	144,000,000	0.4039	58,161,600
Net Sales	9	152,000,000	0.3606	54,811,200
				142,092,000

Benefit Cost ratio = $\frac{\text{NPV}}{\text{Initial Capital}}$

= $\frac{142,092,000}{200,000,000}$

= 0.71

RANKING OF THE PROJECTS

PROJECTS	NPV	INITIAL CAPITAL	BENEFIT COST RATIO	RANKING
WIN NIG. LTD	93,906,000	140,000,000	0.67	2ND
VIDEO CLUB	32,247,000	180,000,000	0.18	3RD
PALM OIL PLANTATION	142,092,000	200,000,000	0.71	1ST

	N
Funds Available	410,000,000
Less Cost of Palm Oil Plantation	200,000,000
	210,000,000
Less Cost of WIN NIG. LTD.	140,000,000
	70,000,000

Fraction Part of Video Club = $\frac{70,000,000}{180,000,000}$ = $\frac{7}{18}$

Recommendation

Based on the above computation, Palm Oil plantation and purchases of WIN Nigeria Ltd. shares should be fully financed and the balance N70,000 should be used to finance part (7/18) of Video Club since it is assumed that the projects are divisible.

SOLUTION 5

COMPUTATION OF NET PRESENT VALUE OF EACH PROJECT A, B, C and D

PROJECT A

Year	CASHFLOW N	DCF 20%	PV N
1999	(10,000,000)	1.0000	(12,000,000)
2000	4,000,000	0.8333	3,333,200
2001	5,000,000	0.6944	3,472,000
2002	6,000,000	0.5787	3,472,200
2003	6,500,000	0.4823	3,134,950
2004	4,000,000	0.4019	1,607,600
			3,019,950

PROJECT B

Year	CASHFLOW N	DCF 20%	PV N
1999	(8,000,000)	1.0000	(8,000,000)
2000	2,000,000	0.8333	1,666,600
2001	2,500,000	0.6944	1,736,000
2002	3,000,000	0.5787	1,736,100
2003	4,000,000	0.4823	1,929,200
2004	10,000,000	0.4019	4,019,000
			3,086,900

PROJECT C

Year	CASHFLOW	DCF 20%	PV
1999	(3,000,000)	1.0000	(3,000,000)
2000	1,000,000	0.8333	833,300
2001	1,100,000	0.6944	763,840
2002	1,300,000	0.5787	752,310
2003	1,800,000	0.4823	868,140
2004	2,000,000	0.4019	803,800
			1,021,300

PROJECT D

Year	CASHFLOW	DCF 20%	PV
	N		N
1999	(20,000)	1.0000	(20,000,000)
2000	10,000,000	0.8333	8,333,000
2001	8,000,000	0.6944	5,555,200
2002	7,500,000	0.5787	4,340,250
2003	6,000,000	0.4823	2,893,800
2004	8,500,000	0.4019	3,416,150
			4,538,400

Project	Net Present Value	Initial Outlay	BCR	RANKING
A	3,019,950	12,000,000	0.25	3RD
B	3,086,900	8,000,000	0.39	1ST
C	1,021,390	3,000,000	0.34	2ND
D	4,538,400	20,000,000	0.23	4TH

	N
Funds Available	20,000,000
Less Cost of Project B	8,000,000
	12,000,000
Less Cost of Project C	3,000,000
	9,000,000

Fractional part of Project A = $\frac{9,000,000}{12,000,000}$ = 0.75

PROJECT	FRACTION UNDERTAKEN	INITIAL OUTLAY	NPV
B	1.0	8,000,000	3,086,960
C	1.0	3,000,000	1,021,390
A	¾	9,000,000	2,264,963
		20,000,000	6,373,313

Comment

The company should accept Project B, C and ¾ of A with a total net present value of N6,373,313.00.

SOLUTION 6

Five management adjustable causes of Capital Rationing are:-

- When Management decides to pay dividend instead of ploughing back the profit
- Budget constraints
- Management decides not to go for additional equity
- When management decides to limit the growth of its organisation
- When management decides not to borrow or dilute shareholding.

SOLUTION 7

Limitations of Benefit Cost Ratio

- When projects are not divisible, there may be idle capital
- Other capital requirement aside the year of capital rationing is not recognised
- The relationship of returns and capital injection is not linear.

SOLUTION 8

Computation of Net Present Value

PROJECT A

Year	Cashflow	DCF = 10%	PV
	N		N
0	(100,000)	1.0000	(100,000)
1	(40,000)	0.9091	(36,364)
2	80,000	0.8264	66,112
3	60,000	0.7513	45,078
4	40,000	0.6830	27,320
			2,146

PROJECT B

Year	Cashflow N	DCF 10%	PV N
0	(60,000)	1.0000	(60,000)
1	(80,000)	0.9091	(72,728)
2	60,000	0.8264	49,584
3	62,000	0.7513	46,581
4	64,000	0.6830	43,712
			7,149

PROJECT C

Year	Cashflow N	DCF 10%	PV N
0	(40,000)	1.0000	(40,000)
1	(60,000)	0.9091	(54,546)
2	40,000	0.8264	49,584
3	20,000	0.7513	15,026
4	46,000	0.6830	31,418
			1,482

PROJECT D

Year	Cashflow N	DCF 10%	PV N
0	(60,000)	1.0000	(60,000)
1	(30,000)	0.9091	(27,273)
2	40,000	0.8264	33,056
3	42,000	0.7513	31,555
4	42,000	0.6830	28,686
			6,024

PROJECT E

Year	Cashflow N	DCF 10% N	PV
0	(10,000)	1.0000	(10,000)
1	(20,000)	0.9091	(18,182)
2	20,000	0.8264	16,528
3	10,000	0.7513	7,513
4	15,000	0.6830	10,245
		NPV_E =	6,104

Let X_1 represent the proportion of Project A to undertake.
Let X_2 represent the proportion of Project B to undertake.
Let X_3 represent the proportion of Project C to undertake.
Let X_4 represent the proportion of Project D to undertake.
Let X_5 represent the proportion of Project E to undertake.

The Objective function is
Maximize NPV = $N2{,}146X_1 + N7{,}149X_2 + N1{,}482X_3 + N6{,}024X_4 + N6{,}104X_5$
Subject to the following constraints

$100{,}000X_1 + 60{,}000X_2 + 40{,}000X_3 + 60{,}000X_4 + 10{,}000X_5 \leq 160{,}000$
$40{,}000X_1 + 80{,}000X_2 + 60{,}000X_3 + 30{,}000X_4 + 20{,}000X_5 \leq 140{,}000$
$X_1, X_2, X_3, X_4, X_5 \leq 1$
$X_1, X_2, X_3, X_4, X_5 \geq 0$

SOLUTION 9

Computation of Net Present Value

PROJECT A

Year	CASHFLOW N	DCF 12%	PRESENT VALUE N
0	(2,000,000)	1.0000	(2,000,000)
1	(4,000,000)	0.8929	(3,571,600)
2	3,000,000	0.7972	2,391,600
3	4,060,000	0.7118	2,889,908
4	3,600,000	0.6355	2,287,800
			1,997,708

PROJECT B

Year	CASHFLOW N	DCF 12%	PRESENT VALUE N
0	(10,000,000)	1.0000	(10,000,000)
1	4,000,000	0.8929	3,571,600
2	4,600,000	0.7972	3,667,120
3	4,800,000	0.7118	3,416,640
4	4,400,000	0.6355	2,796,200
			3,451,560

PROJECT C

Year	CASHFLOW N	DCF 12%	PRESENT VALUE N
0	(8,000,000)	1.0000	(8,000,000)
1	3,600,000	0.8929	3,214,440
2	3,500,000	0.7972	2,790,200
3	3,000,000	0.7118	2,135,400
4	3,400,000	0.6355	2,160,700
			2,300,740

PROJECT D

Year	CASHFLOW N	DCF 12%	PRESENT VALUE N
0	(3,500,000)	1.0000	(3,500,000)
1	(2,500,000)	0.8929	(2,232,250)
2	3,200,000	0.7972	2,551,040
3	3,400,000	0.7118	2,420,120
4	3,000,000	0.6355	1,906,500
			1,145,410

Let X1 represent the proportion of Project A to undertake.
Let X2 represent the proportion of Project B to undertake.
Let X3 represent the proportion of Project C to undertake.
Let X4 represent the proportion of Project D to undertake.

The Objective function is

Maximise NPV = $1,997,708X_1 + 3,451,560X_2 + 2,300,740X_3 + 1145410X_4$

Subject to the following constraints

$2,000,000X_1 + 10,000,000X_2 + 8,000,000X_3 + 3,500,000X_4 < 12,000,000$

$4,000,000X_1 + 2,800,000X_4 \leq 4,000,000 + 4,000,000X_2 + 3,600,000X_3$

$X_1, X_2, X_3, X_4 \leq 1$

$X_1, X_2, X_3, X_4 \geq 0$

SOLUTION 10

Computation of Net Present Value

PROJECT A

Year	CASHFLOW N	DCF = 15%	PRESENT VALUE N
1998	(8,000,000)	1.0000	(8,000,000)
1999	(3,000,000)	0.8696	(2,608,800)
2000	2,000,000	0.7561	1,512,200
2001	3,000,000	0.6575	1,972,500
2002	5,000,000	0.5718	2,859,000
2003	10,000,000	0.4972	4,972,000
			706,900

PROJECT B

Year	CASHFLOW N	DCF = 15%	PRESENT VALUE N
1998	(10,000,000)	1.0000	(10,000,000)
1999	(4,000,000)	0.8696	(3,478,400)
2000	6,000,000	0.7561	4,536,600
2001	7,000,000	0.6575	4,602,500
2002	8,000,000	0.5718	4,574,400
2003	3,000,000	0.4972	1,491,600
			1,726,700

PROJECT C

Year	CASHFLOW N	DCF 15%	PRESENT VALUE N
1998	(20,000,000)	1.0000	(20,000,000)
1999	(2,000,000)	0.8696	(1,739,200)
2000	10,000,000	0.7561	7,561,000
2001	8,000,000	0.6575	5,260,000
2002	8,600,000	0.5718	4,917,480
2003	6,000,000	0.4972	2,983,200
			(1,017,520)

PROJECT D

Year	CASHFLOW N	DCF 15%	PRESENT VALUE N
1998	(4,000,000)	1.0000	(4,000,000)
1999	(100,000,000)	0.8696	(869,600)
2000	800,000	0.7561	604,880
2001	1,400,000	0.6575	920,500
2002	1,600,000	0.5718	914,880
2003	200,000,000	0.4972	994,440
			(1,434,900)

PROJECT E

Year	CASHFLOW N	DCF = 15%	PRESENT VALUE N
1998	(15,000,000)	1.0000	(15,000,000)
1999	(6,000,000)	0.8696	(5,217,600)
2000	7,000,000	0.7561	5,292,700
2001	8,000,000	0.6575	5,260,000
2002	9,000,000	0.5718	5,146,200
2003	9,600,000	0.4972	4,773,120
			254,420

Since the Net Present Value of Project C and D are negative, the two projects should be rejected.

Let X1 represent the proportion of Project A to undertake.
Let Y represent the proportion of Project B to undertake.
Let Z represent the proportion of Project E to undertake.

∴ The Objective function is
Maximise NPV = N706,900X + N1,726,700Y + N254,420Z
Subject to the following constraints

8,000,000X + 10,000,000Y + 15,000,000Z	≤	20,000,000
3,000,000X + 4,000,000 Y + 6,000,000Z	≤	5,000,000
X, Y, Z	≤	1
X, Y, Z	≥	0

REFERENCES AND FURTHER STUDIES

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CHAPTER 5

CAPITAL BUDGETING DECISION WITH RISK & UNCERTAINTY

Question 1
Distinguish between risk and uncertainty.

Question 2
Briefly outline methods of treating risk in capital budgeting.

Question 3
Explain the term "Sensitivity Analysis".

Question 4
Bakassi Ltd. considering introducing a new product to be called sobo, details of which are analyzed below.

Production of sobo will require the purchase and installation of a giant machine at a cost of N750,000 payment immediately. This machine will spend 6 years before being scrapped at a nil value. Sales of sobo will fetch an annual of N1.2m over the 6 years value. Productions cost are estimated at N960,000 per annum. The appropriate discounting rate is considered to be 16%.

Required:
For each of the estimate listed below, calculate the

Sensitivity margin
(i) Sales per annum
(ii) Production cost per annum
(iii) Cost of giant machine
(iv) Useful life of the giant machine

Question 5
The estimated cost of a project is N600,000. The selling price per unit and the cost per unit of the product produced are given as N60 and N35 respectively. The output sold for the project's five years of existence are indicated below:

Year 1	6,000 units
Year 2	8,000 units
Year 3	9,000 units
Year 4	10,000 units
Year 5	11,000 units

(i) You are required to calculate the Project's Net Present Value (please ignore depreciation) given that the project's cost of capital is 12 percent.

(ii) To what extent can the project's margin falls to enable the project NPV remain positive?

Question 6
Discuss the usefulness and the limitations of sensitivity margin as an aid to project appraisal under uncertainty.

Question 7
Bisola Hassan Ltd. is considering two alternative projects (project X and Y respectively). Project X will involve a capital cost of N7.5 million and will have a life of 8 years. The resale value of the capital equipment of project X at the end of the operating life is N50,000. Project will involve a capital equipment cost of N7.6 million and will also have an 8 years life. The finance director of Bisola Hassan Limited has provided the following revenue and cost estimates for both projects:

	PROJECT X		PROJECT Y	
Year	Revenue N'000	Variable Cost N'000	Revenue N'000	Variable Cost N'000
1	1,000	500	1,900	1,000
2	1,000	900	2,500	1,000
3	3,500	900	3,600	1,100
4	3,500	1,000	3,600	1,200
5	3,500	1,400	3,600	1,800
6	4,000	1,600	4,000	2,000
7	6,500	1,900	6,100	2,400
8	6,500	2,000	7,600	2,400

The risk premium applicable to project X is 7% while that of project Y is 8.5%. The risk free rate has been 12 percent and is expected to remain constant throughout the duration of the project. The company's risk adjusted discount rate will be constant throughout the project's duration.

Required:

(i) Calculate the NPV for both Project X and Y.

(ii) Advise the company on the project to select.

Question 8

Uche and Sons Ltd. is considering investing in a project costing N150,000 which has a life of 4 years.

The project's future cash flows are not certain. However, the probability distribution of the future cash flows are given as follows:

Year 1		**Year 2**	
Cash flow	Probability	Cash flow	Probability
N40,000	0.3	N45,000	0.4
N55,000	0.3	N60,000	0.4
N60,000	0.4	N62,000	0.2

Year 3		**Year 4**	
Cash flow	Probability	Cash flow	Probability
N55,000	0.5	N58,000	0.6
N60,000	0.3	N65,000	0.4
N65,000	0.2		

The company's cost of capital is 20% and treasury bill rate return is 12%.

Required:

(i) Calculate the expected value of net present value and standard deviation of the project's net present value.

(ii) Calculate the co-efficient of variation of the project.

Question 9

Emeka Limited is considering between project A and B. Project A cost N3 million while project B cost N4.5 million. The net cash profits from the two projects are uncertain but the likelihood of their occurrence is given as follows:

PROJECT A				PROJECT B			
Year 1		Year 2		Year 1		Year 2	
Cash		Cash		Cash		Cash	
Profit		Profit		Profit		Profit	
N'000	PROB	N'000	PROB	N'000	PROB	N'000	PROB
1,500	0.3	3,000	0.2	3,500	0.25	4,300	0.45
2,500	0.4	2,900	0.3	4,000	0.30	5,400	0.15
1,900	0.2	2,800	0.4	4,600	0.35	6,000	0.20
2,300	0.1	3,600	0.1	5,200	0.10	2,500	0.20

The cost of capital applicable to the risk of project A is 18% while the cost of capital applicable to the risk class of project B is 19%. Emeka Limited currently has some investment in treasury bills which currently yield 12% per annum.

Required:

(i) Estimate the expected net present value and standard deviation of the two project.

(ii) Advise the company as much as possible on the project to select.

QUESTION 10

The probability distribution of expected returns for four projects A, B, C and D are as follows:

Probability Distribution	Project A	Project B	Project C	Project D
0.3	N100	N180	N90	N80
0.5	N140	N170	N160	N120
0.2	N150	N110	N200	N160

(i) Compute a measure of the relative risk of each project per N1 of its expected value. Which of the projects involves the least risk.

(b) Apply the E. V. rule to determine which project in each of the pairs of project A & D, B & C and B & D will be preferred.

Question 11

EDSU Enterprises has concluded the arrangement of embarking on a project whose cash flow is given below:

Year	0	1	2	3	4
Cash flow (N'000)	(1850)	600	950	1250	1750

The estimate risk premium of the project for the first two years is 6½. The risk premium was

however adjusted to 8% percent and 10 percent for the third and fourth years respectively. The current yield on government security is 12% and this is expected to remain so throughout the duration of the project. Using the appropriate measure of adjusting for the risk, advise the company of the desirability of the project.

Question 12

James is considering investment in the following securities whose random returns are given as follows:

Security A		Security B	
Return	Probability	Return	Probability
4000	0.4	6000	0.7
5000	0.5	4000	0.2
6000	0.1	3000	0.1

Required:

(a) Calculate the
- (i) Expected return
- (ii) Variance
- (iii) Standard deviation
- (iv) Coefficient of variation of each security.

(b) Which security would you recommend to James, who is a risk-averse investor? Why?

Question 13

Adebisi Company can invest in one of three mutually exclusive projects. The three proposals have the following probability distributions of net cashflows:

PROJECT X		PROJECT Y		PROJECT Z	
Cash flow	Probability	Cash flow	Probability	Cash flow	Probability
N		N		N	
15,000	0.1	20,000	0.4	27,000	0.5
22,000	0.3	25,000	0.2	30,000	0.2
27,000	0.2	30,000	0.3	35,000	0.1
30,000	0.4	35,000	0.1	40,000	0.2

(a)
- (i) Calculate the project's mean expected cash flow.
- (ii) Calculate the variance of each project.
- (iii) Calculate the coefficient of variation of each project.

(b) Which project is less risky? Explain

Question 14

The Management of Akala Plc has made the following estimates of sales for a new product to be launched in the Nigeran market that has an expected life of 5 years. The firm estimates the initial cost of the product's production machinery at N18,000.

		High Sales	Average Sales	Low Sales
Probability		0.4	0.3	0.3
		N	N	N
Year	1	12,000	6,000	4,000
	2	20,000	10,000	8,000
	3	28,000	14,000	8,000
	4	34,000	17,000	10,000
	5	40,000	20,000	10,000

Variable costs are uncertain but have been estimated at 60% of sales. Fixed costs will be N1,000 per annum if the product is manufactured. The firm uses a cost of capital of 12% per annum to discount its projects.

Calculate the expected Net Present Value (ENPV of making and selling the product over the next 5 years.

QUESTION 15

Udeh and Sons Ltd is considering investing in a project costing N150,000 which has a life of 3 years. The projects future Cash flows are not certain. However, the probability distribution of the future cashflows are given as follows:

YEAR ONE

CASH FLOW	PROBABILITY
N50,000	0.3
60,000	0.3
70,000	0.4

YEAR TWO

CASH FLOW	PROBABILITY
N	
60,000	0.4
70,000	0.4
80,000	0.2

YEAR THREE

CASH FLOW N	PROBABILITY
80,000	0.5
90,000	0.3
95,000	0.2

The company's cost of capital is currently 20% an treasury bill rate is 12%.

Required:
(a) Calculate the expected value of net present value of the project.
(b) Calculate the standard deviation of the project.
(c) Calculate the coefficient of variation of the project.

QUESTION 16

Peter Agaren Ltd. is considering the prospect of investing N4.0 million in either of projects X or Y which are mutually exclusive.

Project X is investment in Mining with the following data

State of nature	Probability	Return in Million Naira	
		Year 1	Year 2
Raining	0.7	5.0	3.5
Draught	0.3	2.0	2.5

Prospect Y is investment in Foods and Beverages with the following data

State of Nature	Probability	Return in Million Naira		
		Year 1	Year 2	Year 3
Boom	0.6	3.5	2.0	3.0
Normal	0.3	2.5	1.5	2.0
Recession	0.1	2.0	4.0	1.0

If the opportunity cost of capital for project X is 13% and for project Y is 10%, determine which project is preferable.

CAPITAL BUDGET DECISION WITH RISK & UNCERTAINTY

SOLUTION 1

Risk is the variability that likely to be associated with future returns from a project. Risk is a situation where the future outcome is unknown, but the likelihood of various possible future outcomes may be assessed with some degree of confidence, probably based on knowledge of past or existing events. In other words, probabilities of possible outcomes can be estimated.

Whereas uncertainty is a situation where the outcome cannot be predicted with any degree of confidence from a knowledge of past or existing events, so that no probability estimate are available.

SOLUTION 2

(a) Methods of treating risk in capital budgeting

Non-probabixlity method

- Sensitivity Analysis
- Risk Adjusted method
- Simulation.
- Certainty Equivalent Method

Probability method

- Expected value of NPV
- Standard Deviation
- Variance
- Coefficient of Variation.

SOLUTION 3

Sensitivity Analysis: is defined as the maximum tolerable unfavourable changes in the key variable factors affecting the project under consideration. It is the maximum negative change permitted by the project otherwise it will render the project unaccepted.

SOLUTION 4

B(i)

Items	Year	Cash flow	DCF 16%	PV
Initial cost	0	(750,000)	1.0000	(750,000)
Sales	1-6	1,200,000	3.6847	4,421,640
Production cost	1-6	(960,000)	3.6847	(3,537,312)
			NPV	134,238

(i) Sensitivity margin of sales per annum $= \frac{NPV}{PV \text{ of sales}} \times 100 = \frac{134328}{44216640} \times 100$

$= 3\%$

Sales must not be reduced by more than 3%.

(ii) Sensitivity margin of production cost per annum $= \frac{NPV}{PV \text{ of production cost}} \times 100$

$= \frac{134328}{3537312} \times 100 = 4\%$

Production cost must not be allowed to increase by more than 4% per annum.

(iii) Sensitivity margin of cost of giant machine $= \frac{NPV}{PV \text{ of the giant machine}} \times 100$

$= \frac{134328}{750000} \times \frac{100}{x1}$

$= 18\%$

The cost of the giant machine should not be allowed to rise by more than 18%.

(iv) Sensitivity margin of useful life of the giant machine

Present Value of cost of giant machine + Present Value of Production cost + Present Value of Sales = O

Let Y = DCF factor at 16%

$\therefore$ -750,000 + (-960000 x y) + 1,200,000 x y = 0

-750,000 - 960,000y + 1,200,000y = 0

240,000y = 0 + 750,000

y = 750,000 / 240,000

y = 3.125

Using present value of ordinary annuity formula $\frac{1-(1+i)^n}{i}$

$\therefore 1-(1.16)^n = 3.125 \times 0.16$

$1-(1.16)^n = 0.5$

$1-0.5 = (1.16)^n$

$0.5 = (1.16)^n$

$\text{Log}_{10}(0.5) = \text{Log}_{10}(1.16)^n$

$\text{Log}_{10}(0.5) = -n \text{Log}_{10} 1.16$

$-n = \frac{\text{Log}_{10} 0.5}{\text{Log}_{10} 1.16} = -4.67$

$\therefore n = 4.67\text{yrs} \simeq 5 \text{ yrs}$

The usefulness of the giant machine should not be less than 5 yrs

SOLUTION 5

(i) Revenue =

				N
YR 1	6,000	x N60	=	360,000
YR2	8,000	x N60	=	480,000
YR 3	9,000	x N60	=	540,000
YR 4	10,000	x N60	=	600,000
YR5	11 ,000	x N60	=	660,000

Cost of goods sold

				N
YR 1	6,000	x N35	=	210,000
YR2	8,000	x N35	=	280,000
YR 3	9,000	x N35	=	315,000
YR 4	10,000	x N35	=	350,000
YR5	11 ,000	x N33	=	385,000

Year	Cost	Revenue	Cost of Goods Sold	Net Cash Profit	DCF = 12%	PV
0	(600,000)			(600,000)	1.0000	(600,000)
1		360,000	(210,000)	150,000	0.8929	133,935
2		480,000	(280,000)	200,000	0.7972	159,440
3		540,000	(315,000)	225,000	0.7118	160,155
4		600,000	(350,000)	250,000	0.6355	158,875
5		660,000	(385,000	275,000	0.5674	156,035
						168,440

The project NPV = N 168,440.

(ii) The extent to which the Project's profit margin may fall to enable the project NPV remain positive

$$= \frac{NPV}{Cost} \times \frac{100}{1}$$

$$= \frac{168440}{600000} \times \frac{100}{1}$$

The project profit margin must not be reduced by more than 28%.

SOLUTION 6

Usefulness

- It enables us to know how far a project can cope within certain environment especially when costs are rising daily.
- It helps to identify the variables that must be carefully watched when investing in uncertain environment.

Limitation

- It fails to recognize that a change in one of the key variables for examples: a small reduction in the life span can cause a great reduction in the NPV.

SOLUTION 7

PROJECT X

Cost/Residential Value	Revenue	Variable Cost	Net Cash Flow	DCF= 19%	PV
(7,500,000)			(7,500,000)	1.000	(7,500,000)
	1,000,000	(500,000)	500,000	0.8403	420,150
	1.060,000	(900,000)	700,000	0.7062	494,340
	3,500,000	(900,000)	2,600,000	0.5934	1,542,840
	3,500,000	(1,000,000)	2,500,000	0.4987	1,246,750
	3,500,000	(1,400,000)	2,100,000	0.4190	879,900
	4,000,000	(1,600,000)	2,400,000	0.3521	845,040
	6,500,000	(1,900,000)	4,600,000	0.2959	1,361,140
50000	6,500,000	(2,000,000)	4,550,000	0.2487	1,131,585
					(421,745)

DCFx = 0.12 + 0.07 = 0.19 = 19%

PROJECT Y Cost	Revenue	Variable Cost	Net Cash Flow	DCF= 20.5%	PV
(7,600,000)			(7,600,000)	1000	(7,600,000)
	1,900,000	1,000,000	900,000	0.8299	1,033,050
	2,500,000	1,000,000	1,500,000	0.6887	1,428,750
	3,600,000	1,100,000	2,500,000	0.5715	1,138,320
	3,600,000	1,200,000	2,400,000	0.4743	708,480
	3,600,000	1,800,000	1,800,000	0.3936	747,840
	4,000,000	2,000,000	2,000,000	0.3266	653,200
	6,100,000	2,400,000	3,700,000	0.2711	1,003,070
	7,600,000	2,400,000	5,200,000	0.2250	1,170,000
					(281,780)

DCFy = 0.12 + 0.205 = 20.5%

ADVICE

The company should not select any of the two projects, because their NPV are negative.

SOLUTION 8

YEAR I

Ct N'000	*Prob.*	Ct x *Prob.*	(Ct-Ev)	(Ct-Ev)^2 *prob.*
40	0.3	12	-12.5	46.875
55	0.3	16.5	2.5	1.875
60	0.4	24	7.50	22.25
	Ev	52.5	σr^2 =	71.25
			SD =	8.441

YEAR II

Ct N'000	*Prob*	CT x *Prob*	(Ct-Ev)	(Ct-Ey^2 *prob*
45	0.4	18	-9.4	35.344
60	0.4	24	5.6	12.544

62 0.2 12.4 7.6 11.552

Ev 54.4 σ_2^2 = 59.44

SD = 7.71

YEAR III

Ct	*Prob*	CT x *Prob*	(Ct-Ev)	(Ct-Ev2 *prop*
55	0.5	27.5	-3.5	6.125
60	0.3	18	1.5	0.675
65	0.2	13	6.5	8.45
	Ev	58.5	σ_3^2 =	15.25
			SD =	3.905

YEAR IV

Ct	Prob	CT x Prob	(Ct-Ev)	(Ct-Ev2 prop
58	0.6	34.8	-28	4.704
65	0.4	26	4.2	7.056
	Ev	60.8	σ_4^2 =	1176
			SD =	3.429

$$ENPV = \sum_{t=1}^{n} \frac{EV_1}{(1+RF)^t} - Co$$

$$\frac{52500}{1+0.12} + \frac{54400}{(1+0.12)^2} + \frac{58500}{(1+0.12)^3} + \frac{60800}{(1+0.12)^4} - 150{,}000$$

46875 + 43367 + 41639 + 38639 – 150000

EV of NPV = N20,520

The standard deviation of the project's net present value.

$$SD = \sqrt{\sum_{t=1}^{n} \frac{\sigma_1^2}{(1+RF)^{2t}}}$$

$$= \frac{\sigma_1^2}{(1+R_F)^{2x1}} + \frac{\sigma_2^2}{(1+R_F)^{2x2}} + \frac{\sigma_3^2}{(1+R_F)^{2x3}} + \frac{\sigma_4^2}{(1+R_F)^{2x4}}$$

$$= \sqrt{\frac{71250}{(1+0.12)^2} + \frac{59440}{(1+0.12)^4} + \frac{15250}{(1+0.12)^6} + \frac{1176}{(1+0.12)^8}}$$

$$= \sqrt{56800 + 37775 + 7726 + 4750}$$

$$= \sqrt{107051}$$

S. D. of NPV = N327

ii. Coefficient of variation of the project = $\frac{\text{S.D of NPV}}{\text{EV NPV}} \times \frac{100}{1}$

$$= \frac{327}{20{,}520} \times \frac{100}{1}$$

$$= 1.59\%$$

SOLUTION 9

Project A

Year 1

Cash Profit	Probability	CF.*prob*	(Cf – EV1)	(Cf – EV)² *prob*
1,500,000	0.3	450,000	-610,000	111,630,000,000
2,500,000	0.4	100,000	390,000	60,840,000,000
1,900,000	0.2	380,000	-210,000	8,820,000,000
2,800,000	0.1	280,000	690,000	47,610,000,000
		EV1 → 2,110,000		σ_1^2 = 228,900,000,000

Year II

Cash Profits	Probability	Cf.*prob*	(CF – EV)	(CF – EV)² *prob*
3,000,000	0.2	600,000	50,000	500,000,000
2,900,000	0.3	870,000	-50,000	750,000,000
2,800,000	0.4	1,120,000	-150,000	9,000,000,000
3,600,000	0.1	360,000	650,000	42,250,000,000
		EV2 → 2,950,000		σ_2^2 = 52,500,000,000

Project B

Year I

Cash Profits	Prob.	Cf.*prob*	(CF – EV)	(CF – EV)2 *prob*
3,500,000	0.25	875,000	-705	124,256,250,000
4,000,000	0.30	1,200,000	-205	12,607,500,000
4,600,000	0.35	1,610,000	395	54,608,750,000
5,200,000	0.1	520,000	995	99,002,500,000
		EV → 4,205,000		σ_1^2 = 290,475,000,000

Year II

Cash Profit	Prob.	Cf.*prob*	(CF – EV)	CF – EV)2 *prob*
4,800,000	0.45	2,160,000	130,000	7,605,000,000
5,400,000	0.15	810,000	730,000	79,935,000,000
6,000,000	0.20	1,200,000	1,330,000	353,780,000,000
2,500,000	0.20	500,000	-2,170,000	941,780,000,000
		EV → 4,670,000		σ_2^2 = 1,383,100,000,000

$$EVNPV_A = (3,000,000) + \frac{2,110,000}{1+0.12} + \frac{2,950,000}{(1+0.12)^2}$$

$$EVNPV_A = (3,000,000) + 1,883,928.57 + 2,351,721.94$$

$$= N1,235,650.51$$

$$EVNPV_B = (4,500,000) + \frac{4,205,000}{1+0.12} + \frac{4,670,000}{(1+0.12)^2}$$

$$EVNPV_B = (4,500,000)\ 3,754,464.286 + 3,722,895.408$$

$$= N2,977,359.69$$

$$\sigma_A = \sqrt{\frac{228,900,000,000}{(1.12)^2} + \frac{52,500,000,000}{(1.12)^2}} = \sqrt{215,842,377,700}$$

SD_A = N464,588.40

$$\sigma_B = \sqrt{\frac{290,475,000,000}{(1.12)^2} + \frac{1,383,100,000,000}{(1.12)^4}} = \sqrt{111,054,946,000}$$

SD_B = 1,053,826.34

(ii) Coefficient of Variation (COV)

$$COV_A = \frac{SD_A}{EVNPV_A} \times \frac{100}{1} =$$

$$= \frac{464,588.40}{1,235,650.51} \times \frac{100}{1} = 38\%$$

$$COV_B = \frac{SD_B}{EVNPV_B} \times \frac{100}{1}$$

$$= \frac{1,053,826.34}{2,977,359.69} \times \frac{100}{1} = 35\%$$

The Company should select Project B because its coefficient of variation is lower than that of Project A. Thus Project B is less risky compare to Project A.

SOLUTION 10

5a. Project A

Ct	Prob	EV	(Ct-Ev)	$(Ct-Ev)^2$ *prob*
100	0.3	30	-30	270
140	0.5	70	10	50
150	0.2	30	20	80
		Ev 130		$\sigma_A^2 = 400$

$SD_A = \sqrt{400} = 20$

Project B

Ct	Prob	Ev	(Ct-Ev)	$(Ct-Ev^2)$ *prob*
180	0.3	54	19	108.3
170	0.5	85	9	40.5
110	0.2	22	-51	920.2
		Ev 161		$\sigma_B^2 = 1189.2$

$SD_B = \sqrt{1189.2} = 34.48$

Project C

Ct	Prob	Ev	(Ct-Ev)	$(Ct-Ev)^2$ *prob*
90	0.3	27	-57	974.7
160	0.5	80	13	84.5
200	0.2	40	53	561.8
		Ev 147		$\sigma_C^2 = 1621$

$SD_C = \sqrt{1621} = 40.26$

Project D

Ct	Prob	Ev	(Ct-Ev)	$(Ct-Ev)^2$ *prob*
80	0.3	24	-36	388.8
120	0.5	60	4	8.0
160	0.2	32	44	387.2
		Ev 116		$\sigma_D^2 = 784$

$SD_D = \sqrt{784} = 28$

Project A has the least risk i.e. $SD_A = 20$

5B. E-V A + D = 130 + 116 = 246
" B + C = 161 + 147 = 308
" B + D = 161 + 116 = 277

The pair of B & C preferred to other combination, because it has the highest expected value.

SOLUTION 11

	$R_F + R_P = R_A$		
Year 1	0.12 + 0.065	=	0.185
2	0.12 + 0.065	=	0.185
3	0.12 + 0.0875	=	0.2075
4	0.12 + 0.1	=	0.22

Where R_F = Risk free rate
R_P = Risk premium
R_A = Risk Adjusted

$$NPV = -\sum_{t=1}^{n} \frac{CF_t}{(1+R_A)^t} - IO$$

$$= \frac{600,000}{1.185} + \frac{950,000}{(1.185)^2} + \frac{1,250,000}{(1.2075)^3} + \frac{1,750,000}{(1.22)^4} - 1,850,000$$

= 506,329.11 + 676,529.76 + 709,984.05 + 789,948.40 – 1,850,000

= 2,682,791.32 – 1,850,000

= N832,791.32

Comment:
Since the NPV is positive, I will advise the company to accept the project.

SOLUTION 12

SECURITY A

Return (N)	Probability		(Xi – EV)	$(Xi - Ev)^2 x Prob$
4,000	0.4	1,600	-700	196,000
5,000	0.5	2,500	300	45,000
6,000	0.1	600	1,300	169,000
		EV = 4,700		$\sigma^2 = 410,000$

Expected return = N4,700
Variance = 410,000
Standard deviation = 410,000 = N640
Coefficient of variation = $\frac{640}{4,700} \times \frac{100}{1}$
= 14%

SECURITY B

Return	Probability		(x1 – Ev)	(X1 – Ev)²*Prob
6,000	0.7	4,200	700	343,000
4,000	0.2	800	-1300	338,000
3,000	0.1	300	-2300	529,000
		Ev = 5,300		σ^2 = 1,210,000

Expected Return = N5,300
Variance = N1,210,000
Standard deviation = $\sqrt{1,210,000}$ = N1,100
Coefficient of variation = $\frac{1100}{5300} \times 100 = 21\%$

(b) I would recommend security A because its coefficient of variation is lower than that of security B. (i.e. it has less risk).

SOLUTION 13

PROJECT X

CFi	Prob	Ev	(CF – Ev)	(CF – Ev)² x *Prob*
15,000	0.1	1,500	-10,500	11,025,000
22,000	0.3	6,600	-3,500	3,675,000
27,000	0.2	5,400	1,500	450,000
30,000	0.4	12,000	4,500	8,100,000
		25,500		σ^2 = 23,250,000
				σ = 4822

PROJECT Y

CFi	Prob	Ev	(CF – Ev)	(CF - Ev)² x *Prob*
20,000	0.4	8,000	- 5,500	12,100,000
25,000	0.2	5,000	- 500	50,000
30,000	0.3	9,000	4,500	6,075,000
35,000	0.1	3,500	9,500	9,025,000
		25,500		σ^2 = 27,250,000
				σ = 5220

PROJECT Z

Cfi	Prob	Ev	(CF – Ev)	(CF – Ev)² x *Prob*
27,000	0.5	13,500	– 4,000	8,000,000
30,000	0.2	6,000	–1,000	200,000
35,000	0.1	3,500	4,000	1,600,000
40,000	0.2	8,000	9,000	16,200,000
		31,000		σ^2 = 26,000,000
				σ = 5099

PROJECT

	X N	Y N	Z N
Project mean expected cashflow	25,500	25,500	31,000
Variance	23,250,000	27,250,000	26,000,000
Standard deviation	4,822	5,220	5,099
Coefficient of variation	$\frac{4822}{25,500} \times \frac{100}{1} = 19\%$	$\frac{5220}{25,000} \times \frac{100}{1} = 20\%$	$\frac{5099}{31,000} \times \frac{100}{1} = 16\%$

(b) Project Z is less risky, because it has the least coefficient of variation.

SOLUTION 14

Computation of Expected Sales

Year 1 = 12,000 (0.4) + 6,000 (0.3) + 4,000 (0.3) = N7,000
2 = 20,000 (0.4) + 10,000 (0.3) + 8,000 (0.3) = N13,400
3 = 28,000 (0.4) + 14,000 (0.3)+ 8,000 (0.3) = N17,800
4 = 34,000 (0.4) = 17,000 (0.3) + 10,000 (0.3) = N21,700
5 = 40,000 (0.4) + 20,000 (0.3) + 10,000 (0.3) = N25,000

Computation of Variable Costs

Year 1 = 0.6 x N7,800 = N4,680
2 = 0.6 x N13.400 = N8,040
3 = 0.6 x N17,800 = N10,680
4 = 0.6 x N21,700 = N13,020
5 = 0.6 x N25,000 = N15,000

Computation of Expected Net Present Value

Year	Cost	Expected Sales N	Variable Costs N	Fixed Cost N	Net Cashflow N	DCF = 12%	
0	(18,000)				(18,000)	1.0000	(18,000)
1		7,800	(4,680)	(1,000)	2,120	0.8929	1,892.95
2		13,400	(8,040)	(1,000)	4,360	0.7972	3,475.79
3		17,800	(10,680)	(1,000)	6,120	0.7118	4,356.22
4		21,700	(13,020)	(1,000)	7,860	0.6355	4,880.64
5		25,000	(15,000)	(1,000)	9,000	0.5677	5,106.06
						ENPV	1,712.20

Since the Expected Net Present Value is positive, the company should embark on the project and its Expected Net Present Value = N1,712.20.

SOLUTION 15

YEAR ONE

CASHFLOW N	PROBABILITY	CF x PROB N	(CF – EV)	(CF – EV)² x Prob N
50,000	0.3	15,000	-11,000	36,300,000
60,000	0.3	18,000	-1,000	300,000
70,000	0.4	28,000	9,000	32,400,000
	EV_1 =	61,000		69,000,000 = σ_1^2

YEAR TWO

CASHFLOW N	PROBABILITY	CF x PROB N	(CF – EV)	(CF – EV)² x Prob N
60,000	0.4	24,000	-8,000	25,600,000
70,000	0.4	28,000	2,000	1,600,000
80,000	0.2	16,000	12,000	28,800,000
	EV_2 =	68,000		56,000,000 = σ_2^2

YEAR THREE

CASHFLOW N	PROBABILITY	CF x PROB N	(CF – EV)	(CF – EV)² x Prob N
80,000	0.5	40,000	-6,000	18,000,000
90,000	0.3	27,000	4,000	4,800,000
95,000	0.2	19,000	9,000	16,200,000
	EV_3 =	86,000		39,000,000 = σ_3^2

(a) Expected Value of Net Present Value (EVNPV) of the project can be calculated using the formula stated below:

$$EVNPV = \sum_{t=1}^{n} \frac{EV_t}{(1 + R_F)^t} - IO$$

Where EVt = Expected value in year t
RF = Risk free rate of return = 12% = 0.12
IO = Initial outlay = N150,000
n = Estimated life of the project = 3

$$\therefore EVNPV = \frac{61,000}{(1+0.12)^1} + \frac{68,000}{(1+0.12)^2} + \frac{86,000}{(1+0.12)^3} - 150,000$$

$$= \frac{61,000}{(1.12)} + \frac{68,000}{(1.12)^2} + \frac{86,000}{(1.12)^3} - 150,000$$

$$= 54,464.29 + 54,209.18 + 61,213.10 - 150,000$$

$$= 169,886.57 - 150,000$$

$$EVNPV = N19,886.57$$

The standard deviation (σ_P) of the Expected Value of NPV can be determined using the formula below:

$$= \sqrt{\sum_{t=1}^{n} \frac{\sigma_A^2}{(1 + RF)2t}}$$

$$= \sqrt{\frac{69,000,000}{(1+0.12)^{2x1}} + \frac{56,000,000}{(1+0.12)^{2x2}} + \frac{39,000,000}{(1+0.12)^{2x3}}}$$

$$= \sqrt{\frac{69,000,000}{(1.12)^2} + \frac{56,000,000}{(1.12)^4} + \frac{39,000,000}{(1.12)^6}}$$

$$= \sqrt{55,006,377.55 + 35,589,012.39 + 19,758,613.73}$$

$$= \sqrt{110,354,003.7}$$

σ_P = N10,504.95

The coefficient of Variation (Cov) of the Project can be determined using the formula stated below:

$$C.O.V. = \frac{\text{Standard deviation}}{\text{Expected Value of Net Present Value}} \times \frac{100}{1}$$

$$= \frac{\sigma^P}{ENPV} \times \frac{100}{1}$$

$$= \frac{10504.95}{19886.57} \times \frac{100}{1}$$

$$= 52.82\%$$

SOLUTION 16

PROJECT X

$$\text{Expected Net Present Value (ENPV)} = \sum_{t=1}^{n} \frac{EV_t}{(1+K)^t} - IO$$

Expected Value (EV) for year 1 = (0.7) (5.0) + (0.3) (2.0)

= 3.5 + 0.6

= N4.1 million

Expected Value for year 2 (EV2) = (0.7) (3.5) + (0.3) (2.5)

= 2.45 + 0.75

= N3.20 million

$$\therefore ENPV = \frac{4,100,000}{(1.013)^1} + \frac{3,200,000}{(1+0.13)^2} - 4,000,000$$

$$= \frac{4,100,000}{1.13} + \frac{3,200,000}{(1.13)2} - 4,000,000$$

= N3,628,318.58 + 2,506,069.39 – 4,000,000

= N2,134,387.97

PROJECT Y

Expected Value for year 1 (EV_1) = (0.6) (3.5) + (0.3) (2.5) + (0. 1) (2.0)

= 2.1 + 0.75 + 0.2

= N3.05 million

Expected Value for year 2 (EV_2) = (0.6) (2.0) + (0.3) (1.5) + (0. 1) (4.0)

= 1.2 + 0.45 + 0.4

= N2.05 million

Expected Value for year 3 (EV_3) = (0.6) (3.0) + (0.3) (2.0) + (0.1) (1.0)

= 1.8 + 0.6 + 0.1

= N2.5million

$$ENPV = \frac{3,050,000}{(1+0.1)} + \frac{2,050,000}{(1+0.1)^2} + \frac{2.500.000}{(1+0.1)^3} - 4,000,000$$

$$= \frac{3,050,000}{(1.1)} + \frac{2,050,000}{(1.1)^2} + \frac{2,500,000}{(1.1)^3} - 4,000,000$$

= 2,772,727.27 + 1,694,218.88 + 1,878,287.00 – 4,000,000

= N2,345,233.15

When two mutually investments have different lives, the Annual Equivalent value must be calculated before a sound decision can be made. Furthermore, if they also differ in risk, they must be discounted to infinity for a good decision.

	Project X	**Project Y**
NPV	2,134,387.97	2,345,233.15
Cumulative DCF	1.6681	2.4869
Annual Equivalent Value	$\frac{2,134,387.97}{1.6681}$	$\frac{2,345,233.15}{2.4869}$
	= N1,279,532.28	= N943,034.76
Risk factor	13%	10%

Annual Equivalent Value $\frac{1,279,532.28}{0.13}$ $\frac{943,034.76}{0.1}$

= N9,842,556 = N9,430,348

Comment

Looking at the above two mutually exclusive investments, they have different lives, hence we have bought them at par by getting their Annual Equivalent Value. Also, they have different risks. Project X has a risk of 13% whereas Project Y has a risk of 10%, this is why we have discounted to infinity.

Therefore, Project X should be selected because it maximises income than the Project Y.

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CHAPTER 6

CAPITAL BUDGETING WITH INFLATION

Question 1

The expected annual rate of inflation is 10% and the money cost of capital is 20%. Calculate the real cost of capital.

Question 2

Nene Nig. Ltd. is considering a project with initial cost of N23,000, the life of the project is expected to be 4 years while the expected scrap value is nil. The annual cash inflow including the effect of inflation are as stated below:

Year 1	N15,000
Year 2	N21,000
Year 3	N23,000
Year 4	N17,000

The market discount rate is 18% per annum and all cash flow are affected by constant inflation at 5% per year.

Required:

(a) Calculate the NPV of the project based on :
  (i) Money cash flow
  (ii) Real cash flow

(b) Calculate the real Internal rate of return.

Question 3

Nana Ltd. is a car company in the commercial transport industry. The company uses special brand of Nissan Primera for its operation. The company maintains fleet of 10 of these special brand of cars, which has a life span of 5 years. At the end of the expired life, which is 5 years, the company replaces all the ten cars at a time. The company cost of capital is 20%.

Today 01/01/04 makes another period of replacement for the cars. Assuming all the cars are replaced at N1,900,000 altogether and that 2008 is 5 years way from today. Also from the experience the cars have no scrap values at the end of their life span. Inflation has predominated the market prices and from the economic survey the inflationary rate has been constant at 25% per annum. Each of cars is capable of generating N100,000 income from the passengers fares yearly.

The annual total maintenance cost for the vehicles have been put at:

Year 1 N200,000
Year 2 N350,000
Year 3 N500,000
Year 4 N650,000
Year 5 N800,000

Other costs have been estimated to be 20% of the annual maintenance cost. Note that inflation affects all cost and revenue.

Required:
Determine the profitability of this business in the current period 2004–2008 which is 5 years.

Question 4

Agbor Limited is considering a project costing N600,000. The project is expected to Have a life of 4 years with a scrap value of N48,000, annual cash revenue from the project is expected to be N400,000 in year 1 rising by 6% per annum for inflation running cost are expected to be N180,000 in 1 year rising by 14% per annum because of inflating labour cost. The general rate of inflation is expected to be 7% and that company money cost of capital is 20%. Advise the company on the acceptability of the project.

Question 5

EKPOMA VENTURES Ltd. is considering a project costing N300,000. The project is expected to have life span of 4 years with residual value of N5,000, the selling price per unit of the product will be N10 in year one and rise by 10% pe annum as a result of inflation. The project also involves labour cost of N5 per unit in year one and rise by 9% per annum as a result of inflation. Other fixed running expenses is expected to be N80,000 in year one and rise by 8% per annum as a result of inflation. The expected annual sales demand is 35,000 units. The general price level increase in the country is 8% and the company cost of capital is 18%.

Required:
Advise the company whether or not to accept the project using the Net present value criteria.

QUESTION 6

Ogunmola Johnson Ltd. is contemplating on two mutually exclusive projects, you as the finance manager is now invited to give your advice as to the better choice between the two using Net Present Value criteria.

INVESTMENT ONE

The company intends to computerise its operations which will boost efficiency and increase in sales.

	N'000
Cost of computerisation	26,000
Annual Service charge	3,800
Annual Sales Increase expected	10,000

The company is expected to phase out the computerisation at the end of 6th year in order to go with the expected latest technology known as magnetic operation.

INVESTMENT TWO

Instead of going computerisation, the company can employ two foreigners who specialise in production and marketing respectively. However, a new production machine will be acquired:

Details are as follows:

	N'000
Annual salaries of the two foreigners	9,200
Cost of new machine	6,000
Annual maintenance cost of the machine	2,800

The existing annual sales of N8 million will be boosted by 25 percent. The life span of this alternative is also 6 years.

The interest rate on capital is put at 15 percent and the annual sales of both alternatives are expected to rise by 10 percent per annum after year 1 because of inflation.

CAPITAL BUDGETING WITH INFLATION

SOLUTION 1

$(1 + K) = (1 + R)(1 + Inf)$
$(1 + 0.2) = (1 + R)(1 + 0.1)$
$1.2 = (1 + R)(1.1)$
$1 + R = \frac{1.2}{1.1}$

$1 + R = 1.09$
$R = 1.09 - 1 = 0.09 = 9\%$
Real cost of capital = 9%

SOLUTION 2

YR	CASH FLOW	DCF 18%	PV
0	(23,000)	1.0000	(23,000)
1	(15,000)	0.8475	12,713
2	21,000	0.7182	15,082
3	23,000	0.6086	13,998
4	17,000	0.5158	8,769
			27,.562

The NPV of the project based on money cash flow = N27,562.

Real cash flow year 1 $= \frac{15,000}{(1.05)} = 14,286$

Real cash flow year 2 $= \frac{(21,000)}{(1.05)^2} = 19,048$

Real cash flow year 3 $= \frac{23,000}{(1.05)^3} = 19,868$

Real cash flow year 4 $= \frac{17,000}{(1.05)^4} = 13,986$

Real cost of capital $= \frac{1.18}{1.05} - 1 = 1.12 - 1 = 0.12$

YEAR	CASH FLOW	DCF 12%	
0	(23,000)	1.00000	(23,000)
1	14,286	0.8929	12,756
2	19,048	0.7972	15,185
3	19,868	0.7118	14,142
4	13,986	0.6355	8,888
			27,971

The NPV of the project based on real cash flow = N27,971

(b)

YEAR	CASHFLOW	DCF52%	PV	CASHFLOW	DCF62%	PV
0	(23,000)	1.0000	(23,000)	(23,000)	1.0000	(23,000)
1	14,286	0.6579	9,399	14,286	0.6173	8,819
2	1,048	0.4328	8,244	19,048	0.3810	7,257
3	19,048	0.2848	5,658	19,868	0.2352	4,673
4	13,986	0.1873	2,620	13,968	0.1452	2,031
				2,921		(220)

Real IRR $= DCF + \frac{NPV}{NPV_+ - NPV_-}(DCF_- - DCFX)$

$= 0.52 + \frac{2921}{2921 + 220}(0.62 - 0.52)$

$= 0.52 + \frac{2921}{3141}(0.1)$

$= 0.52 + 0.092$
$= 0.612$
$= 61\%$

SOLUTION

SALES

The annual sales generated by the cars are regarded as real cash flow and must be converted to money cash flow. 10 cars generate N1 million yearly.

YEAR	N	SALES
1	1,000,000 $(1.25)^1$	1,250,000
2	1,000,000 $(1.25)^2$	1,562,500
3	1, 000,000 $(1.25)^3$	1,953,125
4	1, 000,000 $(1.25)^4$	2,441,406
5	1, 000,000 $(1.25)^5$	3,051,758

Maintenance Cost + Other Cost

Year	Maintenance Cost	Other cost (20%) of Maintenance Cost	Total cost		
1	200,000	40,000	240,000 (1.25)	=	300,000
2	330,000	66,000	396,000 $(1.25)^2$	=	618,750
3	500,000	100,000	60,000,000 $(1.25)^3$	=	1,171,875
4	670,000	134,000	804,000 $(1.25)^4$	=	1,962,891
5	800,000	160,000	960,000 $(1.25)^5$	=	2,929,688

Year	Initial cost	Sale	Total cost	Net cost	DCF 20%	
0	(1,900,000)			(1,900,000)	1.0000	(1,900,000)
1		1,250,000	(3,000,000)	950,000	0.8333	791,635
2		1,562,500	618,750	943,750	0.6944	655,340
3		1,953,125	1,171,875	781,250	0.5787	452,109
4		2441406	1,962,891	478,515	0.4823	230,788
5		3,051,758	2,929,688	122,070	0.4019	49,060
						278,932

Comment: The project is acceptable based on the positive NPV of N278,932.

SOLUTION 4

Year	Cash Revenue	Running Cost	Cash Profit
1	400,000	180,000	220,000
2	400,000 $(1.06)^1$	180,000 $(1.14)^1$	218,800
3	400,000 $(1.06)^2$	180,000 $(1.14)^2$	215,512
4	400,000 $(1.06)^3$	180,000 $(1.14)^3$	209,723

The NPV the project is as follows:

Year	Cash/Scrap Value	Cash Profit	Net flow	Dcf 20%	PV
0	(600,000)		(600,000)	1.0000	600,000
1		220,000	220,000	0.8333	183,326
2		215,800	218,800	0.6944	151,935
3		215512	215,512	0.5787	124,718
4	48,000	209728	257,728	0.4823	124,302
					(15,719)

Comment: The NPV is negative and thus, Agbor Limited should reject the project.

SOLUTION 5

YEAR	SALES		N
1	35000 (10)	=	350,000
2	35000 (10)$(1.1)^1$	=	385,000
3	35000 (10)$(1.1)^2$	=	423,500
4	35000 (10) $(1.1)^2$	=	465,850

YEAR	LABOUR		
1	35000 (5)	=	175.000
2	35000 (5) $(1.09)^1$	=	190,750
3	35000 (5) $(1.09)^2$	=	207,918
4	35000 (5) $(1.09)^3$	=	226.630

YEAR	RUNNING EXPENSES		
1	80000 $(1.08)^0$	=	80,000
2	80000 $(1.08)^1$	=	86,400
3	80000 $(1.08)^2$	=	93,312
4	80000 $(1.08)^3$	=	100,777

Year	Sales	Labour Cost	Running Expense	Cash Profit N
1	350,000	(175,000)	(80,000)	95,000
2	385,000	(190,750)	(86,400)	107,850
3	423,500	(207,918)	(93,312)	122,270
4	46,850	(226,630)	(100,777)	138443

Year	Cost/Residual	Cash	Net Cash	DCF = 18%	PV
0	(300,000)		(300,000)	1.0000	(300,000)
1		95,000	95,000	0.8475	80,513
2		107,850	107,850	0.7182	77,458
3		122,270	122,270	0.6086	74,414
4	5,000	138,443	143,443	0.5158	73,088
					6,373

Since the NPV is positive, EKPOMA VENTURES Ltd. should accept the project.

SOLUTION 6

ALTERNATIVE ONE

Year Annual sales due to 10% inflation

			N
1	10,000,000	=	10,000,000
2	$10,000,000(1.1)^1$	=	11,000,000
3.	$10,000,000(1.1)^2$	=	12,100,000
4	$10,000,000(1.1)^3$	=	13,310,000
5	$10,000,000(1.1)^4$	=	14,641,000
6	$10,000,000(1.1)^5$	=	16,105,100

Year	Cost N	ANNUAL SALES N	SERVICE CHARGES N	NET CASHFLOW N	DCF = 15%	PV N
0	(26,000,000)			(26,000,000)	1.0000	(26,000,000)
1		10,000,000	3,800,000	6200,000	0.8696	5,391,520
2		11,000.000	3,800,000	7200,000	0.7561	5,443,920
3		12,100,000	3,800,000	8,300,000	0.6575	5,457,250

4	13,310,000	3,800,000	9,510,000	0.5718	5,437,818
5	14,641,000	3,800,000	10,841,000	0.4972	5,390,145
6	16,105,100	3,800,000	12,305,100	0.4323	5,319,495
					6,440,148

∴ The Net Present value of alternative one is N6,440,148.

ALTERNATIVE TWO

Year

		Annual Sales due to 10% inflation N
1	10,000,000(1.25)	12,500,000
2	$10,000,000\ (1.25)\ (1.1)^1$	13,750,000
3	$10,000,000\ (1.25)\ (1.1)^2$	15,125,000
4	$10,000,000\ (1.25)\ (1.1)^3$	16,637,500
5	$10,000,000\ (1.25)\ (1.1)^4$	18,301,250
6	$10,000,000\ (1.25)\ (1.1)^5$	20,131,375

Year	ANNUAL SALES	ANNUAL SALARIES	ANNUAL MAINTENANCE	NET SALES
1	12,500,000	(9,200,000)	(2,800,000)	500,000
2	13,750,000	(9,200,000)	(2,800,000)	1,375,000
3	15,125,000	(9,200,000)	(2,800,000)	3,125,000
4	16,637,500	(9,200,000)	(2,800,000)	4,637,500
5	18,301,250	(9,200,000)	(2,800,000)	6,301,250
6	20,131,375	(9,200,000)	(2,800,000)	8,131,375

Year	COST N	NET SALES N	NET CASHFLOW N	DCF = 15%	PV N
0	(6,000,000)		(6,000,000)	1.0000	(6,000,000)
1		500,000	500,000	0.8696	434,800
2		1,375,000	1375,000	0.7561	1,039,638
3		3,125,000	3,125,000	0.6575	2,054,688
4		4,637,500	4,637,300	0.5718	2,651,723
5		6,301,250	6,301,250	0.4972	3,132,982
6		8,131,375	8,131,375	0.4323	3,315,193
					6,829,024

Comment

Looking at the above two mutually exclusive investments, Alternative two should be selected because it has higher positive Net Present Value than Alternative one, i.e. N6,829,024, N6,440,148.

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CHAPTER 7

CAPITAL BUDGETING WITH TAX / BUY OR LEASE DECISION

Question 1

Explain briefly and state 3 merits and 2 limitation of the following terms:

i. Leasing
ii. Hire purchase

Question 2

Make a clear distinction between Operating lease and finance lease.

Question 3

Briefly outline the basic step involves in buy outright with cash, Borrow to buy, lease and hire purchase decision.

Question 4

Mama Tee Limited is considering whether or not to invest in a new project which will involve the production of Lady's Tops, the project will involve capital initial outlay as stated below:

	N000
Machine and Equipment	6,000
Motor Vehicles	2,500
Forklift	1,500
	10,000

The machine and Equipment is expected to have a life of 5 years with a scrap value of N250,000. The Motor Vehicles are expected to have a life of 4 years with a scrap value of N100,000. The forklift is expected to have a life of 5 years with a scrap value of N60,000. The Expected cash revenue and running expenses of the project is as follows;

Year	Cash Revenue	Cash Expenses
	N	N
1	12,000,000	8,500,000
2	10,000,000	7,500,000

3	9,500,000	7,000,000
4	9,000,000	6,000,000
5	8,000,000	5,500,000

Revenue are expected to increase at the rate of 12 percent per annum after year one while expenses are expected to increase at the rate of 10% per annum after year one.

Capital Allowances are available on a reducing balance basis as follows;

Machine and Equipment	25% per annum
Motor Vehicles & Forklift	20% per annum

The general rate of inflation in the country is 11% while the after tax cost of capital is 15%.

You are required to advise Mama Tee on the acceptability of the project using Net Present Value Criteria.

Question 5

Nene Limited is considering purchasing an equipment costing N1 million. The equipment is expected to have a life span of 5 years with residual value of N160,000.

The company financial manager have projected that the equipment will yield annual cash inflow of N400,000.

The inland revenue have confirmed that capital allowance will be allowed on this equipment at the rate of 25% per annum on reducing balance basis. The current rate of corporate tax is 35% payable one year in arrears. The company cost of capital is 20% you are require to advise Nene Ltd on the acceptability of the equipment.

Question 6

Nwata John Limited has decided to install a new Equipment Costing N650,000. The equipment has an estimated useful life of 6 years with a residual value of N100,000. The Equipment will generate additional cash profits of N220,000 per annual for six years, the company is now trying to decide the method of financing the projects. The following methods of financing are being considered.

(i) Purchase the equipment using the company cash resources.

(ii) Lease the equipment on agreement that would entail payment of N140,000 at the end of each year for the next six years. The company can currently borrow at the rate of interest of 18% before tax from its bank. If the company purchases the equipment capital allowances on a reducing balance basis will be available at the rate of 20% per annum. The current corporate tax rate is 35% payable one years in arrears. The company weighted average cost of capital is 14% after tax.

You are required to advise the management of Nwata John Limited on:

(a) Whether or not to purchase the equipment.

(b) The most Economic Method of Finance.

Question 7

You work as Finance Manager in a company that intends to acquire modernised Production Unit but which is not certain whether to lease or borrow and buy. The following information pertain to the choices available to the company:

(i) Borrowing would cost 12% per annum before tax. The assets would cost N9.2 million with a guarantee salvage value of N900,000 at the end of 5 years. The plant will qualify for first year allowance of 100%.

(ii) Two financial leases are being considered, the company is to choose the best of the alternatives.

OUTFLOWS

Year	A N,000	B N,000
1	2,000	500
2	2,000	800
3	2,000	1,600
4	2,000	3,000
5	2,000	5,000

If the asset is leased, the entire salvage value and the allowances will accrue to the lessor.

Required:

(a) Advice on the best method of financing the acquisition of the asset if:

(i) The firm is subject to tax at 25% claimable the following year.

(ii) The firm is permanently in a non taxable position.

Question 8

Odogwu Paul Ltd is considering whether or not to invest in the development of a new product. The product which would involve buying a new equipment costing N7.6 million is expected to have a life of five years, the estimated residual value of the equipment at the end of its life is N700,000. The finance Manager of Odogwu Paul Ltd has estimated that the new product will generate sales of 1.4 million units per annum. The estimated sales price per unit in year 1 is N10.00 rising by 5% per annum for inflation. Material cost per unit is expected to

be N3.00 in year 1, but would increase by 10% per annum because of inflation labour cost per unit is expected to be N4.00 in year 1, but would increase by 12% per annum because of inflation.

Overhead Cost per Unit (Cash item only) is expected to be N2.00 per unit but would increase by 8% per annum because of inflation.

Odogwu Paul Limited can claim capital allowance at the rate of 25% on the reducing balance basis on this project. Tax is currently at the rate of 35% payable one year in arrears.

The company's real after tax weighted average cost of capital is estimated to be 13% per annum and its nominal after tax weighted average cost of capital is 18%.

Odogwu Paul Limited can currently borrow at the rate of 21% per annum from its bank fortune Real Bank Ltd.

Required:

i. Using the Net Present Value criteria, advise the company on the project.

ii. Suppose Odogwu Paul Limited has the opportunity to lease the machine for the project under a finance lease arrangement at a rental of N2.3 million per annum for 5 years payable at the end of each year. Should the company lease or buy the machine?

Question 9

Roasted Chicken Ltd. which operates in the catering industry is considering increasing its production capacity. To achieve the new capacity level the company is considering a machine costing N10 million the machine has a life of five years with a residual value of N500,000. The estimated cash revenue and expenses from the project are as follows:

Year	CASH REVENUE N million	CASH EXPENSES N million
1	7.6	3.6
2	8.6	4.1
3	8.8	4.8
4	9.2	6.6
5	9.4	6.9

Capital Allowance will be available against taxable profit of the project at 35% per annum on a reducing balance basis. The rate of corporation tax is 35% payable one year in arrears.

Madam Rossy, the managing director of the company believes that buying the machine is

always better than leasing because the cost of funds which the lessor implicitly built into a lease is almost always higher than the commercial interest rate on loanable funds. Madam Rossy can currently borrow at the rate of 21% per annum from Zeinab Bank Plc.

Lanky Leasing Agency has offered to lease the machine to the company under a finance lease arrangement at an annual rental of N3.2 million for five years payable at the end of each year.

The above cashflow estimates have already taken into consideration annual sales price increase of 12 percent and annual increase in operating expenses of 10 percent.

The annual general rate of inflation is 8 percent while the company's cost of capital is 20 percent.

Required:

Give a quality advice to the Rested Chicken Limited Management on whether to buy or lease the machine (show all computations).

QUESTION 10

Oyewale block industry has decided to install a new block moulding machine . The machine cost N40 million and are expected to have a life of 5 years with an estimated scrap value of N950,000.

The management of Oyewale Block Industry have recommended that selling price per unit of block will be N60 in the first two years of the machine after which it will be increase to N65 per unit. The sales volumes based on these recommended selling price are as follow.

Year		
	1	5.0 million units
	2	6.5 million units
	3	7.2 million units
	4	7.5 million units
	5	8.0 million units

The variable cost per unit of producing block is N55 while annual cash fixed operation expenses of 24 million will also be incurred capital allowance on a reducing balance basis will be available at the rate of 35% per annum against taxable profit and the first capital allowance will be claimed in year 1.

The following methods of finance are being considered

(a) Purchase the machine using the company cash resources.

(b) Lease the machine under an agreement that would entailed payment of N10.5 million per annum for 5 years payable at the end of each year from softy leasing company.

The company can currently borrow at the rate of 21% before tax. The current rate of corporations tax is 35% payable one year in arrears. The company's nominal after tax weighted average cost is 20% and all the above cashflow estimates are in nominal terms.

Required:

Should Oyewale Block Industry buy or lease the machine?

Question 11

The Lagoon State of Nigeria is considering whether or not to invest in a new project which will involve the manufactured of AIDS preventive pills. The project would involve purchase of equipment costing N10.6 million and are expected to have life of 5 years with an estimated trade-in-value of N800,000. Other project costs that will be incurred at the beginning of the project include forklift cost amounting to N2.6 and tools cost amounting to N1.8 million.

The commissioner of health has recommended that selling price per pack of 10 pills will be N20.00 in the first two years of the project after which it will be increased to N22.50 per pack of 10 pills. The project sales volume based on these recommended selling price are as follows:

Year	Pack of 10 pills
1	5,000,000
2	6,000,000
3	7,500,000
4	8,000,000
5	9,000,000

The variable cost per pack of 10 pills is N18.50 while annual cash fixed expenses of 6 million will also be incurred the above selling prices and variables cost have taken into specific inflation rate of 13% and 11% respectively.

The state government has the options of using the cash resources to finance this project or leasing the equipment from World Health Organisation Agency. World Health Organisation Agency has offered to lease the equipment to the state government on a finance lease arrangement at a rental of N4.2 million per annum for 5 years, payable at the end of each year.

Capital allowance will be available against taxable profits on the equipments of the projects the rate of 25% per annum on a reducing balance basis.

All the above cash flow estimates are in nominal terms while the current rate of corporate tax is 35% payable one year in arrears.

The current nominal rate of return on treasury bill is 10% while the nominal return on the market is 19%. The beta of this project is the same with company's beta of 2.0 Lagoon state can currently borrow at the rate of 20 percent per annum.

You are required to advise the lagoon state government on:

(a) . Whether to acquire the equipment.

(b) Whether to buy or lease the equipment.

QUESTION 12

Biodun Oyekanmi Limited has decided to install a new equipment costing N650,000 the equipment has an estimated useful life of 6 years with a residual value of N100.000.

The equipment will generate additional cash profit of N220,000 per annum for six years. The company is now trying to decide the method of financing the project. The following methods of finance are being considered.

(i) Purchase the equipment using the company cash resources.

(ii) Lease the equipment under an agreement that would entail payment of N140,000 at the end of each year for the next six years.

(iii) Purchase the equipment under hire purchase agreement. This would require an initial deposit of N210,000 and payment of N123,000 per annum at the end of each of the next six years. The interest part of the payment for tax purposes, would be N70,000, N60,000, N46,000, N30,000, N21,000 and N15,000 at the end of year 1, 2, 3,4, 5 and 6 respectively.

The company can currently borrow at the rate of interest of 18% before tax from its bank. If the company provides the equipment Capital Allowances on a reducing balance basis will be available at the rate of 25% per annum.

The current corporate tax rate is 35% payable one years in arrears. The company weighted average cost of capital is 14% after tax.

Required:

Give a quality advice to Oyekanmi Abiodun Limited management on the most economic method of finance that will maximise the wealth of the company.

QUESTION 13

M.M. Hassan Nig. Ltd. has decided to acquire an equipment which costs N9.45 million and has an operation life of 5 years with a scrap value of N697,084. Tax is payable at 35% on operating cashflows one year in arrears.

Capital Allowances are available at 5% a year on a reducing balance basis. Suppose that the company has the opportunity either to purchase the machine or to lease it under a finance lease arrangement, at an annual rent of N3 million for 5 years, payable at the end of each year.

The company can borrow to finance the acquisition at 12%.

Required:

Should the company buy or lease the equipment.

CAPITAL BUDGET WITH TAX/ BUY OR LEASE DECISION

SOLUTION 1

(i) **LEASE:-** A lease is an arrangement where a person (Lessor) having an asset gives it to another person (lessee) for usage on a rental basis and in most cases the lessee has the option of continuing to lease the asset for an indefinite period after the first initial period at a lower rate.

MERITS

- It is a medium term source of finance for firm that is unable to raise funds to purchase and acquiring an asset for immediate use.
- Lease serves as a bail-out for company that having persistence liquidity problem.
- Lease arrangement between the owner of an asset (lessor) and the users of an asset (lessee) serve as a tax benefit to both party.
- Since the rent amount on lease is fixed throughout the lease period, it could be a hedge against persistence increase in the price of an asset (inflation).

LIMITATION

- The rent payment are compulsory for the users of an asset (lessee) irrespective of the profit level.
- Non payment of agreed rent may jeopardise the business of the user of an asset (lessee) in the sense that the owner (lessor) may repossess the asset from the user (lessee) and this may disrupt the user (lessee) production process.
- There is mixed fortune for the lessee and lessor when the interest rate charged by banks fluctuates in the sense that when the interest rate charge by banks goes up and rental payment on lease is fixed, the lessor loses and the lessee gains. On the other hand, when the interest rate charged by banks goes down, the lessor gains and the lessee loses.

(ii) **HIRE PURCHASE:-** Is a means of purchasing an asset on a piece meal payment after the initial down payment. This implies that the balance after the initial down payment are made through regular instalmental payment at a regular periodic interval (quarterly, monthly, bi-annually, annually) in accordance with the contractual agreement between the owner (hiror) and the purchaser (hiree) of an asset. It must be noted that the hiree becomes the owner of the asset immediately he made the last instalmental payment to the original owner of the asset.

MERITS OF HIRE PURCHASE

- It is a means of purchasing an asset on a piece-meal payment basis, therefore it reduces pressure on cash resources of firms compare to outrightly buy with cash.
- It also allows company to use their cash resources for another investment opportunity.
- There is a benefit of Capital Allowance to the purchaser of an asset (hiree).

LIMITATION OF HIRE PURCHASE

- The interest rate in hire purchase agreement may make the cost of acquiring an asset higher than any other means of acquiring an asset.
- If the hiree default in completing the instalmental payment, the ownership status of the asset remain with the original owner (hiror).

SOLUTION

	OPERATING LEASE	**FINANCE LEASE**
1.	It is short term cancellable lease arrangement between the owner (lessor) and the user (lessee) of an asset.	It is medium term non-cancellable lease arrangement between the owner (lessor) and the user (lessee) of an asset.
2.	The lease agreement period does not cover the useful economic life of the lease asset.	The lease agreement period covers the whole useful economic life of the lease asset.
3.	The owner of an asset (lessor) is responsible for maintenance, servicing and insurance of the lease asset.	The user of an asset (lessee) is responsible for maintenance, servicing and insurance of the lease asset.
4.	The user of the asset (lessee) does not have the right to purchase the lease asset after the initial lease period.	The users of the asset (lessee) have the right to purchase the lease asset after the initial lease period.
5.	The lessor remains the owner of the lease asset throughout its life whether in substantive or in law.	The lessee remain the owner of the lease asset throughout its life in substance but not in law.

SOLUTION 3

The basic steps involves are as follows:

(A) ACQUISITION DECISION

The relevant cashflows involves are as follows:

- Initial outlay of the asset.
- Annual Cash profit or inflow of the asset.
- Annual tax liability after claiming Capital Allowances.
- Residual value of the asset if any.

* The above listed cashflows will be compressed to become one which is called AFTER TAX NET CASHFLOW.

* The After tax Net Cashflow will be discounted by using the company after tax weighted average cost of capital.

* If the Net Present Value is positive, then the decision is to purchase or acquire the asset irrespective of the method use to finance the purchase.

(B) FINANCING DECISION

(i) BUY OUTRIGHTLY WITH CASH RESOURCES OPTION

The relevant differential cashflow are as follow:

- The initial outlay of the asset.
- The residual value of the asset, if any.
- Tax Savings on Capital allowance.

* The above cashflows will be discounted using the company's After tax cost of borrowing.

(ii) BORROW TO BUY OPTION

The relevant differential cashflow are as follow:

- Tax Savings associated with Capital Allowance claimed.
- Tax Savings associated with annual interest paid on loan.
- Residual value of the asset, if any.
- Annual loan repayment.

* The above listed relevant cashflow will be compressed to become one which is called AFTER TAX NET CASHFLOWS.

* The After tax net cashflow will be discounted by using the company after tax cost of borrowing.

(iii) LEASE OPTION

The relevant differential cashflows are as follow:

- Tax Savings on lease payment.
- Lease payment.
- The above listed relevant differential cashflow will be discounted by using the company after tax cost of borrowing.

(iv) HIRE PURCHASE OPTION

The relevant differential cashflows are as follow:

- Tax Savings associated with Capital Allowance claimed.
- Tax Savings associated with hire purchase interest.
- Residual value of the asset, if any.
- Hire purchase payment.
- * The above relevant differential cashflow will be compressed to become one which is called After Tax Net Cashflow.
- * The After tax net cashflow will be discounted by using the company after tax cost of borrowing.

(C) Select the best financing decision by selecting the option that has the lowest Present value of cost.

SOLUTION 4

COMPUTATION OF CAPITAL ALLOWANCE (MACHINE & EQUIPMENT)

Year		CAPITAL ALLOWANCE	BALANCE c/d
		N	N
1	0.25 (6,000,000)	1,500,000	4,500,000
2	0.25 (4,500,000)	1,125,000	3,375,000
3	0.25 (3,375,000)	843,750	2,531,250
4	0.25 (2,531,250)	632,813	1,898,437
5	N1,898,437 – 250,000	1,648,437	0

COMPUTATION OF CAPITAL ALLOWANCE (MOTOR VEHICLES

Year		CAPITAL ALLOWANCE	BALANCE c/d
		N	N
1	0.2 (2,500,000)	500,000	2,000,000
2	0.2 (2,000,000)	400,000	1,600,000
3	0.2 (1,600,000)	320,000	1,280,000
4	1,280,000 – 100,000	1,180,000	0

COMPUTATION OF CAPITAL ALLOWANCE (FORKLIFT)

Year		ALLOWANCE	c/d
		N	N
1	0.2 (1,500,000)	300,000	1,200,000
2	0.2 (1,200,000)	240,000	960,000
3	0.3 (960,000)	192,000	768,000
4	0.2 (768,000)	153,600	614,400
5	614,400 – 60,000	554,400	0

COMPUTATION OF TOTAL CAPITAL ALLOWANCE

Year		N
1	1,500,000 + 500,000 + 300,000	2,300,000
2	1,125,000 + 400,000 + 240,000	1,765,000
3	843,750 + 320,000 + 192,000	1,355,750
4	632,813 + 1,180,000 + 153,600	1,966,413
5	1,648,837 + 554,400	2,203,237

COMPUTATION OF CASH REVENUE

Year		CASH REVENUE
		N
1		12,000,000
2	$10,000,000(1.12)^1$	11,200,000
3	$9,500,000\ (1.12)^2$	11,916,800
4	$9,000,000(1.12)^3$	12,644,352
5	$8,000.000\ (1.12)^4$	12,588,155

COMPUTATION OF CASH EXPENSES

Year		CASH EXPENSES (N)
1		8,500,000
2	7,500,000 (1.1)[1]	8,250,000
3	7,000,000 (1.1)[2]	8,470,000
4	6,000,000 (1.1)[3]	7,986,000
5	5,500,000 (1.1)[4]	8,052,550

COMPUTATION OF CASH PROFIT (CASH REVENUE − CASH EXPENSES)

Year	CASH REVENUE	CASH EXPENSES	CASH PROFIT
1	12,000,000	8,500,000	3,500,000
2	11,200,000	8,250,000	2,950,000
3	11,916,800	8,470,000	3,446,800
4	12,644,352	7,986,000	4,658,352
5	12,588,155	8,052,550	4,535,605

COMPUTATION OF TAX LIABILITY

Year	1 N	2 N	3 N	4 N	5 N
CASH PROFIT	3,500,000	2,950,000	3,446,800	4,658,352	4,535,605
Less: Capital Allowances	(2,300,000)	(1,765,000)	(1,355,750)	(1,966,413)	(2,203,237)
Taxable Profit	1200,000	1,185,000	2,091,050	2,691,939	2,332,368
Tax at 35%	(420,000)	(414,750)	(731,868)	(942,179)	(816,329)

COMPUTATION OF NET CASHFLOWS

Year	CASH COST/SCRAP VALUE N	TAX PROFIT N	NET LIABILITY N	CASH FLOW N
0	(10,000)			(10,000,000)
1				
2		3,500,000		3,500,000
3		2,950,000	(420,000)	2,530,000
		3,446,800	(414,750)	3,032,050
4	100,000	4,658,352	(731,868)	4,026,484
5	310,000	4,535,605	(942,179)	3,903,426
6			(816,329)	(816,329)

COMPUTATION OF DISCOUNTED CASH FLOW

Year	NET CASH FLOW N	DCF at 15%	PRESENT VALUE N
0	(10,000,000)	1.0000	(10,000,000)
1	3,500,000	0.8696	3,043,600
2	2,530,000	0.7561	1,912,933
3	3,032,050	0.6575	1,993,573
4	4,026,484	0.5718	2,302,344
5	3,903,426	0.4972	1,940,783
6	(816,329)	0.4323	(352,899)
			840,334

COMMENT

Since the Net Present Value of the project is positive (i.e. NPV = N840,334) the company should embark on the project.

SOLUTION 5

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE	Balance c/d
1	0.25 (1,000,000)	250,000	750,000
2	0.25 (750,000)	187,500	562,300
3	0.25 (562,500)	140,625	421,875
4	0.25 (421,875)	105,469	316,406
5	316,406 – 160,000	156,406	0

COMPUTATION OF TAX LIABILITY

Year	1	2	3	4	5
	N	N	N	N	N
CASH PROFIT	400,000	400,000	400,000	400,000	400,000
Less:					
Capital Allowances	(250,000)	(187,500)	(140,625)	(105,469)	(156,406)
Taxable Profit	150,000	212,500	259,375	294,531	243,594
Tax at 35%	(52,500)	(74,375)	(90,781)	(103,086)	(85,258)

COMPUTATION OF NET CASH FLOW

Year	COST/ SCRAP VALUE	CASH PROFIT	TAX LIABILITY	NET CASH FLOW
	N	N	N	N
0	(100,000,000)			1,000,000
1		400,000		400,000
2		400,000	(52,500)	347,500
3		400,000	(74,375)	325,625
4		400,000	(90,781)	309,219
5	160,000	400,000	(103,086)	296,914
6			(85,258)	(85,258)

COMPUTATION OF AFTER TAX COST OF CAPITAL (K after Tax)

K after tax = K before tax (I – Corporate Tax Rate)
∴ After tax Cost of Capital = 0.20 (1-0.35)
Note T = Corporate tax rate. = 0.13

COMPUTATION OF DISCOUNTED CASHFLOW

Year	NET CASHFLOW	DCF at 13%	PRESENT VALUE
	N		N
0	(1,000,000)	1.0000	(1,000,000)
1	400,000	0.8850	354,000
2	347,500	0.7831	272,127
3	325,625	0.6931	225,691
4	309,219	0.6133	189,644
5	296,914	0.5428	161,165
	(85,258)	0.4803	(40,949)
			161,678

COMMENT

Since the Net Present Value is positive 161,678, the company should purchase the project equipment.

SOLUTION 6

COMPUTATION OF CAPITAL ALLOWANCE

Year		Capital Allowance	Balance c/d
		N	N
1	0.2 (650,000)	130,000	520,000
2	0.2 (520,000)	104,000	416,000
3	0.2 (416,000)	83,200	332,000
4	0.2 (332,800)	66,560	265,440
5	0.2 (265,440)	53,088	212,352
6	212,352 – 100,000	112,352	0

COMPUTATION OF TAX LIABILITIES

Year	1	2	3	4	5	6
	N	N	N	N	N	N
CASH PROFIT	220,000	220,000	220,000	220,000	220,000	220,000
Less: Capital Allowances	(130,000)	(104,000)	(83,200)	(66,560)	(53,088)	(112,352)
Taxable Profit	90,000	116,000	136,800	153,440	166,912	107,648
Tax at 35%	(31,500)	(40,600)	(47,880)	(53,704)	(58,419)	(37,677)

COMPUTATION OF NET CASHFLOW

Year	COST/ SCRAP VALUE N	CASH PROFIT N	TAX LIABILITY N	NET CASHFLOW N
0	(650,000)			(650,000)
1		220,000		220,000
2		220,000	(31,500)	18,500
3		220,000	(40,600)	179,400
4		220,000	(47,880)	172,120
5		220,000	(53,704)	166,296
6	100,000	220,000	(58,419)	161,581
7			(37,677)	(37,677)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	Net Cashflow	DCF 14%	PRESENT VALUE
0	(650,000)	1.0000	(650,000)
1	220,000	0.8772	192,984
2	188,500	0.7695	145,051
3	179,400	0.6750	121,095
4	172,120	0.5921	101,912
5	166,296	0.5194	86,374
6	161,581	0.4556	73,616
7	(37,677)	0.3996	(15,1056)
			55,976

COMMENT

Since the Net Present value of the new equipment is positive, the company should purchase the new equipment irrespective of the method used to finance the purchase.

Computation of After Tax Cost of Borrowing

= Before tax cost of borrowing (1 – Corporate tax rate)

= 0.18 (1 – 0.35)

= 0.18 (0.65)

= 0.117 = 11.7%

Present Value of Lease Option

Year	Item	CASHFLOW N	DCF at 11.7%	PRESENT VALUE N
1-6	Lease rent	(140,000)	4.1466	(580,524)
2-7	Tax saved on Lease rent	49,000	3.7122	181,898
				(398,626)

WORKING

Tax saved on lease rent = 0.35 x 140,000

= N49,000

PRESENT VALUE OF BUY OPTION

Computation of tax saved on Capital Allowance

Year		Tax Saved N
1	0.35 (130,000)	45,500
2	0.35 (104,000)	36,900
3	0.35 (83,200)	29,120
4	0.35 (66,560)	23,296
5	0.35 (53,088)	18,581
6	0.35 (112,352)	39,323

Year	Cost Scrap Value N	Tax Saved on Capital Allowance N	Net Cashflow N	DCF= 11.7%	Pv
0	(650,000)		(650,000)	1.0000	(650,000)
2		45,500	45,500	0.8015	36,468
3		36,400	36,400	0.7175	26,117
4		29,120	29,120	0.6424	18,707
5		23,296	23,296	0.5751	13,398
6	100,000	18,581	118,581	0.5149	61,057
7		39,323	39,323	0.4669	18,124
					(476,129)

COMMENT

Since the lease option has the lower Present value of cost than the buy option. The best financing decision is to lease the equipment.

SOLUTION 7

COMPUTATION OF AFTER TAX COST OF BORROWING

= Before tax cost of Borrowing (1– tax rate)
= 0.12 (1 – 0.25)
= 0.12 (0.75)
= 0.09 = 9%

(i) **TAXABLE POSITION (25%)**

(a) Borrow and buy

Year	Items	CASHFLOW N	DCF 9%	PRESENT VALUE N
0	Cost	(9,200,000)	1.0000	(9,200,000)
2	Tax saved on Capital Allowance	2,250,000	0.8417	1,893,825
5	Scrap Value	900,000	0.6499	584,910
6	Balancing charge	(225,000)	0.5963	(134,168)
				(6,755,433)

Workings

(i) Tax saved on Capital Allowance = 0.25 x 9,000,000
= N2,250,000

(ii) Balance Charge = 0.25 x 900,000 = N225,000.

(b) **LEASE A**

Year		CASHFLOW N	DCF = 9%	PRESENT VALUE N
1-5	Lease Payment	(2,000,000)	3.8897	(7,779,400)
2-6	Tax saved on Lease Payment	500,000	3.5685	1,784,250
				(5,995,150)

WORKINGS

Tax saved on lease payment = 0.25 x 2,000,000 = N500,000

LEASE B

Year	LEASE PAYMENT N	TAX SAVING N	NET CASHFLOW N	DCF 9%	PRESENT VALUE N
1	(500,000)		(500,000)	0.9173	(458,650)
2	(800,000)	125,000	(675,000)	0.8417	(568,148)
3	(1,600,000)	200,000	(1,400,000)	0.7722	(1,081,080)
4	(3,000,000)	400,000	(2,600,000)	0.7084	(1,841,840)
5	(5,000,000)	750,000	(4,250,000)	0.6499	(2,762,075)
6		750,000	750,000	0.5963	447,225
					(6,264,568)

WORKINGS

Tax saved on lease payment

Year			
2	0.25 (500,000)	=	125,000
3	0.25 (800,000)	=	200,000
4	0.25(1,600,000)	=	400,000
5	0.25 (3,000,000)	=	750,000
6	0.25 (3,000,000)	=	750,000

COMMENT

Where the firm is subject to tax at 25%, it is better to lease the asset using Option A as the option has the lowest Present value of cost

(ii) **NON TAXABLE POSITION**

Borrow and buy

Year	Items	Cashflow N	DCF of 12%	PRESENT VALUE N
0	Cost	(9,200,000)	1.0000	(9,200,000)
5	Salvage Value	900,000	0.5674	510,660
				(8,689,340)

Lease A

Year	Items	Cashflow N	DCF of 12%	PRESENT VALUE N
1-5	Lease Payment	(2,000,000)	3.6048	(7,209,552)

LEASE B

Year	Lease Payment N	DCF at 12%	PRESENT VALUE N
1	(500,000)	0.8929	(446,450)
2	(800,000)	0.7972	(637,760)
3	(1,600,000)	0.7118	(1,138,880)
4	(3,000,000)	0.6355	(1,906,500)
5	(5,000,000)	0.5674	(2,837,000)
			(6,966,590)

Comment

Where the firm is permanently in a non-taxable position, leasing under Option B is the most attractive option because it has the lowest Present value of financing cost.

SOLUTION 8

COMPUTATION OF SALES VALUE

Year		N
1	1,400,000 (N10)	14,000,000
2	1,400,000 (N10) $(1.15)^1$	16,100,000
3	1,400,000 (N10) $(1.15)^2$	18,515,000
4	1,400,000 (N10) $(1.15)^3$	21,292,250
5	1,400,000 (N10) $(1.15)^4$	25,486,088

COMPUTATION OF MATERIAL COST

Year		N
1	1,400,000 (N3)	4,200,000
2	1,400,000 (N3) (1.1)	4,620,000
3	1,400,000 (N3) $(1.1)^2$	5,082,000
4	1,400,000 (N3) $(1.1)^3$	5,590,200
5	1,400,000 (N3) $(1.1)^4$	6,149,220

COMPUTATION OF LABOUR COST

Year		N
1	1,400,000 (N4)	5,600,000
2	1,400,000 (N4) (1.12)	6,272,000
3	1,400,000 (N4) $(1.12)^2$	7,024,640
4	1,400,000 (N4) $(1.12)^3$	7,867,597
5	1,400,000 (N4) $(1.12)^4$	8,811,708

COMPUTATION OF OVERHEAD COST

Year		N
1	1,400,000 (N2)	2,800,000
2	1,400,000 (N2) $(1.08)^1$	3,024,000
3	1,400,000 (N2) $(1.08)^2$	3,265,920
4	1,400,000 (N2) $(1.08)^3$	3,527,194
5	1,400,000 (N2) $(1.08)^4$	3,809,369

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE N	BALANCE c/d N
1	0.25 (7,600,000)	1,900,000	5,700,000
2	0.25 (5,700,000)	1,425,000	4,275,000
3	0.25 (4,275,000)	1,068,750	3,206,250
4	0.25 (3,206,250)	801,563	2,404,687
5	2,404,687 – 700,000	1,704,687	0

COMPUTATION OF CASH PROFIT

Year	SALES N	MATERIAL COST N	LABOUR COST N	OVERHEAD COST N	CASH PROFIT N
1	14,000,000	(4,200,000)	(5,600,000)	(2,800,000)	1,400,000
2	16,100,000	(4,620,000)	(6,272,000)	(3,024,000)	2,184,000
3	18,515,000	(5,082,000)	(7,024,640)	(3,265,920)	3,142,440
4	21,292,250	(5,590,200)	(7,867,597)	(3,527,194)	4,307,259
5	29,486,088	(6,149,220)	(8,811,708)	(3,809,369)	5,715,791

COMPUTATION OF TAX LIABILITY

Year	1 N	2 N	3 N	4 N	5 N
CASH PROFIT	1,400,000	2,184,000	3,142,440	4,307,259	5,715,791
Less: Capital Allowance	(1,900,000)	(1,425,000)	(1,068,750)	(801,563)	(1,704,687)
Taxable Profit	(500,000)	759,000	2,073,690	3,305,696	4,011,104
Tax at 35%	175,000	(265,650)	(725,792)	(1,226,994)	(1,403,886)

COMPUTATION OF NET CASHFLOW

Year	Cost/Scrap Value N	CASH PROFIT N	TAX LIABILITY N	NET CASHFLOW N
0	(7,600,000)			(7,600,000)
1		1,400,000		1,400,000
2		2,184,000	175,000	2,359,000
3		3,142,440	(265,650)	2,876,790
4		4,307,259	(725,792)	3,581,467
5	700,000	5,715,791	(1,226,994)	5,188,797
			(1,403,886)	(1,403,886)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	NET CASHFLOW (N)	DCF at 18%	PRESENT VALUE (N)
0	(7,600,000)	1.0000	(7,600,000)
1	1,400,000	0.8475	1,186,500
2	2,359,000	0.7182	1,694,234
3	2,876,790	0.6086	1,750,814
4	3,581,467	0.5158	1,847,321
5	5,188,797	0.4371	2,268,023
6	(1,403,886)	0.3704	(519,999)
			626,893

Since the Net Present value of the project is positive, Odogwu Paul Limited would embark on the development of a new product.

(b) Computation of After tax Cost of Borrowing

= Before tax cost of borrowing
= 0.21(1-0.35)
= 0.21 (0.65)
= 0.1365 = 13.65%

PRESENT VALUE OF LEASE OPTION

Year	ITEM N	CASHFLOW N	DCF at 13.65%	PRESENT VALUE N
1-5	Lease Payment	(2,300,000)	3.4622	(7,963,0600)
2-6	Tax saved on Lease payment	805,000	3.0463	2,452,272
				5,510,788

Working
Tax saved on Lease payment = 0.35 x 2,300,000 = N805,000

PRESENT VALUE OF BUY

Year	Cost/ Scrap Value N	TAX SAVED ON CAPITAL ALLOWANCE N	NET CASH FLOW N	DCFat 13.65%	PRESENT VALUE N
0	(7,600,000)		(7,600,000)	1.0000	(7,600,000)
2		665,000	665,000	0.7742	514,843
3		498,750	498,750	0.6812	339,749
4		374,063	374,063	0.5994	224,213
5	700,000	280,547	980,547	0.5274	517,140
6		596,640	596,640	0.4641	276,901
					(5,727,154)

Working

Tax Saved on Capital Allowance

Year		N
1	0.35 x 1,900,000	665,000
2	0.35 x 1,425,000	498,750
3	0.35 x 1,068,750	374,063
4	0.35 x 801,563	280,547
5	0.35 x 1,704.687	596,640

Comment
Since the Lease Option has the lower Present value cost than the Buy option, then the company should lease the machine.

SOLUTION 9

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE N	Balance c/d N
1	0.35 (10,000,000)	3,500,000	6,500,000
2	0.35 (6,500,000)	2,275,000	4,225,000
3	0.35 (4,225,000)	1,478,750	2,746,250
4	0.35 (2,746,250)	961,188	1,785,069
5	1,785,069 – 500,000	1,285,069	0

COMPUTATION OF CASH PROFIT

Year	CASH REVENUE N	CASH EXPENSES N	CASH PROFIT N
1	7,600,000	(3,600,000)	4,000,000)
2	8,600,000	(4,100,000)	4,500,000
3	8,800,000	4,300,000	4,000,000
4	9,200,000	(5,600,000)	2,600,000
5	9,400,000	(6,900,000)	2,500,000

COMPUTATION OF TAX LIABILITY

	1 N	2 N	3 N	4 N	5 N
CASH PROFIT	4,000,000	4,500,000	4,000,000	2,600,000	2,400,000
LESS: CAPITAL ALLOWANCE	(3,500,000)	(2,275,000)	(1,478,750)	(961,188)	(1,285,069)
TAXABLE PROFIT	500,000	2,225,000	2,521,250	1,638,812	1,214,931
Tax at 35%	(175,000)	(778,750)	(882,438)	(573,584)	(425,226)

COMPUTATION OF AFTER TAX COST OF CAPITAL

= COST OF CAPITAL BEFORE TAX (1 – tax rate)
= 0.20 (1 – 0.35)
= 0.20 (0.65)
= 0.13 = 13%

COMPUTATION OF NET CASHFLOW

Year	Cost/Scrap Value (10,000,000)	CASH PROFIT	TAX LIABILITY	NET CASH FLOW
0				(10,000,000)
1		4,000,000		4,000,000
2		4,500,000	(175,000)	4,325,000
3		4,000,000	(778,750)	3,221,250
4		2,600,00	(882,438)	1,717,562
5	500,000	2,500,000	(573,584)	2,426,416
6			(425,226)	(425,226)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	NET CASH FLOW N	DCF at 13%	PRESENT VALUE N
0	(10,000,000)	1.0000	(10,000,000)
1	4,000,000	0.8850	3,540,000
2	4,325,000	0.7831	3,386,908
3	3,221,250	0.6931	2,232,648
4	1,717,562	0.6133	1,053,381
5	2,426,416	0.5428	1,317,059
6	(425,226)	0.4803	(204,236)
			1,325,760

COMPUTATION OF AFTER TAX COST OF BORROWING

= Before tax cost of borrowing (1 – tax rate)
= 0.21 (1 – 0.35)
= 0.21 (0.65)
= 0.1365 = 13.65%

PRESENT VALUE OF LEASE OPTION

Year	Item	Cashflow N	DCFat 13.65%	PRESENT VALUE N
1 - 5	Lease Payment	(3,200,000)	3,4622	(11,079,040)
2 - 6	Tax Saved on Lease Payment	1,120,000	3.0463	3,411,856
				(7,667,184)

PRESENT VALUE OF BUY OPTION

Year	Cost/Scrap Value N	TAX SAVED ON CAPITAL ALLOWANCE N	NET CASH FLOW N	DCF at 13.65%	PRESENT VALUE N
0	(10,000,000)		(10,000,000)	1.0000	(10,000,000)
2		1,225,000	1,225,000	0.7742	948,395
3		796,250	796,250	0.6812	542,406
4		517,563	517,563	0.5994	310,227
5	500,000	336,416	836,416	0.5274	441,126
6		449,774	449,774	0.4641	208,740
					(7,549,106)

WORKINGS

Tax Saved on Capital Allowance

Year			N
2	0.35 (3,500,000)	=	1,225,000
3	0.35 (2,275,000)	=	796,250
4	0.35 (1,478,750)	=	517,563
5	0.35 (961,188)	=	336,416
6	0.35 (1,285,069)	=	449,744

COMMENT

Since the buy option has the lower Present value cost than the lease option, Roasted Chicken Limited management should buy the machine.

SOLUTION 10

COMPUTATION OF SALES VALUE

Year		N
1	N60 (5,000,000)	300,000,000
2	N60 (6,500,000)	390,000,000
3	N65 (7,200,000)	468,000,000
4	N65 (7,500,000)	487,500,000
5	N65 (8,000,000)	520,000,000

COMPUTATION OF VARIABLE COST

Year		N
1	N55 (5,000,000)	275,000,000
2	N55 (6,500,000)	357,500,000
3	N55 (7,200,000)	396,000,000
4	N55 (7,500,000)	412,500,000
5	N55 (8,000,000)	440,000,000

COMPUTATION OF CASH PROFIT

Year	SALES VALUE N	VARIABLE COST N	CASH OPERATING EXPENSES N	CASH PROFIT N
1	300,000,000	(275,000,000)	(24,000,000)	1,000,000
2	390,000,000	(357,500,000)	(24,000,000)	8,500,000
3	468,000,000	(396,000,000)	(24,000,000)	48,000,000
4	487,500,000	(412,500,000)	(24,000,000)	51,000,000
5	520,000,000	(440,000,000)	(24,000,000)	56,000,000

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE N	BALANCE c/d N
1	0.35 (40,000,000)	14,000,000	26,000,000
2	0.35 (26,000,000)	9,100,000	16,900,000
3	0.35 (16,900,000)	5,915,000	10,985,000
4	0.35(10,985,000)	3,844,750	7,140,250
5	7,140,250 – 950,000	6,190,250	0

COMPUTATION OF TAX LIABILITY

Year	1 N	2 N	3 N	4 N	5 N
Cash Profit	1,000,000	8,500,000	48,000,000	51,000,000	56,000,000
Less: Capital Allowance	(14,000,000)	(9,100,000)	(5,915,000)	(3,844,750)	(6,190,250)
Taxable Profit	(13,000,000)	(600,000)	42,085,000	47,155,250	49,809,750
Tax at 35%	4,550,000	210,000	(14,729,750)	(16,504,338)	(17,433,413)

COMPUTATION OF NET CASHFLOW

Year	Cost/Scrap Value N	CASH PROFIT N	TAX LIABILITY N	NET CASHFLOW N
0	(40,000,000)			(40,000,000)
1		1,000,000		1,000,000
2		8,500,000	4,550,000	13,050,000
3		48,000,000	210,000	48,210,000
4		51,000,000	(14,729,750)	36,270,250
5	950,000	56,000,000	(16,504,338)	40,445,662
6			(17,433,413)	(17,433,413)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	NET CASHFLOW N	DCF at 20%	PRESENT VALUE N
0	(40,000,000)	1.0000	(40,000,000)
1	1,000,000	0.8333	833,000
2	13,050,000	0.6944	9,061,920
3	48,210,000	0.5787	27,889,485
4	36,270,250	0.4823	17,493,142
5	40,445,662	0.1019	16,255,112
6	(17,433,413)	0.3349	(5,838,450)
			25,694,209

COMMENT

Since the Net Present Value of the machine is positive, the company should acquire the machine.

COMPUTATION OF AFTER TAX COST OF BORROWING

= Before tax cost of borrowing (1 – tax rate)
= 0.21 (1 – 0.35)
= 0.21 (0.65)
= 0.1365 = 13.65%

PRESENT VALUE OF LEASE OPTION

Year	Item	Cashflow N	DCF at 13.65%	PRESENTVALUE N
1-5	Lease Payment	(10,500,000)	3.4622	(36,353,100)
2-6	Tax Saved on Lease Payment	3,675,000	3.0463	11,195,153
				(25,157,967)

WORKING

Tax Saved on Lease Payment = 0.35 x 10,500,000 = N3,675,000.

PRESENT VALUE OF BUY OPTION

Year	Cost/Scrap Value N	TAX SAVED ON CAPITAL ALLOWANCE N	NET CASH FLOW N	DCF at 13.65%	PRESENT VALUE N
0	(40,000,000)		(40,000,000)	1.0000	(40,000,000)
2		4,900,00	4,900,000	0.7742	3,793,580
3		3,185,000	3,185,000	0.6842	2,179,177
4		2,070,250	2,070,250	0.5994	1,240,908
5	950,000	1,345,663	2,295,663	0.5274	1,210,733
6		2,166,588	2,166,588	0.4641	1,005,513
					(30,570,089)

WORKINGS

Computation of Tax Saved on Capital Allowance

Year			N
2	0.35 x 14,000,000	=	4,900,000
3	0.35 x 9,100,000	=	3,185,000
4	0.35 x 5,915,000	=	2,070,250
5	0.35 x 3,844,750	=	1,345,663
6	0.35 x 6,190,250	=	2,166,588

Comment

Since the Present Value of cost of lease option is lower than that of buy option, Oyewale Block Industry should lease the machine from Softy Leasing Company.

SOLUTION 11

COMPUTATION OF ANNUAL SALES VALUE

Year		N
1	N20 (5,000,000)	100,000,000
2	N20 (6,000,000)	120,000,000
3	N22.5 (7,500,000)	168,750,000
4	N22.5 (8,000,000)	180,000,000
5	N22.5 (9,000,000)	202,500,000

COMPUTATION OF ANNUAL VARIABLE COST

Year		N
1	N18.5 (5,000,000)	92,500,000
2	N18.5 (6,000,000)	111,000,000
3	N18.5 (7,500,000)	138,750,000
4	N18.5 (8,000,000)	148,000,000
5	N18.5 (9,000,000)	166,500,000

COMPUTATION OF CASH PROFIT

Year	SALES VALUE N	VARIABLE COST N	CASH EXPENSES N	CASH PROFIT N
1	100,000,000	(92,500,000)	(6,000,000)	1,500,000
2	120,000,000	(111,000,000)	(6,000,000)	3,000,000
3	168,750,000	(138,750,000)	(6,000,000)	24,000,000
4	180,000,000	(148,000,000)	(6,000,000)	26,000,000
5	202,500,000	(166,500,000)	(6,000,000)	30,000,000

COMPUTATION OF TOTAL COST

	N
Equipment	10,600,000
Forklift	2,600,000
Tools	1,800,000
	15,000,000

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE N	Balance c/d N
1	0.25 (15,000,000)	3,750,000	11,250,000
2	0.25 (11,250,000)	2,812,000	8,437,500
3	0.25 (8,437,500)	2,109,375	6,328,125
4	0.25 (6,328,125)	1,582,031	4,746,094
5	4,746,094 – 800,000	3,946,094	0

COMPUTATION OF TAX LIABILITY

Year	1 N	2 N	3 N	4 N	5 N
CASH PROFIT	1,500,000	3,000,000	24,000,000	26,000,000	30,000,000
Less: Capital Allowance	(3,750,000)	(2,812,500)	(2,109,375)	(1,582,031)	(3,946,094)
Taxable Profit	(2,250,000)	187,500	21,890,625	24,417,969	26,053,906
Tax at 35%	787,500	(65,625)	(7,661,719)	(8,546,289)	(9,118,867)

COMPUTATION OF COMPANY COST OF CAPITAL

= $R_F + [E(R_m) - R_F]$ Bi
= 0.10 = [0.19 – 0.10] 2-0
= 0.10 + (0.09) 2.0
= 0.10 + 0.18
= 0.28 = 28%

COMPUTATION OF AFTER TAX COST OF CAPITAL

= Company Cost of Capital (1 – tax rate)
= 0.28 (1 – 0.35)
= 0.28 (0.65)
= 0.182 = 18.2%

COMPUTATION OF NET CASHFLOW

Year	Cost/Scrap Value N	CASH PROFIT N	TAX LIABILITY N	NET CASHFLOW N
0	(15,000,000)			(15,000,000)
1		1,500,000		1,500,000
2		3,000,000	787,500	3,787,500
3		24,000,000	(65,625)	23,934,375
4		26,000,000	(7,661,719)	18,338,281
5	800,000	30,000,000	(8,546,289)	22,253,711
			(9,118,867)	(9,118,867)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	Net Cashflow N	DCF at 18.2%	PRESENT VALUE N
0	(15,000,000)	1.0000	(15,000,000)
1	1,500,000	0.8460	1,269,000
2	3,787,500	0.7158	2,711,093
3	23,934,375	0.6055	14,492,264
4	18,338,281	0.5123	9,394,701
5	22,253,711	0.4334	9,644,758
6	(9,118,867)	0.3367	(3,343,889)
			19,167,927

COMMENT

Since the Net Present Value of the project is positive, Lagoon State of Nigeria should purchase or acquire the equipment.

COMPUTATION OF AFTER TAX COST OF BORROWING

= Before tax Cost of Borrowing (1 – tax rate)
= 0.20 (1 – 0.35)
= 0.20 (0.65)
= 0.13 = 13%

PRESENT VALUE OF LEASE OPTION

Year	Item	Cashflow N	DCF at 13%	PRESENT VALUE N
1-5	Lease payment	(4,200,000)	3.5172	(14,772,240)
2-6	Tax Saved on Lease Payment	1,470,000	3.1125	4,575,375
				(10,196,865)

Working

Tax based on lease payment = 0.35 x 4,200,000 = 1,470,000

PRESENT VALUE OF BUY OPTION

Year	Cost/ Scrap Value N	TAX SAVED ON CAPITAL ALLOWANCE N	NET CASH FLOW N	DCF at 13% N	PRESENT VALUE
0	(15,000,000)		(15,000,000)	1.0000	(15,000,000)
2		1,312,500	1,312,500	0.7831	1,027,819
3		984,375	984,375	0.6931	682,270
4		738,281	738,281	0.6133	452,788
5	800,000	553,711	1,353,711	0.5428	734,794
6		1,381,1331	1,381,133	0.4803	663.358
					(11,438,971)

Workings

Tax Saved on Capital Allowance

Year		N
2	0.35 x 3,750,000	1,312,500
3	0.35 x 2,812,500	984,375
4	0.35 x 2,109,375	738,281
5	0.35 x 1,582,031	553,711
6	0.35 x 3,946,094	1,381,133

COMMENT

Since the Present value of cost of lease option is lower than that of Buy option, Lagoon State of Nigeria should lease the equipment.

SOLUTION 12

COMPUTATION OF CAPITAL ALLOWANCE

Year		CAPITAL ALLOWANCE N	Balance c/d N
1	0.25 (650,000)	162,500	487,500
2	0.25 (487,500)	121,875	365,625
3	025 (365,625)	91,406	274,219
4	0.25(274,219)	68,555	205,664
5	0.25 (205,664)	51,416	154,248
6	154,248 – 100,000	54,248	0

COMPUTATION OF TAX LIABILITY

Year	1	2	3	4	5	6
	N	N	N	N	N	N
CASH PROFIT	220,000	220,000	220,000	220,000	220,000	220,000
LESS: CAPITAL	(162,500)	(121,875)	(91,406)	(68,555)	(51,416)	(54,248)
TAXABLE PROFIT	57,500	98,125	128,594	151,44t	168,484	165,752
TAX at 35%	20,125	34,344	45,008	53,006	59,004	58,013

COMPUTATION OF NET CASHFLOW

Year	Cost/Scrap Value	CASH PROFIT	TAX LIABILITY	NET CASH FLOW
	N	N	N	N
0	(650,000)			(650,000)
1		220,000		220,000
2		220,000	(20,125)	199,875
3		220,000	(34,344)	185,656
4		220,000	(45,008)	174,992
5		220,000	(53,006)	166,994
6	100,000	220,000	(59,004)	260,996
7			(58,013)	(58,013)

COMPUTATION OF DISCOUNTED CASHFLOW

Year	NET CASHFLOW	DCF at 14%	PRESENT VALUE
	N		N
0	(650,000)	1.0000	(650,000)
1	220,000	0.8772	192,984
2	199,875	0.7695	153,804
3	185,656	0.6750	125,318
4	174,992	0.5921	103,613
5	166,994	0.5194	86,737
6	260,996	0.4556	118,910
7	(58,013)	0.3996	(23,182)
			108,184

COMMENT

Since the Net Present Value is positive, the company should acquire the new equipment.

COMPUTATION OF AFTER TAX COST OF BORROWING

= Before tax Cost of borrowing (1 – tax rate)
= 0.18 (1 – 0.35)
= 0.18 (0.65)
= 0.117 = 11.7%

PRESENT VALUE OF BUY OPTION

Year	Cost/ Scrap Value	TAX SAVED ON CAPITAL ALLOWANCE	NET CASH FLOW	DCFat 11/7%	PRESENT VALUE
	N	N	N		N
0	(650,000)		(650,000)	1.0000	(650,000)
2		(56,875)	56,875	0.8015	45,585
3		42,656	42,656	0.7175	30,606
4		31,992	31,992	0.6424	20,552
5		23,994	23,994	0.5751	13,799
6	100,000	17,996	17,996	0.5149	60,756
7		18,987	18,987	0.4609	8,751
					(469,951)

Workings

Tax Saved on Capital Allowance

Year		N
2	0.35 (162,500)	56,875
3	0.35 (121,875)	42,656
4	0.35 (91,406)	31,992
5	0.35 (68,555)	23,994
6	0.35 (51,416)	17,996
7	0.35 (54,248)	18,987

PRESENT VALUE OF LEASE OPTION

Year	Item	Cashflow	DCF at	PRESENT VALUE
		N	11.7%	N
1-6	Lease Payment	(140,000)	4.1466	(580,524)
2-7	Tax Saved on Lease Payment	49,000	3.7122	181,898
				(398,626)

NWORKING

Tax Saved on lease payment = 0.35 x 140,000 = N49.000

PRESENT VALUE OF HIRE PURCHASE OPTION

Year	Item	CASHFLOW N	DCF at 11.7%	PRESENT VALUE N
0	Hire Purchase Deposit	(210,000)	1.0000	(210,000)
1-6	Hire Purchase Payment	(123,000)	4.1466	(510,032)
6	Scrap value	100,000	0.5149	51,490
2	Total tax Saved on Capital Allowance and Interest	81,375	0.8015	65,222
3		63,656	0.7175	45,673
4		48,092	0.6424	30,894
5		34,494	0.5751	19,837
6		25,346	0.5149	13,051
7		24,237	0.4609	11.171
				(482,694)

WORKINGS

Computation of Total tax saved on Capital Allowance and Interest

CAPITAL ALLOWANCE + INTEREST

Year		N
2	0.35 (162,500 + 70,000)	81,375
3	0.35 (121,875 + 60,000)	63,656
4	0.35 (91,406 + 46,000)	48,092
5	0.35 (68,555 + 30,000)	34,494
6	0.35 (51,416 + 21,000)	25,346
7	0.35 (54,248 + 15,000)	24,237

COMMENT

Since the Present Value of cost of lease option is the lowest of all the option, Biodun Oyekanmi Limited should lease the new equipment.

SOLUTION 13

COMPUTATION OF CAPITAL ALLOWANCES

Year		Amount N	Balance c/d N
1	0.05 x 9,450,000	472,500	8,977,500
2	0.05 x 8,977,500	448,875	8,528,625
3	0.05 x 8528,625	426,431	8,102,194
4	0.05 x 8102,194	405,110	7,697,084
5	7,697,084 – 697,084	7,000,000	

COMPUTATION OF TAX SAVED ON CAPITAL ALLOWANCES

Year			N
1	0.35 x 472,500	=	165,375
2	0.35 x 448875	=	157,106
3	0.35 x 426431	=	149,251
4	0.35 x 405,110	=	141,789
5	0.35 x 7,000,000	=	2,450,000

COMPUTATION OF AFTER TAX COST OF BORROWING (K after Tax)

K after Tax = Cost of borrowing before tax (1 - tax rate)
= 0.12 (1 - 0.35)
= 0.078 = 7.8%

BUY OPTION

Year	Cost/Scrap Value N	Tax saved on Capital Allowance N	Net cash Flow	DCFct 7.8%	Present Value N
0	(9450,000)		(9450,000)	1.0000	(9450,000)
2		165375	165375	0.8605	142,305
3		157106	157106	0.7983	125,418
4		149251	149251	0.7405	110,520
5	(9,450,000)	141789	141789	0.6869	576,222
6		2,450,000	2,450,000	0.6372	1,561,140
					(6,934,395)

LEASE OPTION

Year	Item	Cashflow N	Dcf At 7.8%	Present Value N
1 - 5	Lease Payment 220,000	(3,000,000)	4.0138	(12,041,400)
2 - 6	Tax Saved on lease payment	1,050,000	3.7235	3,909,675
				8,131,725

COMMENT

Since the BUY OPTION has the lower present value of cost than the lease option, then the company should buy the equipment.

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CHAPTER 11

PORTFOLIO THEORY

Question 1
Define or explain the following terms
(I) Portfolio
(Ii) Portfolio theory
(Iii) Portfolio return
(Iv) Portfolio risk
(V) Optimal portfolio

Question 2
The securities of company Y and Z have the following expected returns and standard deviations:

	Y	Z
Expected return (%)	20	25
Standard deviation (%)	30	40

Calculate the expected returns for the following portfolio:
(I) 100% Y
(Ii) 70% Y + 30% Z
(Iii) 50% Y + 50% Z
(Iv) 30% Y + 70% Z
(V) 100%Z

Question 3
The securities of companies X and Y have the following expected returns and standards deviation:

	X	Y
Expected return(%)	20	15
Standard deviation (%)	35	24

Assume that the correlation coefficient between the returns of the two securities is 0.4

Required:
(A) Calculate the expected returns and standard for the following portfolio
(I) 100% X
(Ii) 75% X + 25% Y
(Iii) 50% X 50% Y
(Iv) 25% X 75% Y
(V) 100% Y

(B) Which of the portfolio in (a) above is optimal?

Question 4
Make a clear distinction between
(I) Optimal portfolio diversification
(Ii) Positively correlated investment and negatively correlated investment.

Question 5
A risky portfolio has an expected return of 0.33 and a standard deviation of 0.4. The return on the treasury bill is 12%. Your portfolio has 40% investment in the treasury bill and the balance of the fund in the risky portfolio. What is the expected return and standard deviation of your portfolio.

Question 6
Over three years, security X has returns of 10%, 14% and 3%. For the same periods, security Y has returns of 12%, 10% and 5%. What is the variance and standard deviation of returns for these two securities? What is the covariance of returns between these securities. What is the correlation of returns? Is it attractive to have these two securities in a portfolio? Why.

Question 7
The expected return on securities A, B and C are 24%, 22% and 18% respectively. The variance - covariance matrix of the three securities are shown below:

	A	B	C
A	0.4	0.1	0.1
B	0.1	0.1	0.05
C	0.1	0.05	0.12

Hint: The numbers along the diagonal are the variance values while the off-diagonal numbers are the covariance values.

You are required to:

(A) Calculate the expected return, variance and standard deviation of equal-weighted two security portfolio selected from securities A, B, C.

(B) Which of these portfolio are inefficient? Why?

(C) Which of the portfolio would you recommend for a risk-averse investor and why?

Question 8

Given the following information on Securities X and Y

SECURITY X		SECURITY Y	
Returns	Probabilities	Returns	Probabilities
16%	0.25	25%	0.30
28%	0.35	26%	0.35
35%	0.40	27%	0.35

Calculate the return and standard deviation of the following portfolios (a) 55% in X and 45% in Y, and (b) 50% in X and 50% in Y, assuming the correlation between the securities are 1, 0 and -1 respectively.

Question 9

Aisha is considering investment in the following securities whose random returns are given as follows:

SECURITY A		SECURITY B		SECURITY C	
Rates of Returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
39	0.5	65	0.4	58	0.2
45	0.4	44	0.1	49	0.7
47	0.1	21	0.5	40	0.1

The pairwise correlation between the returns of the securities are

$\overline{AB} = 0.64$, $\overline{AC} = 0.46$ $\overline{BC} = 0.42$

Required:

Calculate the means and standard deviation of:

(I) 0.6A + 0.6B

(Ii) 0.7A + 0.3C

(Iii) 0.8B + 0.2C

Question 10

Ismaila is considering investing has N5million in the following securities random returns are given as follows:

SECURITY A		SECURITY B		SECURITY C	
Rates of Returns %	Probability	Rates of Returns %	Probability	Rates of Returns %	Probability
38	0.4	64	0.4	57	0.2
44	0.5	43	0.5	48	0.1
46	0.1	26	0.1	41	0.7

The pairwise correlation between the returns of the securities are $\overline{AC} = 0.5$, $\overline{BC} = 0.3$, $\overline{AB} = 0.4$

(A) Calculate the means and variances of the following portfolios:

(I) P1 = 0.4A + 0.6B

(Ii) P2 = 0.75A + 0.25C

(Iii) P3 = 0.5B + 0.5C

(B) If the risk free rate is 12.65%, expected return on the market is 18% and variance o the return on the market is 2.26%
Calcualte the beta of P1.

Question 11

Mr. Ayinde is considering building up a portfolio consisting of Securities A and B. The random returns of the two securities are as follows:

SECURITY A

Rates of Returns %	Probability of Occurrence
24	0.3
25	0.2
30	0.3
22	0.2

SECURITY B

Rates of Returns%	Probability of Occurrence
38	0.4
45	0.2
28	0.3
18	0.1

Required:
Calculate the means and standard deviation of a portfolio consisting of:
(I) 80% of Security A and 20% of Security B with a correlation between the two securities of 0.2, 0.4, and 0.6.
(Ii) 35% of Security A and 65% of Security B with perfect positive, perfect negative and zero correlation

Question 12
Ade is considering investment in the following securities with random returns as follows:

SECURITY A		SECURITY B		SECURITY C	
Rate of Returns %	Probability of Occurrence	Rate of Returns%	Probability of Occurrence	Rate of Returns %	Probability of Occurrence
26	0.3	16	0.4	30	0.1
32	0.3	20	0.1	45	0.2
35	0.2	36	0.3	50	0.5
40	0.2	45	0.2	60	0.2

The treasury bill rate is currently 12%

Required:
(a) Calculate the means and standard deviation of the following portfolio:
(i) P = $\frac{1}{2}A + \frac{1}{2}B$ with a correlation coefficient of 0.6.
(ii) M = $\frac{1}{3}B + \frac{2}{3}C$ with a correlation coefficient of -0.4
(iii) Q = $0.6A + 0.4C$ with a correlation coefficient of 0.8.
(iv) R = $\frac{7}{8}A + \frac{1}{8}B$ with a correlation, how will it affect the result in (a)

(b) Assuming securities A and B have zero correlation, how will it affect the result.

PORTFOLIO THEORY

SOLUTION 1
(I) Portfolio can be defined as the basket of investment. It can also be defined as a collection of various investment that make up an investor's total investment. For example, Mr. Kuye invested in UBA Plc, Royal Exchange Insurance Plc, Cadbury Nigeria Plc and Nigeria Brewery Plc. The above mentioned investment constitute Mr. Kuye investment portfolio. It must be noted that a portfolio must consist of at least two different investments or securities.

(Ii) **Portfolio theory:** This can be defined as the way and manner in which a portfolio is selected with the aim of reducing, eliminating, decreasing or minimising the risk in the given portfolio.

(Iii) **Portfolio return:** This can be defined as the weighted average return of all securities in a given portfolio, where the weight represent the proportion of each security in a given portfolio. Portfolio return of a given portfolio can be estimated using the formula stated below.

$$E(Rp) = \sum_{t=1}^{n} W_t \; E(Rt)$$

Where E(Rp) = Expected return of a portfolio
Wt = Weight or proportion of total funds invested in security t.
E(Rt) = Expected return of individual security or security t.

n = Number of security in the portfolio.

For a portfolio that consist of two securities X and Y, the expected return of the portfolio can be expressed as:

$$E(Rp) = W_x \; E(Rx) + W_y \; (E(Ry)$$

Where

E(Rp) = Expected return of a portfolio
Wx = Weight or proportion of total funds invested in Security X
Wy = Weight or proportion of total funds invested in Security Y
E(Rx) = Expected return of Security X
E(Ry) = Expected return of Security Y
Wx + Wy = 1

(Iv) Portfolio risk: This can be defined as the relationship between the securities that make up a portfolio. It can also be defined as the risk associated or involved in a portfolio. It must be noted the portfolio risk depends not only on the riskiness of the securities making up a portfolio but also the relationship among those securities. Portfolio risk is measured by determining the standard deviation of a portfolio and can be expressed as stated below:

$$\sigma P = \sqrt{\sum_{i=1}^{n} \sum_{j=j}^{n} W_i W_j \sigma_{ij}}$$

Where σp = Standard deviation of a portfolio.

Wi = Weighted or proportion of total funds invested in security i.
Wj = Weighted or proportion of total funds invested in security j.
σij = Covariance between possible returns for securities i and j.

For a portfolio that contain or consist of two securities X and Y, the portfolio risk is expressed as stated below:

$$\sigma P = \sqrt{W^2 \sigma_x^2 + W_y^2 \sigma_y^2 + 2 W_x W_y \sigma_x \sigma_y \sqrt{xy}}$$

Where:

W_x^2 = Weight or proportion of total funds invested in security X squared

W_y^2 = Weight or proportion of total funds invested in security y squared.

σ_x^2 = Variance of return on security X.

σ_y^2 = Variance of return on security y.

σx = Standard deviation of return on security x.

= Standard deviation of return on security y.

σy = Correlation coefficient between possible returns for security A and B and its values lies between Y and +1.

$\sqrt{xy}$

(Iv) **Optimal Portfolio:-** Is a portfolio of securities in which for a given risk has the highest expected return and for a given expected return has the lowest risk.

SOLUTION 2

Rp = Wy Ry + Wz Rz

I. 100% Y + 0% Z = (1)(0.2) + (0)(0.25) = 0.2 + 0 = 0.2
Ii. 70% Y + 30% Z + (0.7) (0.2) + (0.3) (0.25) = 0.14 + 0.075 + 0.15
Iii. 50% Y + 50% Z = (0.5) (0.2) + (0.5) (0.25) = 0.10 + 0.125 = 0.225
Iv. 30% Y + 70% Z = (0.3) (0.2) + (0.7) (0.25) = 0.06 + 0.175 = 0.235
V. 0% Y + 100% Z = (01) (0.2) + (1) (0.25) = 0 + 0.25 = 0.25

SOLUTION 3

Rp = Wx Rx + Wy Ry

(I) 100% X + 40% Y = (1) (0.2) = (0) (0.15) = 0.2 + 0 = 0.2
(Ii) 75% X + 25% Y = (0.75) (0.2) + (0.25) = 0.14 + 0.075 + 0.15
(Iii) 50% Y + 50% Z = (0.5) (0.2) + (0.50 (0.15) = 0.1 + 0.075 = 0.175
(Iv) 25%X + 75% = (0.25) (0.2) + (0.75) (0.15) = 0.05 + 0.1125 = 0.1625
V) 0% Y + 100%Z = (0) (0.2) + (1) (0.15) = 0 + 0.15 = 0.15

$$\sigma P = \sqrt{W_x^2 \sigma_x^2 + W_y^2 \sigma_y^2 + 2 W_x W_y \sigma_x \sigma_y \sqrt{xy}}$$

$$100\% x + 0\% Y = \sqrt{(1)^2 (0.35)^2 + (0)^2 (0.24)^2 + (2)(1)(0)(0.35)(0.24)(0.4)}$$

$= \sqrt{0.1225 + 0 + 0}$

$= \sqrt{0.1225}$

$= 0.35$

75% X + 25% Y $= \sqrt{(0.75)^2(0.15)^2 = (0.15)^2(0.24)^2 + (2)(0.75)(0.25)(0.35)(0.24)(0.4)}$

$= \sqrt{(0.06890625 + 0.0036 + 0.0126}$

$= \sqrt{0.08510625}$

$= 0.29$

50% X + 50% Y $= \sqrt{(0.5)^2(0.35)^2 = (0.5)^2(0.24)^2 + (2)(0.75)(0.25)(0.35)(0.24)(0.4)}$

$= \sqrt{(0.030625 + 0.0144 + 0.0168}$

$= \sqrt{0.061825}$

$= 0.25$

25% X + 75% Y $= \sqrt{(0.25)^2(0.35)^2 = (0.75)^2(0.24)^2 + (2)(0.25)(0.75)(0.35)(0.24)(0.4)}$

$= \sqrt{(0.0076525 + 0.0324 + 0.0126}$

$= \sqrt{0.052656625}$

$= 0.23$

$= \sqrt{(0)2(0.35)2 + (1)2(0.24)2 + (2)(0)(1)(0.35)(0.24)(0.4)}$

$= \sqrt{0+0.0576}$

$= \sqrt{0.0576}$

$= 0.24$

(B) The optimal portfolio determined from the table below:

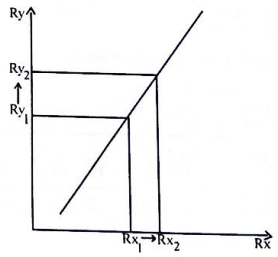
Portfolio	Portfolio Return	Portfolio Risk
100% X + 0% Y	0.2	0.35
75% X + 25% Y	0.1875	0.29
50% X + 50% Y	0.175	0.25
25% X + 75% Y	0.1625	0.23
0% X + 100% Y	0.15	0.24

The Optimal Portfolio occurs at a given point where the returns on a portfolio is maximum and the risk is minimum. So, the optimal portfolio is 25% X + 75% Y, because it has the least risk.

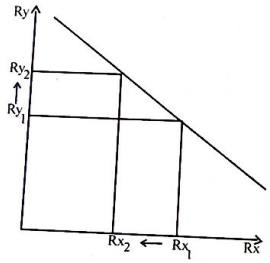
SOLUTION 4

(I) Optimal Portfolio occurs at a given point where the returns on a portfolio is maximum and the risk is minimum while ***Portfolio diversification*** is combination of various investments in a portfolio with the primary aim of eliminating or decreasing or minimising or reducing risk.

(Ii) Positively correlated investment implies that the returns of the securities in the portfolio moves in the same direction, i.e the increase in the returns of the security leads to the increase in the returns of the other. The correlation coefficient is always positive, but if the correlation coefficient is +1, between two securities in a portfolio, then the two securities is perfectly positively correlated investments.



Negatively correlated investments implies that the returns of the securities in the portfolio moves in opposite direction. The increase in the returns of one security leads to the decrease in the return of the other. The correlation coefficient is always negative, but if the correlation coefficient is - 1 between two securities in a portfolio, the the two securities is perfectly negatively correlated investment.



SOLUTION 5

Let x represent a risky portfolio
Let y represent the treasury bill

$E(Rx) = 0.33, \quad x = 0.4$
$E(Rx) = 0.12, \quad y = 0$

40% in treasury bill + 60% in risky Portfolio

$\therefore E(Rp) = W_x E(Rx) + W_y E(Ry)$
$= (0.6)(0.33) + (0.4)(0.12)$
$= 0.198 + 0.048$

$\sigma P = \sqrt{W_x^2 \sigma_x^2 + W_y^2 \sigma_y^2 + 2 W_x W_y \sigma_x \sigma_y \overline{rxy}}$

$= \sqrt{(0.6)^2(0.4)^2 + (0.4)^2(0)^2 + (2)(0.6)(0.4)(0.4)(0(0)}$

$= \sqrt{0.0576 + 0 + 0}$

$= \sqrt{0.0576}$

$= \underline{0.24}$

SOLUTION 6

SECURITY X

Xi	Xi - $\overline{X}$	$(X - \overline{X})2$
0.10	0.01	0.004
0.14	0.05	0.0025
0.03	-0.06	0.0036
0.27		0.0065

Variance of security X $= \sigma_x^2 = \frac{\sum(X - \overline{X})^2}{n-1}$

$\overline{X} = \frac{0.27}{3} = 0.09$

$= \sigma_x^2 = \frac{0.0065}{3 - 1}$

$= \frac{0.0065}{2}$

Variance of X $= 0.00325$

Standard deviation of X: $= \sqrt{0.000325}$

$= 0.057$

SECURITY Y

Yi	Yi - $\overline{Y}$	$(Yi - \overline{Y})^2$
0.12	0.03	0.0009
0.10	0.01	0.0001
0.05	-0.04	0.0016
0.27		0.0026

$$Y = \frac{0.27}{3} = 0.09$$

Variance of security $Y = \frac{(Yi - \bar{Y})^2}{h - 1}$

$$= \frac{0.0026}{3 - 1}$$

$$= \frac{0.0026}{2}$$

$$= 0.0013$$

Standard deviation of security Y

$$= \sqrt{0.0013}$$

$$= 0.036$$

COVARIANCE

$X - \bar{X}$	$Y - \bar{Y}$	$(X-\bar{X})$ $(Y- \bar{Y})$
0.001	0.03	0.0003
0.05	0.01	0.0005
-0.06	-0.04	0.0024
		0.0032

$$\text{Covariance (xy)} = \frac{(X - \bar{X})(Y - \bar{Y})}{n - 1}$$

$$= \frac{0.0032}{3-1}$$

$$= \frac{0.0032}{2}$$

$$= \frac{0.0032}{2}$$

Cov (x, y) $= 0.0016$

∴ Covariance of returns between securities X and Y is 0.0016

CORRELATION

$$\overline{XY} = \frac{\text{Cov}(X - Y)}{\sigma_X \sigma_Y}$$

$$= \frac{0.0016}{(0.057\ (0.0236)}$$

$$= 0.78$$

^ The correlation of return is 0.78

COMMENT

It is not attractive to have these two securities in a portfolio because the correlation coefficient between the returns of the two securities X and Y is highly positive (i.e the returns of X and Y moves in the same direction) and diversification will be less attractive and effective in reducing overall risk of a portfolio.

SOLUTION 7

A(i) 50% A + 50% B (Equal-weighted) two securities portfolio).

^ $W_A = 0.5$
$W_B = 0.5$
$E(R_A) = 0.24$
$E(R_B) = 0.22$

^ $Rp = (0.5)(0.24) + (0.5)\ (0.22)$
$= 0.12 + 0.11$
Portfolio return $= 0.23$

Variance $= W_A^2\sigma_A^2 + W_B^2\sigma_B^2 + 2W_A\ W_b\ \text{Cov}(A,B)$
$\sigma_A^2 = 0.4$

$\sigma_B^2 = 0.1$

Cov (A,B)= 0.1

^Variance = (0.5) (0.4) +(0.5) (0.1) +(2) (0.5) (0.5) (0.1)
= 0.1 + 0.025 + 0.050
Variance = 0.175

Standard deviation = $\sqrt{\text{Variance}}$
= $\sqrt{0.175}$
Standard deviation = 0.418 = 0.42

(ii) 50% A + 50% C
Portfolio Return = $W_A E(RA) + W_c E(RC)$

W_A = 0.5
W_C = 0.5
$E(R_A)$ = 0.24
$E(R_C)$ = 0.18

Portfolio return = (0.5) (0.24) +(0.5) 0.18)
= 0.12 + 0.09
Portfolio return = 0.21

Variance = $W_A^2\sigma_A^2 + W_c^2\sigma_c^2 + 2W_A W_c$ Cov (A, C)
W_A = 0.5, σ_A^2 = 0.4 Cov (A, C) = 0.1
W_C = 0.5, σ_C^2 = 0.12

Variance = $(0.5)^2 (0.4) + (0.5)^2 (0.12) + (2) (0.5) (0.5) (0.12)$
= 0.1 + 0.03 + 0.06)
Variance = 0.19
Standard deviation = $\sqrt{\text{Variance}}$
= $\sqrt{0.19}$

Standard deviation = $0.436 \simeq 0.44$

(Iii) 50% B + 50%C
Portfolio return = WB E (RB) + WC E(RC)
= WB = 0.5
WC = 0.5
E(RB) = 0.22
E (RC) = 0.18
= 0.11 + 0.09
Portfolio return = 0.20

Variance = WB σ^2_B + W_C^2 σ_C^2 + (2) (W_B) (W_C) Cov (B, C)
WB = 0.5 σ_B^2 = 0.1 Cov (B, C) = 0.05
WC = 0.5, σ_C^2 = 0.12
= (0.5) 2 (0.1) + (0.5)2 (0.12) + 2 (0.5) (0.05)
= 0.025 + 0.03 + 0.025
Variance = 0.08

Standard deviation = $\sqrt{\text{Variance}}$

= 0.08

= 0.28

(B)

Portfolio	Return	Standard deviation
50% A + 50% B	0.23	0.42
50% A + 50% C	0.21	0.44
50% B + 50% C	0.20	0.28

Portfolio that consist of 50% A + B and 50% A + 50% C are inefficient because their standard deviations (Risks) is on high side.

(C) I would recommend portfolio that consist of 50% B + 50% C for a risk-averse investor, because it has the least risk (standard deviation as shown above.

SOLUTION 8

SECURITY X

Return	Prob	Ev	$(Ri - E(R)^2 \times Prob$
0.16	0.25	0.04	0.0036
0.28	0.35	0.098	0
0.35	0.40	0.14	0.00196
		0.278	0.00556 $-\sigma x^2$
		0.28 = E(Rx)	

SECURITY Y

Return	Pi	Ev	$(Ri - E(Ri)^2 \times Prob$
0.25	0.30	0.075	0.00003
0.26	0.35	0.091	0
0.27	0.35	0.0945	0.000035
		0.2605	0.000065 σ_y^2
		0.26 = E(Ry)	

(A) 55% X + 45% Y

Rp = Wx E(Rx) + Wy (Ery)

Wx = 0.55, E(Rx) = 0.28

Wy = 0.45, E(Ry) = 0.26

= Rp = (0.55)(0.28) + (0.45) (0.26)

= 0.154 + 0.17

= 0.271 = 27.1%

Standard deviation, when $\sqrt{xy} = 1$

$$P = \sqrt{W_x^2\sigma_x^2 + W_y^2\sigma_y^2 + (2)\ (Wx)\ (Wy)\sigma_x\ \sigma_y\ \sqrt{xy}}$$

When $\sqrt{xy} = +1$

$Wx = 0.55, \sigma x^2 = 0.00556, \quad \sigma x = \sqrt{0.00556} = 0.746$

$Wy = 0.45, \quad \sigma y^2 = 0.000065, \sigma y = \sqrt{0.000065} = 0.00806$

$$P = \sqrt{(0.55)^2 (0.00556) + (0.45)^2\ (0.000065) + (2)\ (0.55)\ (0.45)\ (0.0746)\ (0.00806)\ (1)}$$

$$= \sqrt{0.0016819 + 0.0000131625 + 0.00029763162}$$

$$= \sqrt{0.0019926935}$$

$$= 0.0446 = 4.46\%$$

When $\sqrt{xy} = 0$

$$\sigma_P = \sqrt{(0.55)^2\ (0.00556) + (0.45)^2\ (0.000065) + (2)\ (0.55)\ (0.45)\ (0.0746)\ (0.00806)\ (0)}$$

$$= \sqrt{0.0016819 + 0.0000131625 + 0}$$

$$= \sqrt{0.001695062}$$

$$= 0.0412 = 4.12\%$$

When $\sqrt{xy} = -1$

$$\sigma_P = \sqrt{(0.55)\ (0.00556) + (0.45)\ (0.000065) + (2)\ (0.55)\ (0.45)\ (0.0746)\ (0.00806)\ (-1)}$$

$$= \sqrt{0.0016819 + 0.0000131625 + 0.00029763162}$$

$$= \sqrt{0.001397431}$$

$$= 0.0374 = 3.74\%$$

50% X + 50% Y

(b) Rp = W x E (Rx) + Wy E (Ry)

Wx = 0.5 E (Rx) = 0.28

Wy = 0.5 E(Ry) = 0.26

Rp = (0.5) (0.28) + (0.5) (0.26)

= 0.14 + 0.13

= 0.27 = 27%

σ_P Standard deviation is calculated as follows

$$\sqrt{W_x^2\sigma_x^2 + W_y^2\sigma_y^2 + 2W_x\ W_y\sigma_x\sigma_y\ \sqrt{xy}}$$

When xy = + 1
Wx = 0.5 σ_x^2 = 0.00556,
Wy = 0.5 σ_x^2 0.0000651
σ_y 0.00806

$$\sigma_P = \sqrt{(0.55)^2 (0.00556) + (0.50)^2 (0.000065) + (2)\ (0.5)\ (0.5)\ (0.0746)\ (0.00806)\ (1)}$$

$$= \sqrt{0.00139 + 0.00001625 + 0.000300638}$$

$$= \sqrt{0.001706888}$$

$$= 0.0413 = 4.13\%$$

When $\sqrt{xy}$ = 0

$$\sigma_P = \sqrt{(0.55)^2 (0.00556) + (0.50)^2 (0.000065) + (2)\ (0.5)\ (0.5,\ 0.0746)\ (0.00806)\ (0)}$$

$$= \sqrt{0.00139 + 0.00001625 + 0}$$

$$= \sqrt{0.00140625}$$

$$= 0.0375 = 3.75\%$$

when $\sqrt{xy}$ = -1

$$\sigma_P = \sqrt{(0.55)^2 (0.00556) + (0.50)^2 (0.000065) + (2)\ (0.5)\ (0.5)\ (0.0746)\ (0.00806)\ (-1)}$$

$$= \sqrt{0.00139 + 0.00001625 + 0}$$

$$= \sqrt{0.0011105612}$$

$$= 0.0333 = 3.33\%$$

SOLUTION 9

SECURITY A

R A	P i	RA Pi	RA - $\bar{R}$A	[RA - $\bar{R}$A] x PC
0.39	0.5	0.195	0.032	0.032
0.45	0.4	0.18	0.028	0.028
0.47	0.1	0.047	0.048	0.048
		0.422 = RA		0.001056 = σ_A^2

$$= \sqrt{0.001056} = \sigma A = 0.0325$$

SECURITY B

R_B	P i	RA Pi	$R_B - \bar{R}_B$	$[R_B - \bar{R}_B]^2$ x Pi
0.65	0.4	0.26	0.0241	0.0232324
0.44	0.1	0.044	0.031	0.0000961
0.21	0.5	0.105	-0.199	0.0198005
		0.409 = R_B		0.043129 = σ_B^2

$$= \sqrt{0.001056} = \sigma_B = 0.2077$$

SECURITY C

R_B	P i	Rc Pc	Rc - $\bar{R}$c	$[R_C - \bar{R}_C]^2$ x Pi
0.58	0.2	0.116	0.081	0.0232324
0.49	0.7	0.343	0-0.009	0.0000961
0.40	0.1	0.040	-0.099	0.0198005
		0.499 = Rc		0.043129 = σ_B^2

$$= 0.001056 = \sigma_B = 0.2077$$

(1) 0.6 A + 0.4 B

Mean = W_A E(R_A) + W_B E(R_B)
= (0.6) (0.422) + (0.4) (0.409)
= 0.2532 + 0.1636
= 0.4168 = 41. 68%

Standard deviation = $\sqrt{W_A^2 \sigma_A + W_B^2 \sigma_B^2 + 2W_A\ W_B \sigma_A \sigma_B\ r_{AB}}$

W_A= 0.6, σ_A^2 = 0.001056, σ_A = 0.0325 $_{AB}$ = 0.64

W_B= 0.4, σ_B^2 = 0.043129, σ_B = 0.2077

$$\sigma_P = \sqrt{(0.6)^2 (0.001056) + (0.4)^2 (0.043129)\ (0.043129) + (2)\ (0.6)\ (0.4)\ (0.0325)\ (0.2077)\ (0.64)}$$

$$= \sqrt{(0.00038016 + 0.00690064 + 0.0020736768}$$

$= \sqrt{0.0093544768}$

$\sigma_P = 0.0967 = 9.67\%$

(Ii) 0.7A + 0.3 C

Mean = $W_A E(R_A) + W_C E(R_C)$
$= (0.7)(0.422) + (0.3)(0.499)$
$= 0.2954 + 0.1497$
$= 0.4451 = 44.50\%$

Standard deviation $= \sqrt{W_A^2 \sigma_A^2 + W_C^2 \sigma_C^2 + 2W_A W_C \sigma_A \sigma_C r_{AC}}$

$W_A = 0.7$, $\sigma_A^2 = 0.001056$, $\sigma_A = 0.0325$ $r_{AB} = 0.46$

$W_B = 0.3$, $\sigma_C^2 = 0.02349$, $\sigma_C = 0.0486$

$\sigma_P = \sqrt{(0.7)^2(0.001056) + (0.3)^2(0.002347) + (2)(0.7)(0.3)(0.0325)(0.0485)(0.46)}$

$= \sqrt{0.00051744 + 0.00021141 + 0.0003045315}$

$= \sqrt{0.0010333815}$

$\sigma_P = 0.0321 = 3.21\%$

(iii) 0.8 B + 0.2 C

Means $= W_B E(R_B) + W_C E(R_C)$
$= (0.8)(0.409) + (0.2)(0.499)$
$= 0.3272 + 0.998$
$= 0.427 = 42.7\%$

Standard deviation $= \sqrt{W_B^2 \sigma_B^2 + W_C^2 \sigma_C^2 + 2W_B W_C \sigma_B \sigma_C r_{BC}}$

$= \sqrt{(0.8)^2(0.043129) + (0.2)^2(0.002399) + (2)(0.8)(0.2)(0.2077)(0.0485)(0.42)}$

$= \sqrt{0.02760256 + 0.00009396 + 0.00135387168}$

$= \sqrt{0.029050391}$

$\sigma_P = 0.1704 = 17.04\%$

SOLUTION 10

SECURITY A

R_B	Pi	(R_A) Pi	$R_A - \bar{R}_A$	$[R_A - \bar{R}_A]^2 \times$ Pi
0.38	0.4	0.152	0.038	0.0005776
0.44	0.5	0.22		0.000242
0.46	0.1	0.046	0.0001764	
		0.418 = R_A		0.000096 = σ_A^2

$= \sqrt{0.001056} = \sigma_A = 0.0316$

SECURITY B

R_B	Pi	(RA) Pi	$R_B - \bar{R}_B$	$[R_B - \bar{R}_B]^2 \times$ Pi
0.64	0.4	0.256	0.143	0.0081796
0.43	0.5	0.215	-0.067	0.0022445
0.26	0.1	0.026	-0.237	0.0056169
		0.497 = R_B		0.016041 = σ_B^2

$= \sqrt{0.0016041} = \sigma_B = 0.01267$

SECURITY C

R_C	Pi	(R_C) Pi	$R_C - \bar{R}_C$	$[R_C - \bar{R}_C]^2 \times$ Pi
0.57	0.2	0.114	0.114	0.0029282
0.48	0.1	0.048	0.048	0.0000961
0.41	0.7	0.287	-0.039	0.0010647
		0.449 = R_C		0.004089 = σ_C^2

$= \sqrt{0.004089} = \sigma_C = 0.065$

(i) P1 = 0.4 A + 0.6 B

Mean = W_A E(R_A) + W_B E (R_B)

= (0.4) (0.418) + (0.6) (0.497)

= 0.1672 + 0.2082

= 0.4654 = 46.54%

Variance = $W_A^2 \sigma_A^2 + W_B^2 \sigma_B^2 + 2W_A W_B \sigma_A \sigma_B r_{AB}$

W_A = 0.4 σ_A^2 = 0.000996, σ_A = 0.0316

W_B = 0.6 σ_B^2 = 0.016041, σ_B = 0.1267 r_{AB} = 0.4

= (0.4) (0.000996) + (0.6) (0.016041) + (2) (0.4) (0.6) 0.316) (0.1267) (0.4)

= 0.00015936 + 0.0057476 + 0.00076871424

= 0.00670283424

σ_P^2 = 0.00670 = 0.67%

(Ii) P2 = 0.75A + 0.25C

Mean = W_A E (R_A) + W_C E(R_C)

= (0.75) (0.418) + (0.25) (0.459)

= 0.3135 + 0.11225

= 0.42575 = 42.58%

Variance = $W_A^2 \sigma_A^2 + W_C^2 \sigma_C^2 + 2W_A$ WC $\sigma_A \sigma_C r_{AC}$

WA = 0.75 σ_A^2 = 0.000996

WC = 0.25 σ_C^2 = 0.004099

σ_A = 0.0316, σ_C = 0.0639

σ_P^2 = (0.75) (0.000996) + (0.25) 0.004089) + (2) (0.75) (0.25) (0.0316) (0.0639) (0.5)

= 0.00056025 + 0.0002555625 + 0.0002555625 + 0.0003786075

= 0.0011942

= 0.0012

Variance = 0.12%

(Iii) P3 = $\sqrt{0.5\ B + 0.5\ C}$

Mean = W_B E (R_B) + W_C E (R_C)

= (0.5) (0.497) + (0.5) (0.49)

= 0.2485 + 0.2245

= 0.473 = 47.3%

Variance = $W_B \sigma_B^2$ + $W_C^2 \sigma_C^2$ + (2) W_B $W_C \sigma_B \sigma_C r_{BC}$

= (0.5)2 (0.016041) + (0.5)2 (0.004089) + (2) (0.5) (0.5) (0.1267) (0.0639) (0.3)

= 0.00401025 + 0.00102225 + 0.0012144195

= 0.0062469195

= 0.0062 = 0.62%

(b) beta of P_I = $\frac{Cov\ (A, B)}{\sigma_m^2}$

But Cov (A, B) = $r_{AB} \sigma_A \sigma_B$

∴ Beta of P_I = $\frac{r_{AB} \sigma_A \sigma_B}{\sigma_m^2}$

r_{AB} = 0.4, σ_A = 0.0326, σ_B = 0.1267, σ_m^2 = 0.0226

Beta of P_I = $\frac{0.4 \times 0.0316 \times 0.1267}{0.0226}$

= $\frac{0.001601488}{0.0226}$

Beta of P_I = 0.07

SOLUTION 11

SECURITY A

R_A	Pi	(R_A) Pi	$R_A - \bar{R}_A$	$[R_A - \bar{R}_A]^2 \times$ Pi
0.24	0.3	0.072	-0.016	0.0000768
0.25	0.2	0.05	-0.006	0.0000072
0.30	0.3	0.09	0.044	0.0005808
0.22	0.2	0.044	0.036	0.0002592
		0.256 - R_A		0.000924 = σA^2

$\sigma A = \sqrt{0.000924} = 0.0304$

SECURITY B

R_B	Pi	RA Pi	$R_B - \bar{R}_B$	$[R_B - \bar{R}_B] \times$ Pi
0.38	0.4	0.152	0.036	0.0005184
0.45	0.2	0.09	0.106	0.0022472
0.28	0.3	0.084	-0.064	0.0012288
0.18	0.1	0.018	-0.164	0.0026896
		0.344 - R_B		0.006684 = σ_B^2

$\sigma_B = \sqrt{0.006684} = 0.0818$

(I) 80% A + 20% B

Mean = $W_A \bar{R}_A + W_B \bar{R}_B$

$W_A = 0.80$, RA = 0.256

$W_B = 0.20$, RB = 0.344

Mean = (0.8) (0.256) + (0.2) (0.344)

= 0.2048 + 0.0688

= 0.2736 = 27.36%

Standard deviation = $W_A^2 \sigma_A^2 + W_B \sigma_B + (2) W_A W_B \sigma_A \sigma_A r_{AB}$

$\sigma_A^2 = 0.000924$, $\sigma_B^2 = 0.006684$, $W_B^2 = 0.8$

$\sigma_A = 0.304$, $\sigma_B = 0.0818$, $W_B = 0.2$

When $r_{AB} = 0.2$

$$\sigma_P = \sqrt{(0.8)^2 (0.000924) + (0.2)^2 (0.006684) + (2) (0.8) (0.2) (0.0304) (0.0818) (0.2)}$$

$$= \sqrt{0.00059136 + 0.00026736 + 0.0001595008}$$

$$= \sqrt{0.00101787008}$$

= 0.0319%

When $r_{AB} = 0.4$

$$\sigma_P = \sqrt{(0.8)^2 (0.000924) + (0.2)^2 (0.006684) + (2) (0.8) (0.2) (0.0304) (0.0818) (0.4)}$$

$$= \sqrt{0.00059136 + 0.00026736 + 0.00031830016}$$

$$= \sqrt{0.001017702016}$$

= 0.0343 = 3.43%

When $r_{AB} = 0.6$

$$\sigma_P = \sqrt{(0.8)^2 (0.000924) + (0.2)^2 (0.006684) + (2) (0.8) (0.2) (0.0304) (0.0818) (0.6)}$$

$$= \sqrt{0.00059136 + 0.00026736 + 0.0004775}$$

$= \sqrt{0.00133617024}$

$= 0.0366 = 3.66\%$

(ii) 35% A + 65%B

Mean $= W_A \bar{R}_A + W_B \bar{R}_B$

$= (0.35)(0.256) + (0.65)(0.344)$

$= 0.0896 + 0.2236$

$= 0.3132 = 31.32\%$

Standard deviation $= \sqrt{W_A^2 \sigma_A^2 + W_B^2 \sigma_B^2 + 2 W_A W_B \sigma_A \sigma_B r_{AB}}$

$W_A = 0.35 \quad \sigma_A^2 = 0.000924, \quad \sigma_A = 0.0304$

$W_B = 0.65, \quad \sigma_B^2 = 0.006684, \quad \sigma_B = 0.818$

With Perfect Positive Correlation $r_{AB} = +1$

$$\sigma_P = \sqrt{(0.35)^2 (0.000924) + (0.65)^2 (0.006684) + (2)(0.35)(0.65)(0.0304)(0.0818)(1)}$$

$$= \sqrt{0.00011319 + 0.000282399 + 0.00011314576}$$

$$= \sqrt{0.002938460196}$$

$= 0.0542 = 5.42\%$

With Perfect negative Correlation $r_{AB} = -1$

$$\sigma_P = \sqrt{(0.35)^2 (0.000924) + (0.65)^2 (0.006684) + (2)(0.35)(0.65)(0.0304)(0.0818)(-1)}$$

$$= \sqrt{0.00011319 + 0.000282399 - 0.00011314576}$$

$= \sqrt{0.0018057224}$

$= 0.0425 = 4.25\%$

With Zero Correlation $r_{AB} = 0$

$$\sigma_P = \sqrt{(0.35)^2 (0.000924) + (0.65)^2 (0.006684) + (2)(0.35)(0.65)(0.0304)(0.0818)(0)}$$

$$= \sqrt{0.00011319 + 0.000282399 + 0}$$

$$= \sqrt{0.00293718}$$

$= 0.0542 = 5.42\%$

SOLUTION 12

SECURITY A

R_A	Pi	(R_A) Pi	$R_A - \bar{R}_A$	$[R_A - \bar{R}_A]^2 \times Pi$
0.26	0.3	0.078	-0.064	0.0012288
0.32	0.3	0.096	-0.004	0.0000048
0.35	0.2	0.07	0.026	0.0001352
0.40	0.2	0.08	0.076	0.0011552
		0.324 - R_A		0.002524 = σ_A^2

$\sigma_A = \sqrt{0.002524} = 0.0502$

SECURITY B

R_B	Pi	RA Pi	$R_B - \bar{R}_B$	$[R_B - \bar{R}_B] \times Pi$
0.16	0.4	0.064	-0.122	0.0059536
0.20	0.1	0.02	-0.082	0.0006724
0.36	0.3	0.108	-0.078	0.0018252
0.45	0.2	0.09	-0.168	0.0056448
		0.282 - R_B		0.014096 = σ_B^2

$\sigma_B = \sqrt{0.014096} = 0.1187$

SECURITY C

Rc	Pi	Rc Pi	Rc - $\bar{R}c$	$[Rc - \bar{R}c]^2$ x Pi
0.30	0.1	0.03	-0.19	0.00361
0.45	0.2	0.09	-0.04	0.00032
0.50	0.5	0.25	0.01	0.00005
0.60	0.2	0.12	0.11	0.00242
		0.49 = Rc		0.0064 = σ_c^2

$\sigma_c = \sqrt{0.0064} = 0.08$

(i) P = $^1/_3$A + $^2/_3$ B with Correlation Coefficient of 0.6

Mean = W_A R_A + W_B R_B

$W_A = {}^1/_3 = 0.33$,

$W_B = {}^2/_3 = 0.67$

RA = 0.324

RB = 0.282

∴ Mean = $^1/_3$ (0.324) + $^2/_3$(0.282)

= 0.108 + 0.188

= 0.296

Standard deviation = $\sqrt{W_A^2\sigma_A^2 + W_B^2\sigma_B^2 + 2W_A\ W_B\ \sigma_A\ \sigma_A\ r_{AB}}$

$W_A = {}^1/_3$, $\sigma_A^2 = 0.0002524$, $\sigma_A^2 = 0.0502$ $r_{AB} = 0.6$

$W_B = {}^2/_3$, $\sigma_B^2 = 0.014096$ $\sigma_B^2 = 0.1187$

$\sigma_P = \sqrt{(^1/_3)^2(0.002524) + (^2/_3)^2(0.014096) + (2)\ (^1/_3)\ (^2/_3)\ (0.0502)\ (0.1187)\ (0.6)}$

$= \sqrt{0.0002804 + 0.006264889 + 0.001588997}$

$= \sqrt{0.008134330667}$

= 0.09 = 9%

(Ii) M = $\frac{7}{8}$ B + $\frac{1}{8}$ C with Correlation Coefficient of -0.4

Mean = W_B $\bar{R}_B$ + W_C $\bar{R}c$

$W_B = \frac{7}{8}\bar{R}_B = 0.282$

$W_C = \frac{1}{8}\bar{R}c = 0.49$

Means = $\frac{7}{8}$ (0.282) + $\frac{1}{8}$ (049)

= 0.24675 + 0.060125

= 0.308 = 30.8%

Standard deviation = $\sqrt{W_A^2\sigma_A^2 + W_B^2\sigma_B^2 + 2W_A\ W_B\ \sigma_A\ \sigma_A\ r_{AB}}$

$W_B = \frac{7}{8}\ \sigma_B^2 = 0.014096$ $\sigma_B^2 = 0.1187$ $r_{BC} = -0.4$

$W_C = \frac{1}{8}\ \sigma_C^2 = 0.0064$ $\sigma_C^2 = 0.08$

$r_{AB} = 0.4$

$\sigma_P = \sqrt{(\frac{7}{8})^2(0.0014096) + (\frac{1}{8})^2(0.0064) + (2)\ (\frac{7}{8})\ (\frac{1}{8})\ (0.1187)\ (0.08)\ (-0.4)}$

$= \sqrt{0.01079225 + 0.0001 - 0.0008309}$

$= \sqrt{0.01006.135}$

= 0.1003 = 10.03%

(iii) Q = 0.6A + 0.4C with a Correlation Coefficient of 0.8

Mean = W_A $\bar{R}_A$ + W_C $\bar{R}c$

W_A = 0.6, $\bar{R}_A$ = 0.324

W_C = 0.4, $\bar{R}c$ = 0.49

∴ Means = (0.6) (0.324) +(0.4) (0.49)

= 0.1944 + 0.196

= 0.3906 = 39.06%

Standard deviation $= \sqrt{W_A^2 \sigma_A^2 + W_C^2 \sigma_C^2 + 2W_A W_C \sigma_A \sigma_A r_{AC}}$

$W_A = 0.6$ $\sigma_A^2 = 0.0002524$, $\sigma_A = 0.0502$ $r_{AC} = 0.8$
$W_B = 0.4$, $\sigma_C^2 = 0.0064$ $\sigma_C = 0.08$

$$\sigma_P = \sqrt{(0.6)^2 (0.002524) + (0.4)^2 (0.0064) + (2)(0.6)(0.4)(0.0502)(0.8)(0.8)}$$

$$= \sqrt{0.00090864 + 0.001024 + 0.001542144}$$

$$= \sqrt{0.003474784}$$

$$= 0.0589 = 5.89\%$$

(Iv) R = ½ A + ½ B with a Correlation Coefficient of 0.1

Mean = $W_A R_A + W_B RB$

$W_A = ½$ $R_A = 0.324$

$W_B = ½$, $R_B = 0.282$

∴ Mean = ½ (0.324) + ½ (0.282)

= 0.162 + 0.141

Mean = 0.303

Standard deviation $= \sqrt{W_A^2 \sigma_A^2 + W_B^2 \sigma_B^2 + 2W_A W_B \sigma_A \sigma_A r_{AB}}$

$W_A = 1/2$ $\sigma_A^2 = 0.002524$ $\sigma_A^2 = 0.0502$ $r_{AB} = 0.1$
$W_B = ⅓$, $\sigma_B^2 = 0.014096$ $\sigma_B = 0.1187$

$$\sigma_P = \sqrt{(½)^2 (0.002524) + (½)(0.014096) + (2)(1/2)(½)(0.0502)(0.1187)(0.1)}$$

$$= \sqrt{0.000631 + 0.003524 + 0.000297937}$$

$$= \sqrt{0.004452937}$$

$$= 0.0667 = 6.67\%$$

(B) If Security A and B have a zero correlation, then the standard deviation computed above (i) & (iv) will change as shown below:

Standard deviation $= \sqrt{W_A^2 \sigma_A^2 + W_B^2 \sigma_B^2 + 2W_A W_B \sigma_A \sigma_B r_{AB}}$

when $r_{AB} = O$, $W_A = 1/3 + W_B 2/3$

$$\sigma_P = \sqrt{(1/3)^2 (0.002524) + (2/3)^2 (0.014096) + (2)(1/3)(2/3)(0.0502)(0.1187)(0)}$$

$$= \sqrt{0.0002804 + 0.00624889 + 0}$$

$$= \sqrt{0.006545289}$$

$$= 0.0809 = 8.009\%$$

(iv) $r_{AB} = 0$, $W_A = ½$, $W_B = ½$

$$\sigma_P = \sqrt{(1/2)^2 (0.002524) + (½)^2 (0.014096) + (2)(1/2)(½)(0.0502)(0.1187)(0)}$$

$$= \sqrt{0.00631 + 0.003524 + 0}$$

$$= \sqrt{0.004155}$$

$$= 0.0645 = 6.45\%$$

CHAPTER 15

FINANCIAL MARKET

Question 1
Mention and discuss FIVE methods of bringing securities to the stock exchange.

Question 2
Explain the term Primary Market Segment and briefly outline its essential characteristics.

Question 3
What are the major functions of secondary security market.

Question 4
Briefly outline the circumstances in which it would be more beneficial for shareholders to take up their rights rather than sell their options in whole or in part.

Question 5
What are the factors to be considered by firm management in setting the price of a right issue.

Question 6
Compare and contrast the listing requirements of the first-tier market with that of the second tier market.

Question 7
Discuss the merits and demerits of right issues as a method of bringing security to the stock exchange.

Question 8
What are the aims of companies seeking quotation on the stock exchange? Briefly outline the benefits and limitation of quotation.

Question 9
Discuss the major functions of Nigeria Stock Exchange

Question 10
Briefly state the functions of Security and Exchange Commission.

Question 11
State the functions of Discount houses in Nigerian money market.

Question 12
Explain the term " efficient capital market" and briefly discuss the various forms of market efficiency.

Question 13
What are the factors that a financial Manager should take into account when investing in marketable securities?

Question 14
What are implications of Efficient Capital Market Hypothesis for Financial managers of a limited laibility companies.

Question 15
Write short notes on the following
(i) Term loans
(ii) Factoring
(iii) Discounting of bills
(iv) Overdraft
(v) Leasing

Question 16
Write short notes on the following to bring out clearly the salient finance issues:
(i) Secured credit and unsecured credit
(ii) Factoring an invoice discounting
ilii) Risk and Uncertainly
(iv) Operating lease and sales and lease back
(v) Hire purchase and sales and lease back
(vi) Ordinary shares and Preference Shares

Question 17
A friend of yours wishes to invest some funds in ordinary shares, preference shares and debenture stock and has sought your advice as to the choice between Ordinary Shares, Preference Share and Debenture Give him a quality advice.

Question 18
Alaba Ltd, is a new company that is making its first public issue of shares. It has decided to make the issue by means of an offer for sale by tender. The intention is to issue up to 2,500,000 shares (the full amount of authorised share capital); the money raised, net of issue cost of N600,000 would be invested in projects which would earn benefits with a present value equal to 120% of the net amount invested. The following tenders have been received. Each applicant has made only one offer, and there are no multiple applications.

Price tendered Per share	Number of Share applied for at this price
350k	125,000
3000k	150,000
275k	350,000
250k	625,000
225k	850,000
200k	1,000,000
175k	850,000
150k	350,000

You are required to:
(a) How many shares would be issued, and how much in total would be raised, if Alaba Ltd. Chooses to:
(I) Issue exactly 2,500,000 shares
(Ii) Maximise the total amount raised?

(b) Barrister Damesi, a private investor, has applied for 15,000 share at a price of 300k and has despatched a cheque for N45,000 to the issuing house that is handling the issue. In both cases (i) and (ii) above, how many shares would be issued to Barrister Damesi, assuming that any partial acceptance of offers would mean allotting shares to each accepted applicant in proportion to the number of shares applied for. How much will Barrister Damesi receive back out of N50,000 she has paid?

(c) Estimate the likely market value of Alaba Ltd. Shares after the issue, assuming that the market price fully reflects the investment information given above and that exactly 2,500,000 shares were issued.

Question 19
Alhaji Ajomale intends to buy 20,000 shares of "Tinubu Straight Plc" currently price on the stock exchange at N25.00 per share. The Brokers Commission is 3% of his total transaction, security and Exchange Commission (SEC) Commission is 1/8 of 10% of total transaction and stamp duty is 14k on every N250.

Required:
(A) Compute his total liability under the transaction.
(B) Assuming he is selling in a different account, what is his proceeds on sale?

Question 20
Miss Soludo who is an investor in CBA Plc has been informed of right issue of 1 of 4 rights issue at N11.50. The market price of the share is N20.00. Mrs. Soludo has a total of 100,000 shares in the company.

Required:
(a) Calculate the theoretical ex-rights rice.
(b) Calculate the value of her right
(c) What will be Miss Soludo position in the following:
  (i) If she takes all rights;
  (ii) If she sells all her rights;
  (iv) If she sells 60% of her rights and acquire the remainder;
  (v) If she does not take any position at all

Question 21
Obodo Charles Company Limited has 25, 000,000 Ordinary Shares outstanding. A right offer is to be used to raise N100,000,000 of new finance. The subscription price is set at N30 per share.

You are required to:
(i) Compute the number of new shares to be sold
(ii) Compute the number of rights needed to purchase one new share.
(iii) How many rights will existing shareholders received?
(iv) If the market price of the share at the beginning of the rights offering is N35, compute the theoretical ex-right price of the share.

Question 22
From the information about Bosede, calculated:
(I) Ex-rights price
(Ii) Value of a right

The company is solely financed with equity. The company needs additional n25,000,000 of finance. The management has decided to make a right of 1 for 5 to raise the fund. The company has issued 5,000,000 share to date. The share price of the company's is currently quoted at N26.52.

FINANCIAL MARKET

SOLUTION 1
METHODS OF BRINGING SECURITIES TO THE STOCK EXCHANGE
Securities may be brought to the Stock Exchange by any of the following methods:

(a) **Offer for Sale**
Here the company offers shares to an issuing house (merchant bank or finance company) which then offers shares to the public at a price set by negotiation between the company and the issuing house. The issuing house handles the administration of the issue in return for the margin between the price paid to the company and the price received from the public, the company is guaranteed that all the shares will be sold.

(b) **Offer for sale by tender**
It is often very difficult to decide upon the price at which the shares should be offered to the general public. One way of trying to ensure that the issue price reflects the value of the shares as perceived by the market is to make an offer for sale by tender. Under this method.

(A) A minimum price will be fixed and subscribers will be invited to tender for shares at prices equal to or above the minimum.
(B) The shares will be allotted at the highest price at which they will all be taken up. This is known as the striking price.

(c) **Public Issue by Prospectus**
In a prospectus issue, or public issue, a company offers its own shares to the general public. An issuing house or merchant bank may act as an agent, but not as an underwriter. This type of issue is therefore risky, and is very uncommon. Well known companies making a large new issue may use this method.

(d) **Placing**
A placing is an arrangement whereby the shares are not all put on offer to the public but instead, the sponsoring market make arranges for most of the issue to be bought by a small number of investors, usually institutional investors such as pension funds and insurance companies. The method is used when it is unlikely that an offer sale will be fully authorised.

Placing are much cheaper, although most of the shares will be placed with relatively small number of (institutional) shareholders, which means that most of the shares are unlikely to be available for buying and selling in an 'active' market after the floatation.

(e) **Stock Exchange Introduction**
By this method of obtaining a quotation no shares are made available to the market neither existing nor newly- created shares; nevertheless, the Stock Exchange grants a quotation. This will only happens where shares in a large company are already widely held, so that a market can be seen to exist. It may therefore be asked why a company should want an introduction. The reasons would be a greater marketability for the shares, a known share valuation for inheritance tax purposes and easier access in the future to additional capital

(f) **Right Issue**
A right issues is a means of raising new equity capital whereby the existing shareholders are invited to subscribe for the new issue in proportion to their existing holding. A right issue enables a firm to raise further equity while ensuring that existing shareholder have the right to maintain their proportional investment in the firm.
When a firm has a stock exchange quotation this is usually the only method of raising new equity permitted. No prospectus is required.

SOLUTION 2
PRIMARY MARKET
Primary market is a market for a new long term securities. It is a market where new or fresh long term securities are initially sold to the public. It must be noted that the firm that issues new or fresh long term securities to the public accrue to the firm purse alone.
The activities in the primary security market is controlled by Securities and Exchange Commission (SEC) while other participants or play makers are issuing houses, stock broking firms, commercial banks, merchant banks, rational investors and corporate bodies and agencies. Only the companies that were quoted on the Stock Exchange were allowed to raise fresh or new funds from the Primary market.

Its essential characteristics are outlined as follows:
(I) It allows companies that were quoted on the stock exchange to raise fresh or new capital or fund.
(Ii) Based on the above, the issue share capital of a firm increases.
(Iii) Primary market segment is only segment of capital market where capital formation occurs
(Iv) New issues of share or stock are initially sold to an interested investor or public.

SOLUTION 3

SECONDARY SECURITY MARKET

Secondary security market is a market that allows existing holders of financial securities (Ordinary security Preference shares or Debenture stock) to sell whole or part of their shares or stock to other interested investors.

The major function of Secondary security market are as follows:

(I) It provides cash for financial securities such as Ordinary shares, Preference shares and Debenture stock that were traded initially in the primary market.

(Ii) It provides means or revenue for financial securities that are initially sold in the primary market segment to the traded on.

(Iii) The prevailing prices of a firm share or stocks in the secondary market segment has a great relationship on the price quoted in the Primary market segment for the firm shares or stock.

(Iv) It offer the market participants or investors the opportunity to invest in a wide range of firm ad thereby spread or reduce their risk.

(V) The prevailing price or current market price of a firm shares or stock in the secondary market segment has a direct relationship on shareholders or investors wealth; that is the higher the current market price of a company share or stock in the Secondary market segment, the higher the wealth of the shareholder or investors of that company and vice versa.

SOLUTION 4

Circumstances in which it would ne more beneficial for shareholder to take up their rights rather than sell their options in whole or in part are:

1. Where they have an existing substantial stake in the company with influential voting power which will fall if they do not take up their rights in full.
2. Where favourable forecasts relating to future profits and dividends of the company will result in the market price exceeding the theoretical ex-rights price.
3. In cases where the shareholders has surplus funds for investment and cannot obtain a higher return for them elsewhere with no greater risk.
4. Where funds can be borrowed at a lower rate than the return offered by the rights issue.
5. Where substantial capital gains tax fall to be paid if any part of the present holding (including the rights) is sold.

SOLUTION 5

Factors to consider in setting the price of right issue

1. The existing share price
2. The current prices of comparable stocks
3. The reasons for the issue
4. The size of the issue an the ability of shareholder to subscribe
5. Any current government cotrols.
6. The future cost of dividends of the same level of dividend is to be maintained on the enlarged share capital.

SOLUTION 6

Listing requirements on the Nigeria Stock Exchange

First Tier Market FEM

(1) A company must registered as Public Limited Company under CAMA '90.
2) The company must submit the financial statement records for the past five years
(3) Date of last audited account must not exceed 9months.
(4) Amount of money that can be raised is unlimited depending on the borrowing power of the Director of the companies.
(5) Annual quotation fee is based on market capitalization.
(6) At least 25% of the nominal value of share capital must be offered to the public.
(7) Number of shareholder must not be less than 300.
(8) After listing company must submit quarter, half yearly, and annual account.
(9) Securities must be fully paid for at the time of allotment.
(10). Unallotted securities must be sold on the floor of stock exchange.

Second Tier Market FEM

(1) A company must be Public Limited Company under CAMA '90.
2) The company must submit the financial statement records for the past 3 years
(3) Date of last audited account must exceed 9months.
(4) Amount of money that can be raised may not exceed N20m.
(5) Annual quotation fee is based on market capitalization.
(6) At least 10% of the nominal value of share capital must be offered to the public.
(7) Number of shareholder must not be less than 100.
(8) After listing company must submit half yearly, and annual account.
(9) Securities must be fully paid for at the time of allotment.
(10). Unallotted securities must be sold on the floor of stock exchange.

SOLUTION 7

Advantage of 'right' issues

(a) Under the Stock Exchange Rules all issues of shares for cash other than rights issues must be approved by the company in general meeting, in contrast to subject to the Articles of Association, a right issues may be made at the discretion of the directors;

(b) Right Issues are cheaper than issues to the general public. This is partly because no prospectus is required partly because the administration is simpler and partly because the cost of underwriting will be less;

(c) Relative voting rights are unaffected:

(d) All issues of new shares including rights issues are made at a price slightly below the current market price so as to induce potential investors to subscribe for the shares. In an issue to the general public, the benefit of this 'discount' will be received by those subscribing for the shares. Under a rights issue it will be reserved for the existing members of the company.

(e) The will usually be a strong demand for the share. If the company is a fast- growing, demand may be such that after adjustment downwards to the ex-rights price, the share price starts moving up rather than down. Certain investors and institutions specialize in new issues and in acquiring shares depressed by a rights issue;

(f) If dividend per share can be maintained at the rate in the effect before the rights issue, the expected fall in ex-rights price will mean that the dividend yield will in fact rise. A rights offer may lead the shareholder of a fast growing company to believe the company is confident and that it will continue to expand and it draws attention to the company's good performance in the past. Factors such as these boost the demand for a company's shares and so cause prices to rise;

Disadvantages

(a) There will usually be a fall in the share price because of:

(i) marking down of price by jobbers in anticipation of some shareholders being forced to sell their rights through lack of money;

(ii) an increase in the supply of the shares on the marekt;

(iii) the earning yield, calculated on the previous year's net profits will probably be reduced;

(b) The cost of a rigts issue is often the same as the cost of equity funds. If the shares are undervalued at the ex-rights price, and the existing sharehodlers do not take up all the rights, the cost of rights funds will be higher than the cost of new equity;

(c) A public issue will spread ownership more widely and so prevent potential bidders getting control over a large number of shares.

SOLUTION 8

Aims of companies seeking quotation:

(i) To ease their raising of capital
(ii) More trading activities in the company's shares
(iii) Free publicity
(iv) Valuation of the company's securities
(v) Less risky in term of marketability.

Advantages of Quotations

(i) It facilitates the issue of securities to raise new finance, making a company less dependent upon retained earnings and banks.
(ii) Te wider share ownership which results will increase the likelihood of being able to make rights issues.
(iii) The transfer of shares becomes easier. Less of a commitment is necessary on the part of shareholders. For this reason the shares are likely to be perceived as a less risky investment hence will have a higher value.
(iv) The greater marketability and hence lower risk attached to a market listing will lead to a lower cost of equity and also to a weighted average cost of capital.
(v) A market determined price means that shareholder will know the value of their investment at all times.
(vi) The share price can be used by management as an indicator of performance, particularly since the share price is forward looking, being based upon expectations, whilst other objective measure are backward looking.
(vii) The shares of a quoted company can be used more readily as consideration in takeover bids.
(viii) The company increases its standing by being quoted and it may obtain greater publicity.
(Ix) Obtaining a quotation provides an entrepreneur with the opportunity to realize part of his holding in a company.

Disadvantages of Quotations

(i) The cost of obtaining a quotation is high, particularly when a new issue of shares is made and the company is small. This is because substantial costs are fixed and hence the relatively greater for small companies. Also, the annually cost of maintaining the quotation may be high due to such things as increase, disclosure, maintaining a larger share register, printing more annual reports etc.

(ii) The market determined price and the greater accountability to shareholders that comes with it concerning the company performance may not be liked by management.

(iii) Control of a particular group of shareholders may be diluted by allowing a proportion of shares to be held by the public.

(iv) The increased prosperity which may be perceived to accompany a quotation could prompt employees, management, auditors etc, to charge more for their services.

(v) The company may not be big enough, or growing fast enough for its shares to perform well on the stock market.

(vi) Listed company status will put extra administrative burdens and responsibilities on to the management'

(vii) The profit and performance of a public company are much more in the 'public eye' and the company cannot afford to show poor results, (unlike a private company which does not have to publish its results) without attracting considerable attention and criticism.

SOLUTION 9

Major Functions of a Stock Exchange

A market is a place where a commodity is bought or sold, it may be a new or second-hand market. Financial markets follow the same pattern. Texts suggest that financial markets discharge five major functions (1) shifting funds from suppliers (surplus units) to users (deficit users); (ii) liquefying securities (iii) pricing securities (iv) discounting the future; and (v) allocating funds and economic resources.

The Stock Exchange discharges each of these functions

(I) It provides a central market where money may be exchanged for securities (government bonds and company shares) thus helping to channel savers; funds to government and business. This new issue activity represents the primary market.

(Ii) Another function is to provide a secondary market (for liquefying securities)in which holders of securities are able to sell them. Confidence in having new issues is enhanced if the buyer knows that they have future marketability.

(iii) At a market place prices are being constantly determined. Supply and demand arises as buyers and sellers come together. The price agreed for a share takes into account the past history of the underlying company as well as the likely future benefits from owning the share. Similarly government bonds are priced as a stream of future payments of interest and a capital redemption sum, suitably discounted, interest rates, inflation and other economic factors play their part in the final price.

(iv) The future then is clearly discounted in the market which anticipates future economic trends and business behaviour. The market reflects people's expectations of the future. Buying and selling will take place according to whether there is likely to be expansion or depression in the economy.

(v) Finally the Stock Exchange offers an opportunity for choice between many investment be they short or long term. Those that are likely to give successful results are chosen.

Finally, it is suggested therefore that all the five functions of a financial market are discharged by the Stock Exchange.

SOLUTION 10

Functions of Security Exchange Commission

(i) Determining the amount of the price and time of which securities are to be sold;
(ii) Registration of all securities proposed to be offered for sale;
(iii) Monitors the capital market to ensure orderliness, fair and equitable;
(iv) Registering Stock Exchanges, their branches registers etc.
(v) Protecting the integrity of the Security Market.
(vi) Act as the Apex regulator for Capital Market;
(vii) Reviewing, approving and regulating mergers acquisitions and all forms of business combinations;
(viii) Creating conducive atmosphere for the orderly growth and development of the Capital Market.
(ix) Enforcing the necessary ordinances.

SOLUTION 11

Functions of Discount Houses

(i) To promote the rapid growth and efficiency of money market in Nigeria.
(ii) They act as an intermediary between the CBN and licensed banks.
(iii) To provide discount s and re-discount facilities for treasury bills, government securities and other eligible financial instrument acquired by banks.

(iv) They facilitate the issue and sale of short term government securities by tender.
(v) Provisions of facilitates for liquidity in the money market.
(vi) Provisions of facilities for a more efficient indirect monetary management.
(vii) Rediscounting treasury bills for banks
(viii) They serves as the lender of second to the last resort to banks while Central Bank plays the role of lender of last resort.
(ix) Enhances the maximization of profit by its clients through the provision of facilitates and instruments for the investment of otherwise idle short term funds.
(x) Provision of safe and secure facilities for client surplus short term funds.
(xi) Facilitating the issue and sale of short term government securities.

SOLUTION12

A market is efficient when:

* the prices of securities bought and sold reflect all the 'relevant' information which is available and therefore known to the buyers and sellers. In other words, share prices change quickly to reflect all new information about future prospects;
* No individual dominates the market.
* transaction costs of buying and selling are not so high as to significantly discourage trading;
* there is no possibility of speculative bubbles; to unrealistically high or low levels.

From the above, it follows that if the Stock Market is efficient we can predict how share prices and the prices of other securities should vary in a rational way; for example:

(a) if a company makes a profitable investment, shareholders will get to know about it, and the market price of its shares will rise in anticipation of future dividend increases;
(b) if a company makes a bad investment shareholders will find out and so the price of its shares will fall;
(c) if interest rates rise, shareholders will want a higher return from their investments, and therefore, market prices will fall (e.g suppose an investor expects the net dividend form his shares to give him a 10% return on the value of his investment, so that he is satisfied that shares offering his a dividend of 25k each should betworth N2.50 on the market. Not if interest rates subsequently rise, he will now expect a higher return himself, of say 12%, so that shares offering a dividend of 25% will now be worth only N2.08 each).

The efficient market hypothesis refers to the proposition that a particular stock market is an efficient market, at least to a greater or lesser extent.

The efficiency of a market has been categorized into three varying degrees or 'forms' of efficiency.

* weak form
* semi-strong form
* strong form

Weak form

The 'weak-form' hypothesis of market efficiency explain changes in share prices as the result of new information which becomes available to investor. In other words, share prices only change when new information about a company and its profits have become available. Share prices do not change in anticipation of new information being announced.

Semi-Stronged Form

Share prices reflect both:
- all relevant information about past price movements and their implications, but also
- all knowledge which is available publicly is relevant to the valuation of the share.

For example, if two companies plan a merger, share prices of the two companies will inevitably change once the merger plans are formally announced. The market would show semi-strong efficiency, however, if it were able to anticipate such an announcement, so that share prices of the companies concerned would change in advance of the merger plans being confirmed.

Strong Form

Share prices reflect all information available:
- from past price changes, and
- from 'public' knowledge/anticipation, and also
- from 'insider' knowledge 'available to specialists or experts (e.g investment managers).

The theory presupposes perfect make assumptions but allows reasonable transaction cost and does not required costless information.

Essentially, the theory implies that every characteristics of a company:
- its strategy
- its standards,
- its productive capabilities,
- its marketing process,
- its relationship with staff,

- its administration
- its earnings,
- its dividends,
- its expectations,
- its taxation
- its relationship with the community,
- its effect on the environment

All can be expressed in terms of potential future cashflows which can be added together and discounted to give the value of the company at a particular moment. Such value should be the presence rise at which bargains are struck and the buyer is presumed to be taking into account all this available information at the time of sale.

Only extra information will cause a buyer and seller to change the price and this is the crucial point.

SOLUTION 13

The Factors that a Financial Manager should take into Account when investing in marketable fixed interest investments. Most short-term investment in marketable securities include:

(i) Default risk: The risk that interest and/or principal will not be paid on schedule on fixed interest investment. Most short-term investment in marketable securities is confirmed to investments with negligible risk of default.

(ii) Price risk. The risk of the value of the investment changing for example when interest rates change. Financial managers normally wish to avoid substantial price risk.

(iii) Marketability. Securities should normally be marketable at short notice at close to the quoted market price.

(vi) Taxation. Are they any special tax effects of the seldom selected marketable securities?

(v) Yield. Managers will usually try to achieve the maximum yield possible consistent with a satisfactory level of risk and marketability.

(vi) Foreign exchange risk. If marketable securities are not denominated in the domestic currency of the investor, foreign exchange risk must be taken into account.

(vii) The amount of funds to be invested. Some types of investment require a minimum size of investment.

(viii) The period for which investment is to be made. The type of investment should be matched with the timing requirements of the future need for funds.

SOLUTION 14

The implications of Efficient capital market Hypothesis for Financial managers of a limited liability company's are enumerated as follows:

(i) **Timing of new Issues**

It is a commonly held view that it is most beneficial for a company to issue shares when the company's share price is relatively high and that a company should not raise finance when the share price is low. The higher the share price then the lower is the cost of the finance raised. This is not true if the market is efficient. It is impossible to say whether the market is relatively high or low, the current share price being the market's best estimate of the worth of the shares.

The timing of the issue of shares thus becomes irrelevant, whether the issue turns out to be cheap or expensive after the event is a matter of chance.

(ii) **Project evaluation**

When evaluating new projects financial managers usually use required rates of return drawn form securities traded on the capital market. For example, the rate of return required on a particular project may be determined by observing the rate of return required by shareholders of firms investing in projects of similar risks. This assumes that securities are fairly priced for the risk they carry - other words that the stock market is efficient. If this is not the case then financial managers could be appraising projects on the wrong basis and therefore making bad investment decisions.

(iii) **Creative accounting**

In an efficient stock market share price are based upon the expected future cash flows offered securities and their level of risk. In turn these expectations reflect all current information. There is little point in firms attempting to distort current information to their advantage as investors will quickly see through any such attempts. One American to test of the semi-strong form of the hypothesis examined the impact on share prices of companies moving from accelerated to straight-line depreciation. This change only increased reported accounting earnings; actual cash flows remained unaltered as tax allowable depreciation remained unchanged. Initially, share prices rose, possibly because the investors were immediately informed of the changes, but within three months share prices fell as investors concluded that the cosmetics alterations to earnings were a sign of weakness rather than strength. Other studies support this conclusion and it seems unlikely that investors can be 'fooled' by the manipulation of accounting profit figures. Eventually (and usually sooner rather than later) investors will realise the cash-flow consequences and later share prices appropriately.

(iv) **Validity of current market price**
If the market if efficient, statements such as 'XYZ shares are underpriced' are meaningless. The market reflects all known information in existing share prices and investors therefore know that if they purchase a security at current market price they are receiving a fair risk and return combination for their money. Financial managers should also known that if they sell new equities at current market prices they are raising funds at a fair cost. Therefore,, there should be no need for substantial discounts on new issues; if the current price is fair investors will need no great extra incentives to purchase the securities as in an efficient capital market they are unlikely to find better buyers elsewhere.

(v) **Merger and takeovers**
One of the main reasons why a takeover bid is made for a company is that its share price is considered to be too lo, enabling it to be bought cheaply. If the market is efficient, however, it is impossible for a share price to be too low. The implication of the efficient market hypothesis is that the purchase of shares is a zero NPV transaction and no abnormal profit is available from taking over a company merely because its shares price has fallen. This does not mea that takeovers cannot be profitable at all; other factor could make a takeover a positive NPV investment, e.g synergy, diversification, poor management, excess cash, under-utilization of assets, etc.

Conclusion
From the above discussions, it should be apparent that the relevance of the Efficient Capital Market hypothesis for the financial manager is that if the hypothesis holds true then the company's real financial position will be reflected in its share price, similarly a 'bad' decision will cause a drop in share price. Therefore in order to maximize shareholder's wealth, financial management need concentrate only on maximizing NPV of investments or projects.

SOLUTION 15

(A) **Term Loans:** It is an arrangement by which a customer is offered financial assistance of a fixed sum for a definite period. Term loans are usually liquidated or repaid in regular instalments covering principal and interest. The regular instalments are called amortization payments. Therefore, the project they are used to finance must be self liquidating.

(B) **Factoring:-** It refers to a contracted relationship whereby a person know as the factor undertakes to buy up a debt being owned another person called the principal creditors by a third party called the principal debtor. Factoring can be with or without recourse depending on the terms of agreement between the parties involved.

(C) **Leasing:-** It can be described as a means of obtaining the services of fixed assets for a fee. Banks do purchase equipment like computers, the services of which they make available to corporate bodies which needs the use of such fixed assets but have the necessary funds to acquire them. Generally, there are two main parties to a lease viz: the lessor, who owns the fixed asset on lease and the lessee who rents the asset from the owner for a fee.

(D) **Discounting of bills:** This is a process by which a customer is allowed to deposit his bill of exchange in his account and drawn on the amount to the credit of his account while paying a fee known as discounting charge. In this way, the customer of a bank has access to short-term source of funds.

(E) **Overdrafts:.** This is a kind of lending by banks to their customers which allows the customers to withdraw an amount which is in excess of the sum standing to the credit of their accounts. Interest is paid according to the amount actually used.

SOLUTION 16

(i) Secured credit is a short term sources of funds that refers to credit obtained from financial institution such as banks which includes overdraft and bank loans; while unsecured credit refers to credit obtained for the finance of a business not from financial institution such as friends, relatives and personal savings.

(ii) Factoring: This involves raising funds based on the debtors of the company so that cash is received earlier than if the company had waited for the debtors to pay; while Invoice Discounting is usually undertaken by firm who have lots of their funds tied up in receivables and have immediate need for the cash.

(iii) Risk can be defined as a situation where the probability of possible future outcome can be estimated; while uncertainty is a situation where the probability of possible future cannot be estimated.

(iv)

Operating Lease	**Finance Lease**
- It is cancellable.	- It is non-cancellable
- It covers a short period of the useful life of the best	- It covers a large period of the useful life of the asset.
- The lessor pays for the upkeep, maintenance and insurance of the asset.	- The lease pays for the upkeep, maintenance and insurance of the asset.

(v) Hire Purchase:- This is a medium source of finance in which an asset is bought and paid for installmentally (usually with interest). Ownership is not transferred until the final payment. While Sales and Lease back is a situation where a firm decides to sell one or more of its assets so as to source finance, and then it back from the company the asset was sold to.

(vi) Ordinary shares owned by an investors show the degree to which he or she owns the company. It is a long-term source of funds. The shareholders are entitled to dividends at the end of the financial year. The shareholders of the company usually bear the most risk and not entitled to dividends until the income claim of others have been settled. While Preference shares have many features of both ordinary Shares and Debentures. Preference shareholders have no voting rights and they cannot participate in extra-ordinary profit earned by the company. Preference share claim is honoured after theat of a debenture and before that of ordinary share.

SOLUTION 17

Ordinary shares:- When you invest in a firm through its ordinary shares, you automatically become a permanent part-owner of that firm. You are entitled to an unlimited dividend declared out of that firm's residual income, that is, profits after all other claims have been settled.

- Limits your liability to the amount of capital you have contributed in the extent of bankruptcy of the firm.
- It exposes you to double taxation

Preference Shares:- Preference shares do not qualify you as an automatic permanent part-owner of the firm. This is because your dividend are at fixed rates and the right to vote for he election of the firms directors at the AGM is not extended to you.

- Entitles you to a prior claim on the firm's income and assets before the ordinary shareholders.
- Are not tax deductible as compared to the ordinary shares.
- Sometimes, Preference shares may be converted into ordinary shares.
- It carries less risk than the ordinary shares.

Debenture Stock: The holders receive fixed interest rates on their investment. The interest on the loan stock is normally against the firm's revenue and not a share of the distributable profit.

- It can be secured with a collateral or unsecured
- It can be redeemable or irredemable
- Debentures can sometimes be convertible to ordinary shares.

Base on the above explanation, I will advise you to invest in the ordinary shares if you are a risk lover and if you are risk averter then invest in the Preference Shares or Debentures.

SOLUTION 18

(a)

Price tendered Price share	Number of Shares applied for at this price	Commutative Total of Share
350k	125,000	125,000
300k	150,000	275,000
275k	350,000	625,000
250k	625,000	1,250,000
225k	850,000	2,100,000
200k	1,000,000	3,100,000
175k	850,000	3,950,000
150k	350,000	4,300,000

The point at which the issue is fully subscribed is 200k (N2.00) also known as the striking price. Subscribers who submitted tenders at or above this price will receive the full number of shares they applied for at the striking price of 200k (N2.00). Those subscribers who tendered prices in excess of 200k (N2.00) will only have to pay the (200k). The subscribers who submitted tenders of 175k and 150k will not be allotted any shares.

(i) If the company chose to issue exactly 2,500,000 shares
:. The number of share issue will be 2,500,000 shares
The total amount raise will be number of share issued x Striking Price
=2,500,000 x N2.00
=N5,000,000

(ii) If the company chose to maximize the total amount raised
:. The number of share issue will be 3,100,000 shares
The total amount raised will be number of share issued x Striking Price
=3,100,000 x N2.00
=N6,200,000

(b) Barrister Daresi will receive 15,000 shares at Striking price of N2.00 per share in both cases.
Her balance or change will be
= Amount Paid - (Number of Share issued x Striking Price)
= N45,000 - (15,000 x N2.00)
= N45,000 - N30,000
= N15,000

(c) The likely market value will be the average price as shown below:-

Price Pi	Number of Shares issued (qi)	Piqi
N3.50	125,000	437,500
N3.00	150,000	450,000
N2.75	350,000	962,500
N2.50	625,000	1,562,500
N2.25	850,000	1,912,500
N2.00	400,000	800,000
	2,500,000	6,125,000

:.Likely market price = Average Price
= 6,125,000
2,500,000
=N2.45

SOLUTION 19

(a) ALHAJI AJOMALE TOTAL LIABILITIES

	N	N
Consideration = 20000 x N25.0		500,000
Add:		
Brokers commission (0.03 x N500,000)	150,000	
Stamps duty $\left(\frac{14}{20,000} \times N500,000\right)$	350	
SEC commission $\left(\frac{1}{8} \times \frac{1}{100} \times N500,000\right)$	625	15,975
TOTAL LIABILITY		515,975

(b) **HIS PROCEED**

	N	N
Consideration (20000 x N25.00		500,000
Less:		
Brokers commission (0.03 x N500,000)	150,000	
Stamps duty $\left(\frac{14}{20,000} \times N500,000\right)$	350	
SEC commission (1/8 x 1/100 x N500,000)	625	(15,975)
TOTAL LIABILITY		484,025

SOLUTION 20

(A) The theoretical ex-rights price can be computed using the following formula:

$$\text{Ex-Right Price} = \frac{\text{Market Price per Share before Right Issue} + \text{Proceed of Right Issue}}{\text{Total number of share with the Right issue}}$$

Market Value of Share
Before Right Issue = N20 x 100,000 = N2,000,000

Proceed of right Issue = 1/4 x 100,000 x N11.50

= N287,500

Total number of Shares
With the Right Issue = 100,000 + (1/4 x 100,000)
= 100,000 + 25,000
= 125,000

Ex-Right Price = $\frac{2,000,000 + 287,500}{125,000}$

= $\frac{2,287,500}{125,000}$

= N18.30

bB) Market Price per Share = N20.000
Rights Price = N11.50
Ex-rights price = N18.30

Her value of rights = Ex-right Price - Cost of taking up the right
= 18.30 - 11.50
= N6.80

(c) If she takes all he rights

	N	N
Value of shareholding ex-rights		2,287,500
Less:		
Value o shareholding before rights issue	2,000,000	
Cost of taking up the rights	287,500	(2,287,500)
Gain in wealth		NIL

Comment:
If she takes all her rights, the wealth and control of Miss Soludo will remain in tact.

Workings

(i) Value of shareholding ex-rights
= Total number of share with the Right Issue x Theoretical Ex-right Price
= 125,000 x N18.30
= N2,287,500

(ii) Value of shareholding before rights issue
= total number of shares x Market price per share before right issue
= 100,000 x N20.00
= N2,000,000

(iii) Cost of raking up the rights =
New share issue with right issue x Right issue price per unit
= 25,000 x N11.50
= N287,500

(c) (ii) If she sells all her rights

	N
Value of shareholding ex-rights	1,830,000
Less:	
Value of shareholding before right issues	(2,000,000)
	(170,000)
Add:	
Proceed on sale of her rights	170,000
Gain in wealth	0

Comment:
If she sell all her right, the wealth of Miss Soludo will remain in tact.

Workings

(i) Value of shareholding ex-rights = Total number of share Before ex-right x Theoretical Ex-right Price
= 100,000 x N18.30
= N1,830,000

(ii) Value of shareholding before right issue
= Total numebr of share Before right issue x Current Market Price per share before right issue
= 100,000 x N20
= N2,000, 000

(iii) Proceed on sale of her rights = Value of her right x New share issued
= N6.80 x 250,000
= N170,000

(c) (iii) If she sells 60% of her rights and acquired the remainder

	N	N
Value of shareholding ex-rights		2,013,000
Add Proceeds on Sales of 60% rights		102, 000
		2,115,000
Less:		
Cost of taking up rights	115,000	
Value of shareholding before Ex-right	2,000,000	(2,115,000)
Gains in wealth		0

Comment:
If she sells 60% of her rights and acquired the remainder, Miss Soludo wealth remain in tact.

Workings

(i) **Value of shareholding ex-right**
= Total number of share with 40% of her right x Theoretical Ex-right price
= [100,000 + (04x25,000)] x 18.30
= [100,000 + 10,000] x 18.30
110,000 x 18.30 = N2,013,000

(ii) **Proceed on Sales of 60% Rights** = 60% of her Rights x Value of a right
= 0.6 x 25,000 x 6.80
= N102,000

(iii) **Cost of taking up rights** = 40% of her right x right issue price per issue
= 0.4 x 25,000 x N11.50
= N115,000

(iv) **Value of Shareholding before Ex-right**
Total number of shares before Ex-right x Market Price per share
100,000 x N20
= N2,000,000

(C) (iv) If she does not take any position at all

	N
Value of shareholding ex-rights	1,830,000
Less value of shareholding before right issues	2,000,000
Loss of wealth	170,000

Comment:
If she does not take any position at all, Miss Soludo wealth will decrease by N170,000.

Workings

(I) **value of shareholding ex-right**
= (Number of share + 0% of her right) x Theoretical Ex-right Price
= (100,000 + 0) x 18.30
= 100,000 x N18.30
= N1,830,000

(Ii) **Value of shareholding before right Issue**
= Number of shares before right issue Market Price per share
= 100,000 x N20
= N2,000,000

SOLUTION 21

(I) Number of new shares to be sold
= Amount Required / Issued price
= N100,000,000 / N 30
= 3,333,333 shares

(Ii) Number of rights needed to purchase one new share =
Number of ordinary shares outstanding / Number of new shares to be sold
= 250,000,000 / 3,333,333
= 7.5 Rights

(Iii) The number of rights existing shareholder will receive

= Number of share to sell / Number of rights to purchase one new shares

= 3,333,333 / 7.5

= 444,444 Rights

(I) Ex-Rights Price = Market value of a share Before right issue + Proceed of Rights issue / Total number of shares with the right issue

= (35 x 25,000,000) + 100,000,000 / 25,000,000 + 3,333,333

= 875,000,000 + 100,000,000 / 28,333,333

= 975,000,000 / 28,333,333

= N34.41

SOLUTION 22

Number of share to sell

= Number of outstanding shares x $\frac{1}{5}$

= 5,000,000 x $\frac{1}{5}$

= 1,000,000 shares

∴ Ex-rights price = Market Value of share Before right issue + Proceed of Rights Issue / Total number of shares with the right issue

= (N26.52 x 5,000,000) + N25,000,000 / 5,000,000 + 1,000,000

= 157,600,000 / 6,000,000

= N26.27

∴ Ex-rights price of Bosede share is N26.27

(Ii) **value of the Right**

= Ex-right price - Cost of taking up the rights

Where

Cost of taking up the rights = Proceeds from right issue / Right issue

= 25,000,000 / 1,000,000

= N25.00

∴ Value of the right = N26.27 - N25.00

= N1.27

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